1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
13	0002231663	ENGINEERING RELEASED	2013-08-10

J16 MLB

LAST_MODIFIED=Sat Aug 10 21:13:36 2013

DRAWING
TITLE=J16
ABBREV=DRAWING
LAST_MODIFIED=Sat Aug 10 21:13:36 2013

DRAWING TITLE: SCHEM,MLB,J16

Apple Inc.

DRAWING NUMBER: 051-9889 | SIZE: D
REVISION: 13.0.0
BRANCH:

DRAWING

8	7	6	5	4	3	2	1

Main BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
985-0035	PCBA,MLB,DEV,J16	DEVELOPMENT,J16_DEVEL
639-4281	PCBA,MLB,107GX,VRAM_HYNIX,BETTER,J16	J16,J16_COMMON,CPU:BETTER,GPU:107GX,FBA,FBB,FB:BOTH_HYNIX,SSD:Y,EEEE:F9RR
639-4286	PCBA,MLB,107GX,VRAM_ELPIDA,BETTER,J16	J16,J16_COMMON,CPU:BETTER,GPU:107GX,FBA,FBB,FB:BOTH_ELPIDA,SSD:Y,EEEE:F9T5
639-4282	PCBA,MLB,107GX,VRAM_HYNIX,CTO,J16	J16,J16_COMMON,CPU:CTO,GPU:107GX,FBA,FBB,FB:BOTH_HYNIX,SSD:Y,EEEE:F9T3
639-4287	PCBA,MLB,107GX,VRAM_ELPIDA,CTO,J16	J16,J16_COMMON,CPU:CTO,GPU:107GX,FBA,FBB,FB:BOTH_ELPIDA,SSD:Y,EEEE:F9RW

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7896	1	MLB LABEL,2D	EEEE_F9RR	CRITICAL	EEEE:F9RR
825-7896	1	MLB LABEL,2D	EEEE_F9T5	CRITICAL	EEEE:F9T5
825-7896	1	MLB LABEL,2D	EEEE_F9T3	CRITICAL	EEEE:F9T3
825-7896	1	MLB LABEL,2D	EEEE_F9RW	CRITICAL	EEEE:F9RW

Schematic / PCB #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-9889	1	SCH,MLB,J16	SCH	CRITICAL	J16
820-3482	1	PCBF,MLB,J16	PCB	CRITICAL	J16

BOM Groups

BOM GROUP	BOM OPTIONS
J16_COMMON	COMMON,ALTERNATE,J16_COMMON1,J16_COMMON2,J16_PROGPARTS,J16_PRODUCTION
J16_COMMON1	XDP,SPEAKERID,TBTHV:P12V,CPUVCC:3PHASE,EXT_GPU:YES,VDDQ:P1V5
J16_PROGPARTS	SMC:PROG,BOOTROM:PROG,T29ROM:PROG,CIVROM:PROG,CAMROM:PROG
J16_DEVEL	XDP_CONN,LPCPLUS,DDRVREF_DAC,DEVEL_SENSORS,DEVEL_AUDIO
DEVEL_SENSORS	AP_ISNS:Y,HDD_IVSNS:Y,TEMPSNSDEV
J16_PRODUCTION	AP_ISNS:N,HDD_IVSNS:N

ADD 'J16_PRODUCTION' AT REVA RELEASE

GPU & VRAM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4427	1	IC,GPU,NV,GK107-GX,PS,926MHZ,2.5GHZ	U8700	CRITICAL	GPU:107GX
333S0630	4	IC,GDDR5,2GBIT,64MX32,5GBPS,GEMMA-DIE	U9200,U9250,U9300,U9350	CRITICAL	FB:BOTH_HYNIX
333S0695	4	IC,GDDR5,2GBIT,64MX32,5GBPS,B-DIE	U9200,U9250,U9300,U9350	CRITICAL	FB:BOTH_ELPIDA

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4610	1	HSW,SR14J,PRQ,C0,2.9G,65W,4+2,1.15,6M,LGA	CPU	CRITICAL	CPU:BETTER
337S4608	1	HSW,SR14H,PRQ,C0,3.1G,65W,4+2,1.2,8M,LGA	CPU	CRITICAL	CPU:CTO

Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB Diode Array
377S0124	377S0057		ALL	TVS
376S0975	376S1081		ALL	P/NCh dual FET
155S0578	155S0367		ALL	120OHM EMI BEAD
128S0368	128S0365		ALL	150UF AL POLY
128S0298	128S0293		ALL	330 UF AL POLY
138S0681	138S0638		ALL	Taiyo 10uf 805 alt
197S0481	197S0480		Y1905	25MHz PCH Xtal
197S0464	197S0477		Y9700	27 MHz GPU Xtal
197S0466	197S0477		Y9700	27 MHz GPU Xtal
197S0479	197S0478		Y4200	12 MHz Cam. Xtal
341S3913	341S3912		U3990	Enet ROM,ATMEL,V1.15
377S0155	377S0104		ALL	USB Diode
138S0860	138S0775		ALL	Single-source 1uF 402
138S0859	138S0788		ALL	Single-source 10uF
107S0251	107S0249		ALL	Sense resistor
138S0648	138S0652		ALL	4.7uF GFX decoupling
138S0746	138S0705		ALL	10uF alt;Audio
138S0715	138S0740		ALL	4.7uF GFX decoupling
107S0254	107S0241		ALL	5mOhm Sense Resistor
107S0250	107S0248		ALL	3mOhm Sense Resistor

R5400,R5520,R5530

CPU Socket

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0080	1	SOCKET,MOLEX,LGA1150,CPU-LF	U0500	CRITICAL	

ASIC Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4541	1	IC,LPT-D,SR198Z87,C2,PRQFCBGA708,23X22MM	U1100	CRITICAL	
338S1113	1	IC,TBT,CR-4C,B1,PRQ,CIO,288 12X12 FC-CSP	U2800	CRITICAL	
343S0616	1	IC,BCM57766A,CIV+,A0,8X8	U3900	CRITICAL	
353S3908	1	IC,LP8561,LED BLKT CTLR,LLP24,B0-F	U8100	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3928	1	IC,EFI,V0112,J16G	U5210	CRITICAL	BOOTROM:PROG
335S0807	1	IC,64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S3903	1	IC,SMC-A3,V2.16A4,PROTO0,J16G	U5000	CRITICAL	SMC:PROG
338S1159	1	IC,SMC12-A3,40MHZ/50MIPS,SCPL FW,157BGA	U5000	CRITICAL	SMC:BLANK
341S3859	1	IC,TBT,EEPROM,CR,V23.10,J16	U2890	CRITICAL	T29ROM:PROG
335S0865	1	IC,EEPROM,SERIAL,256KB,MLP8	U2890	CRITICAL	T29ROM:BLANK
341S3912	1	IC,ENET SPI ROM,NUMONYX,V1.15,J16/J17	U3990	CRITICAL	CIVROM:PROG
335S0862	1	IC,SERIAL FLASH,2MBIT,2.7V,REV F	U3990	CRITICAL	CIVROM:BLANK
341S3778	1	IC,CAMERA,FLASH,V7230,J16/J17	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK

NEED NEW EFI,SMC PROG PARTS

SYNC_MASTER=MASTER SYNC_DATE=04/21/2013

PAGE TITLE

BOM Configuration

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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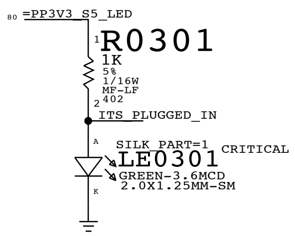
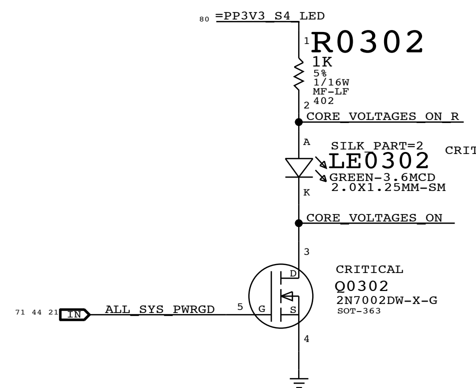
8	7	6	5	4	3	2	1

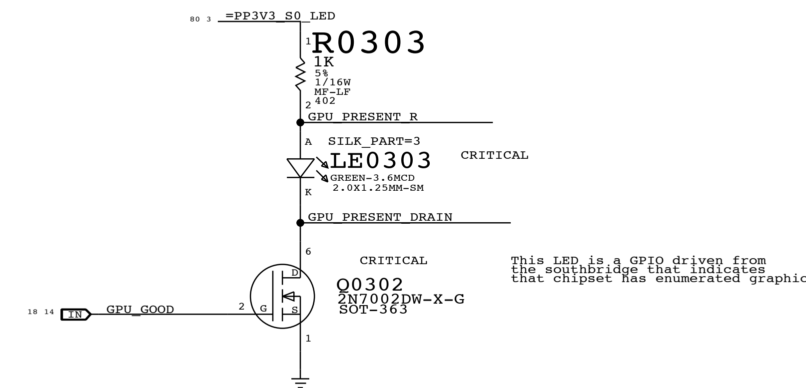
S5 Led



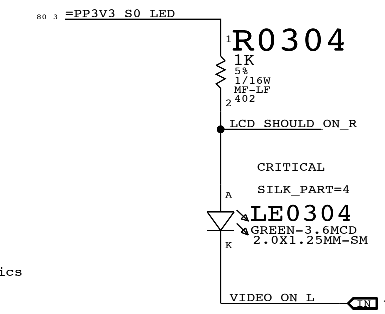
ALL_SYS_PWRGD Led



GPU GOOD Led



VIDEO ON Led



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

DEBUG LEDS

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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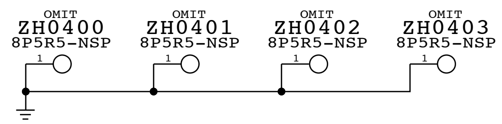
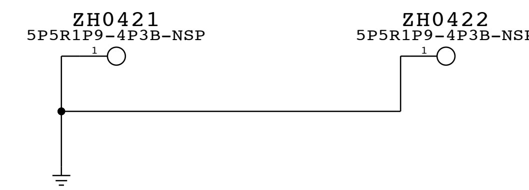
CPU Heatsink

4mm Plated Holes (998-0850)



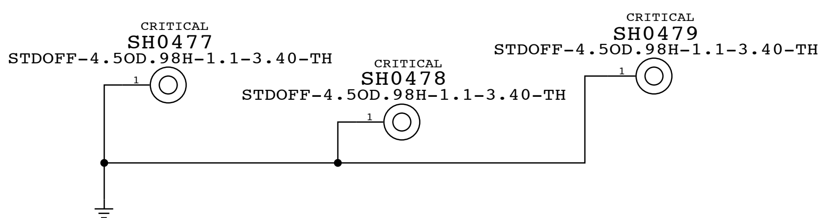
WIRELESS CARD MTG HOLES

998-4560 (Plated holes, 2.3mm inner diameter, 4.3mm pad)



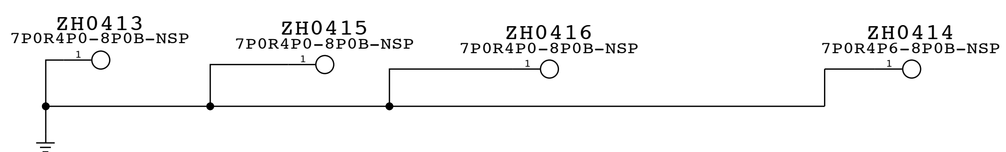
GPU HEATSINK MOUNTING FEATURES

(860-1532)



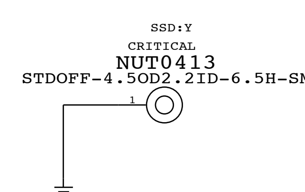
Rear Cover

998-4559 (Plated holes, 4mm inner diameter, 8mm pad)
998-5089 (ZH0414) near BLC has slightly larger hole to allow for grommet



SSD STANDOFF

APN: 860-1624



SYNC MASTER=J16 DG SYNC DATE=04/21/2013

PAGE TITLE
Holes/PD parts

Apple Inc.

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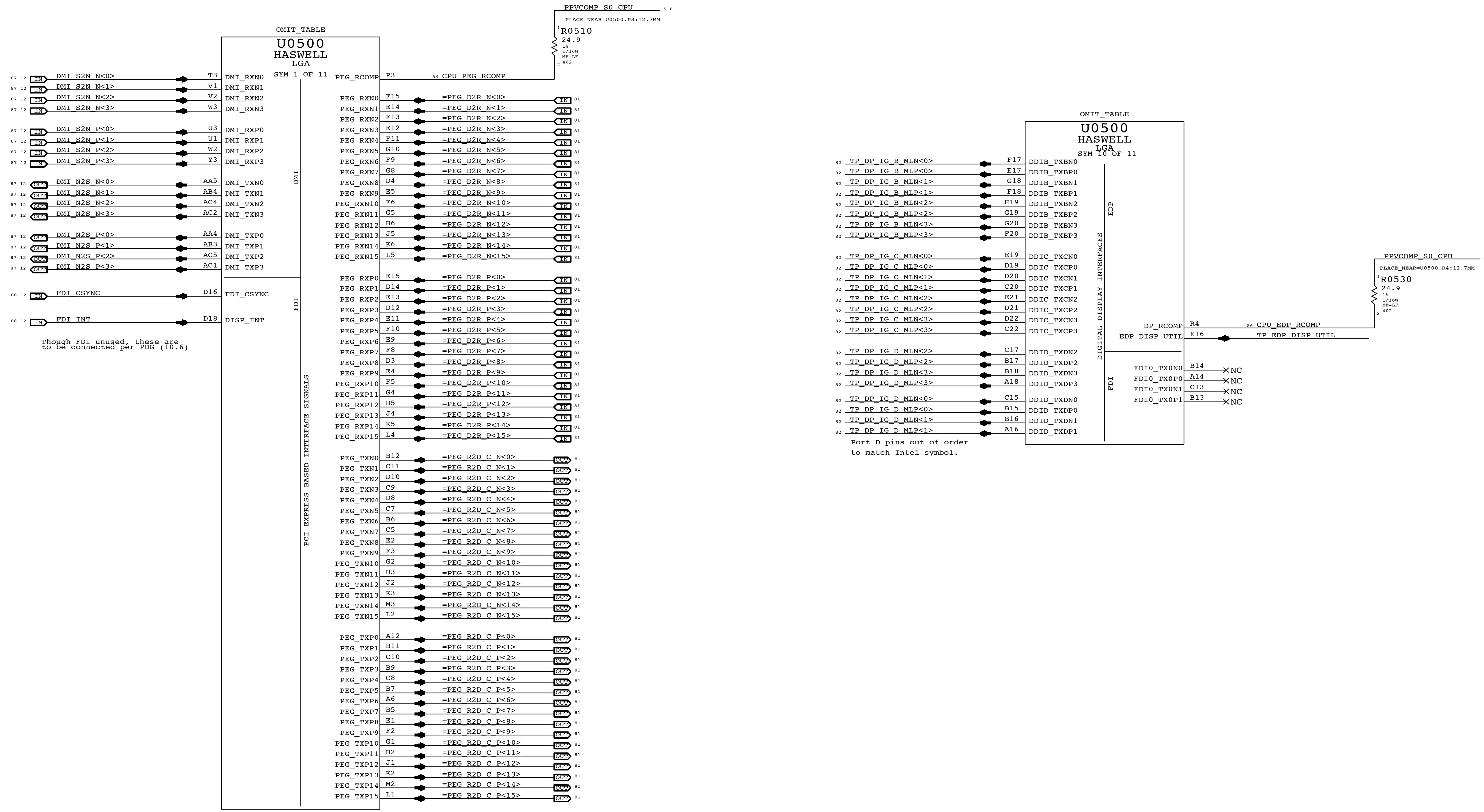

SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

CPU DMI/PEG/FDI/RSVD

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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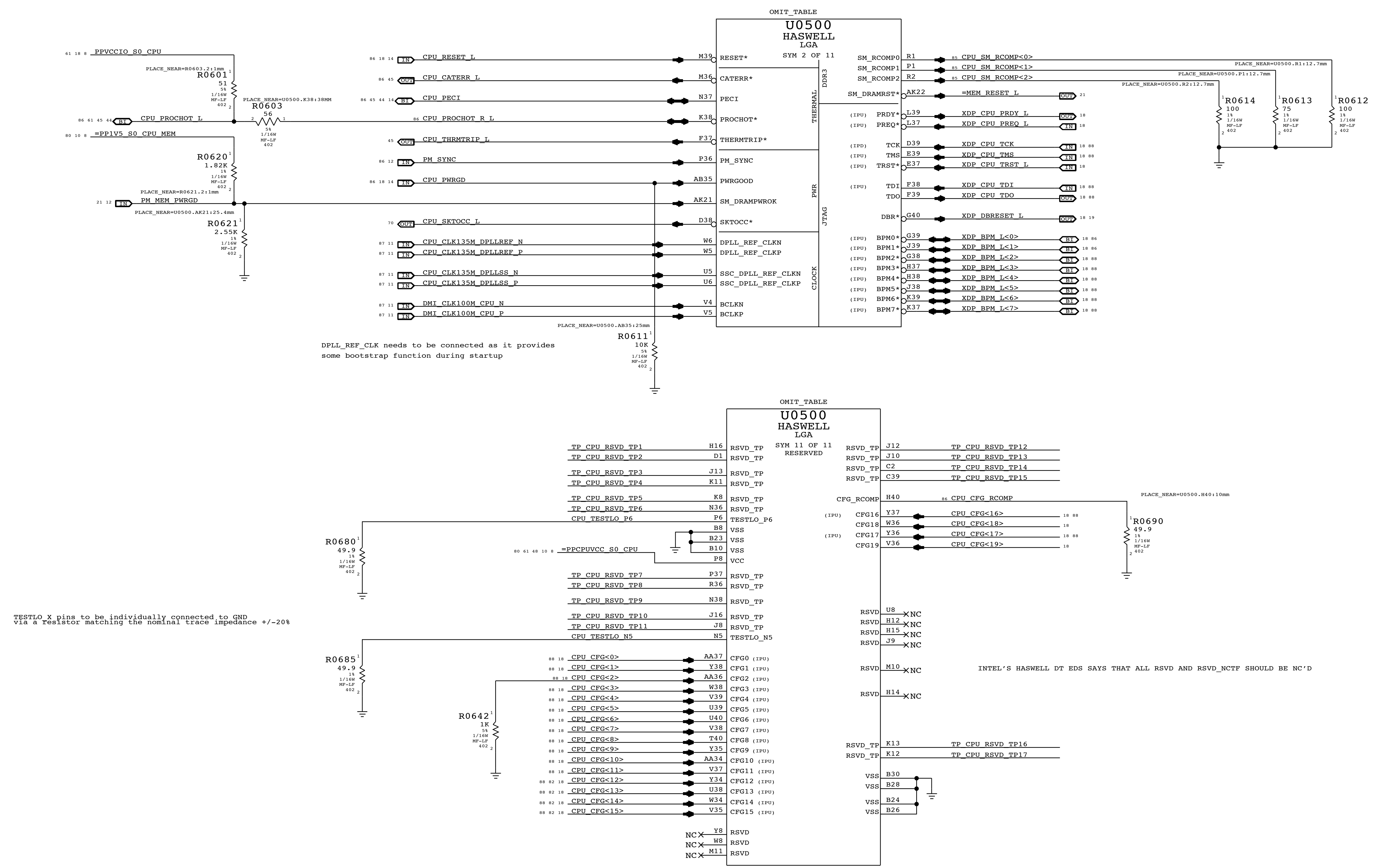

CFG [7] :PEG DEFER TRAINING 1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION 11 = 1 X16 (default) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE 1 = DISABLED(default) 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL 1 = NORMAL OPERATION(default) 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL 1 = NORMAL OPERATION(default) 0 = LANES REVERSED

SYNC MASTER=J17 SYNC DATE=04/21/2013

PAGE TITLE

CPU Clock/Misc/JTAG/CFG

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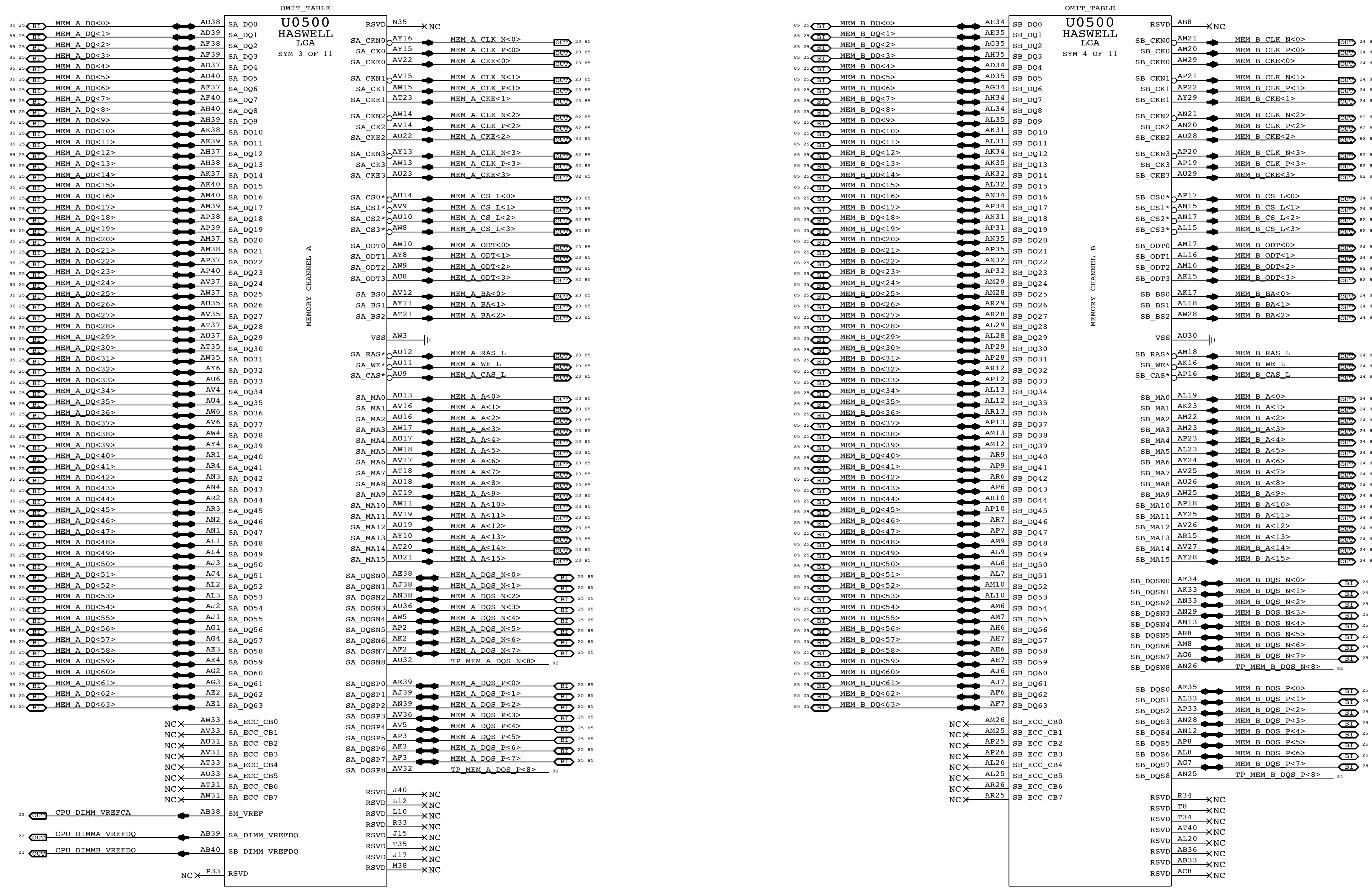

INTEL'S HASWELL DT EDS SAYS THAT ALL RSVD AND RSVD_NCTF SHOULD BE NC'D

SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

CPU DDR3 Interfaces

Apple Inc.

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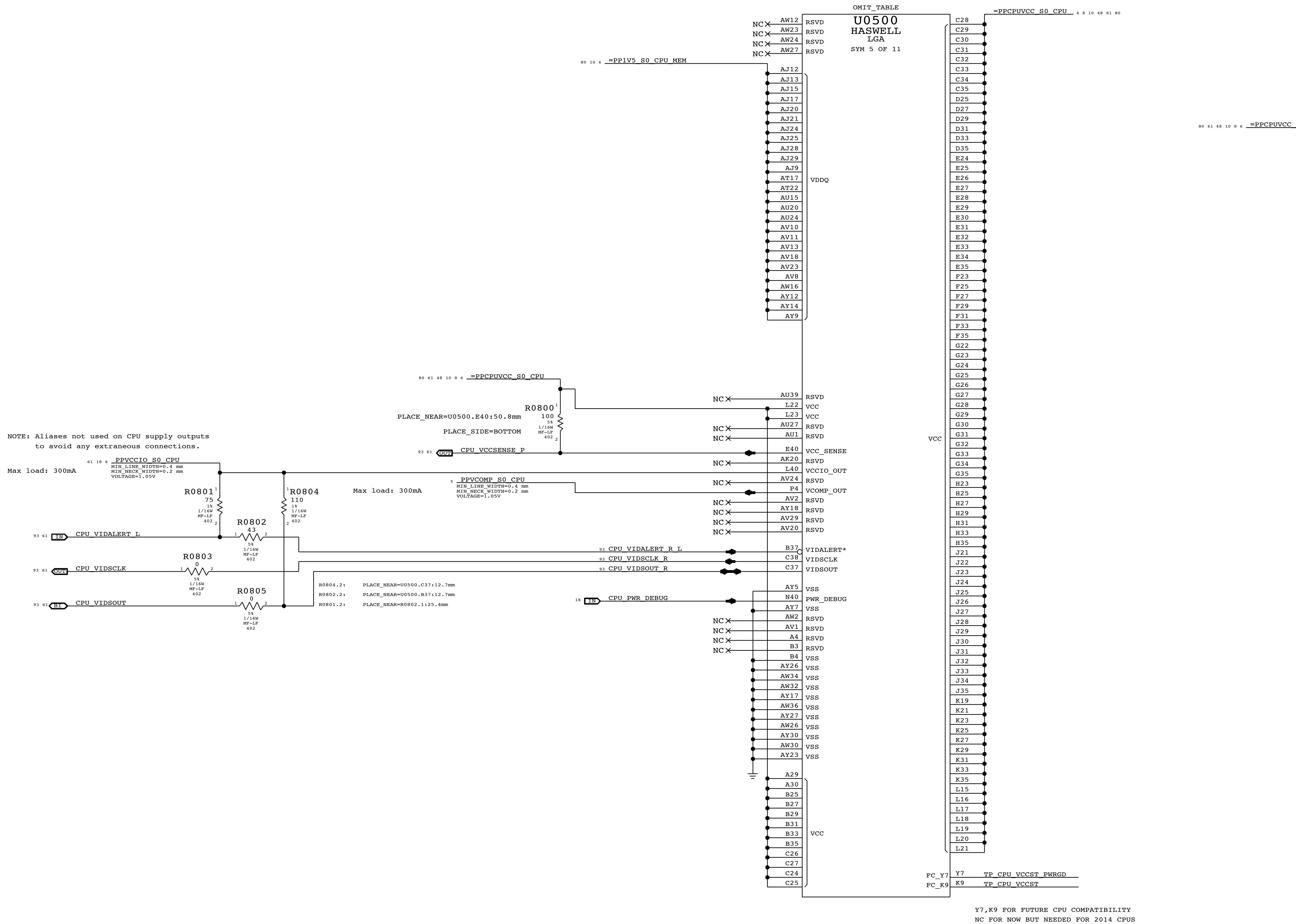

SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

CPU Power

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

BRANCH

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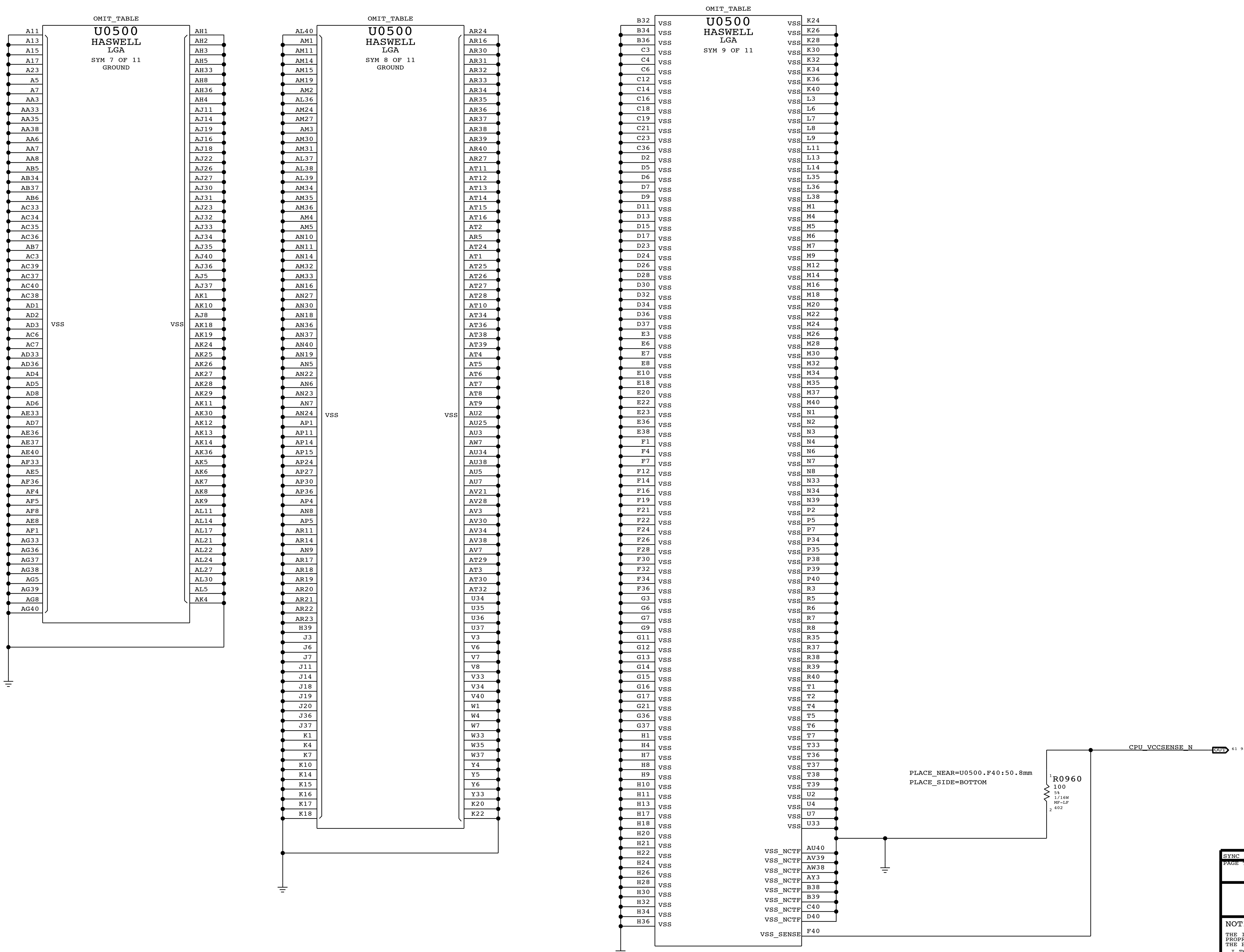

SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

CPU Ground

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

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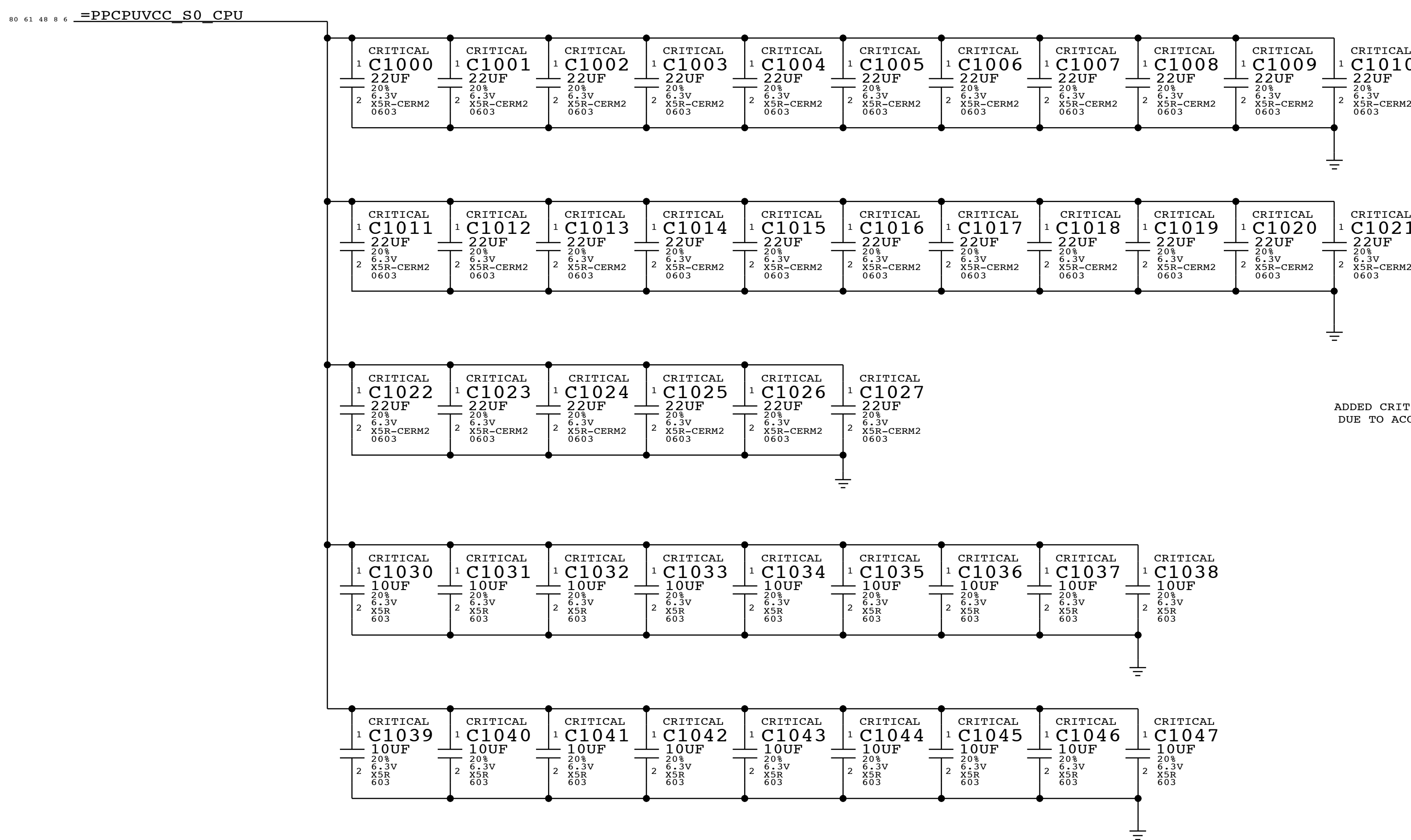
CPU VCORE DECOUPLING

Intel Recommendation:22x 22UF 0805,topside (18 inside cavity, 4 north of processor),8x 470uF bulk caps(5 stuffed,3 no-stuffed)
Apple Implementation:28x 22UF 0603 per Harold
18x 10UF 0603 placed inside socket cavity

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

BULK CAPS ON CPU VREG PAGE 71

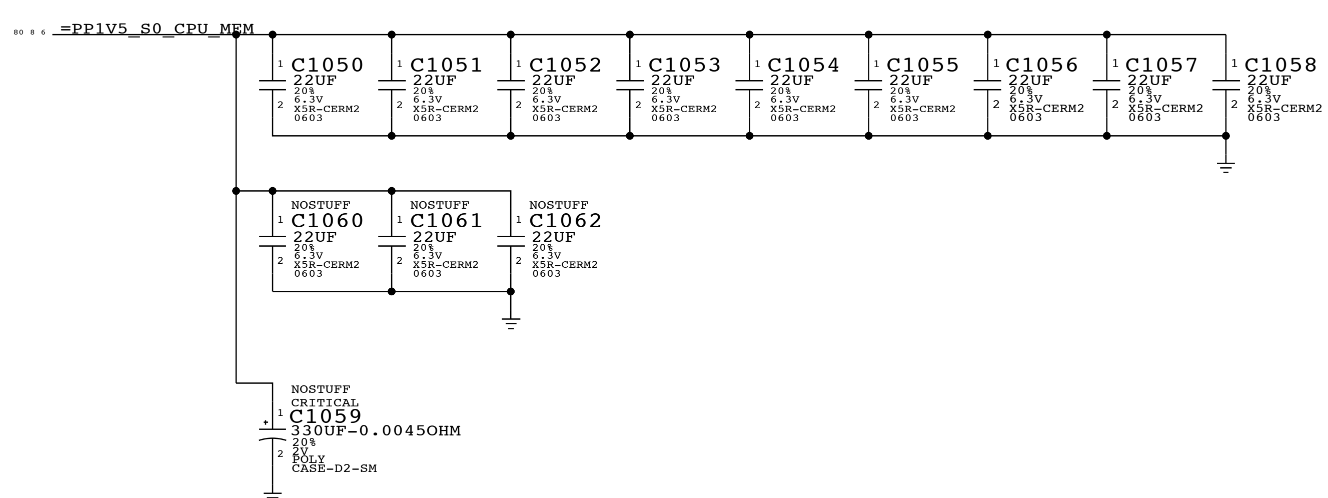


ADDED CRITICAL PROPERTY TO CPU CORE DECOUPLING
DUE TO ACOUSTICS CONCERNS

Memory (CPU VCCDDR) DECOUPLING

Intel Recommendation:9x 22UF 0805 near CPU power pins
Apple Implementation:9x 22UF 0603 per Harold

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE
CPU DECOUPLING

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

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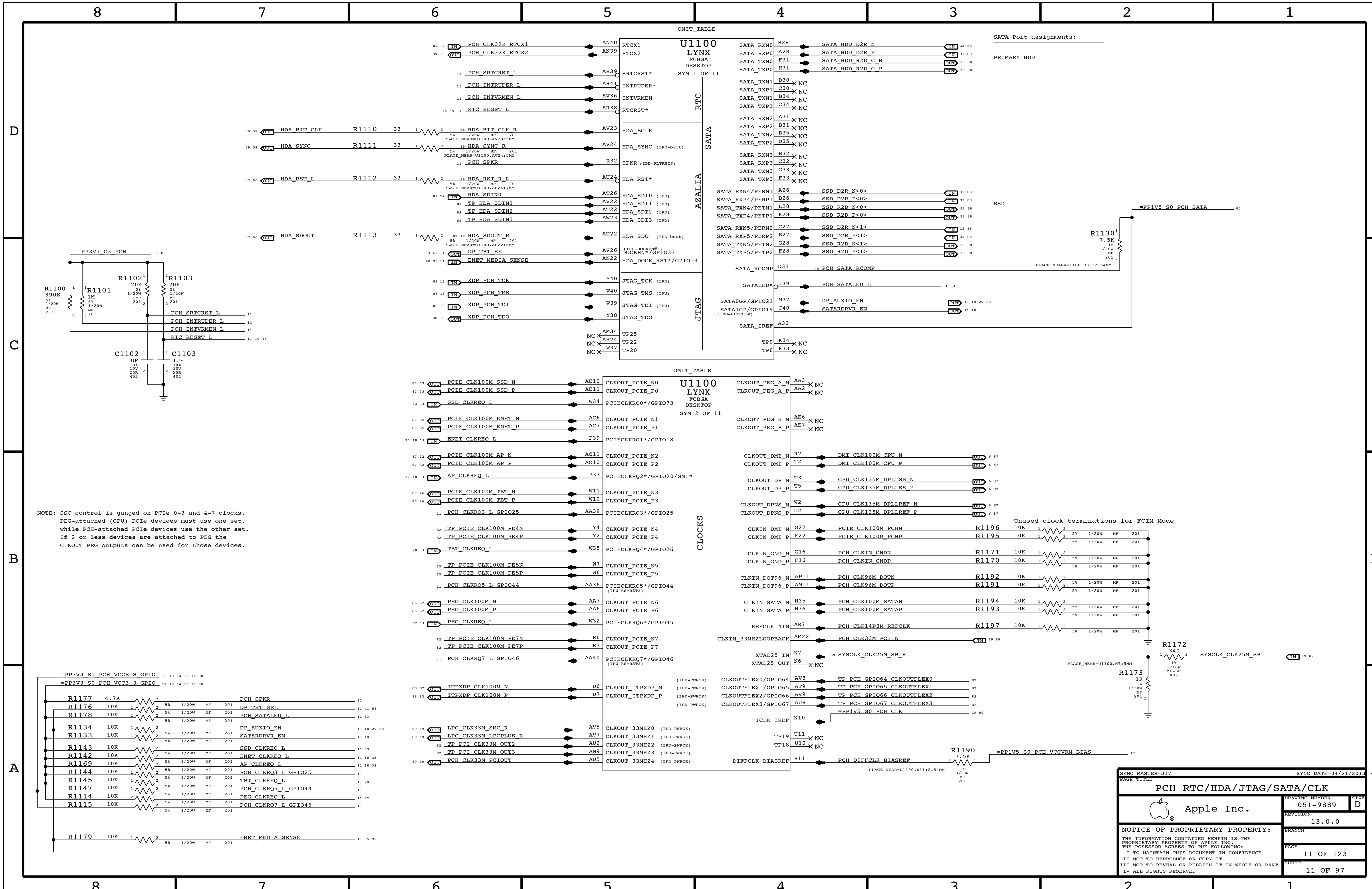

SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

PCH RTC/HDA/JTAG/SATA/CLK

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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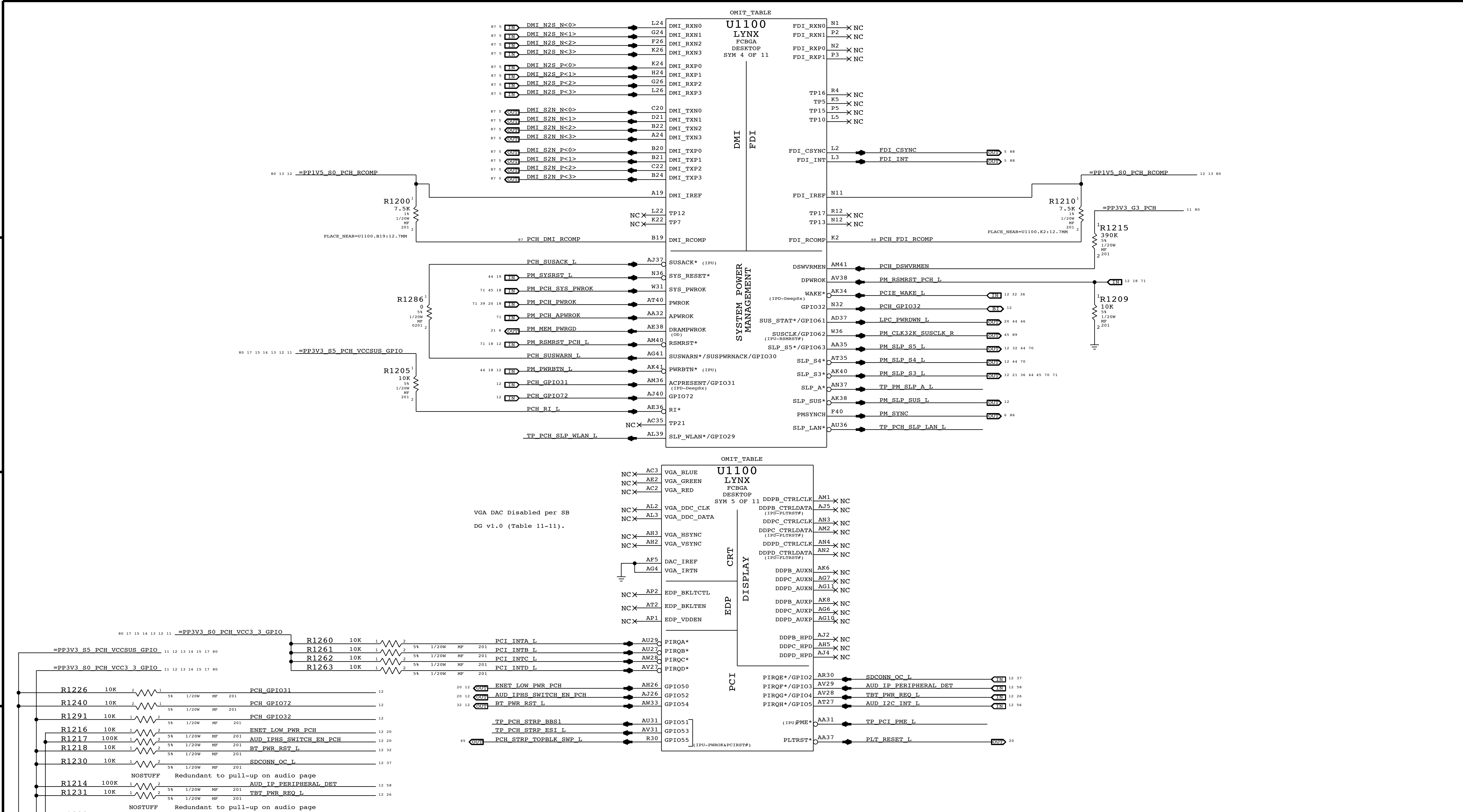


SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

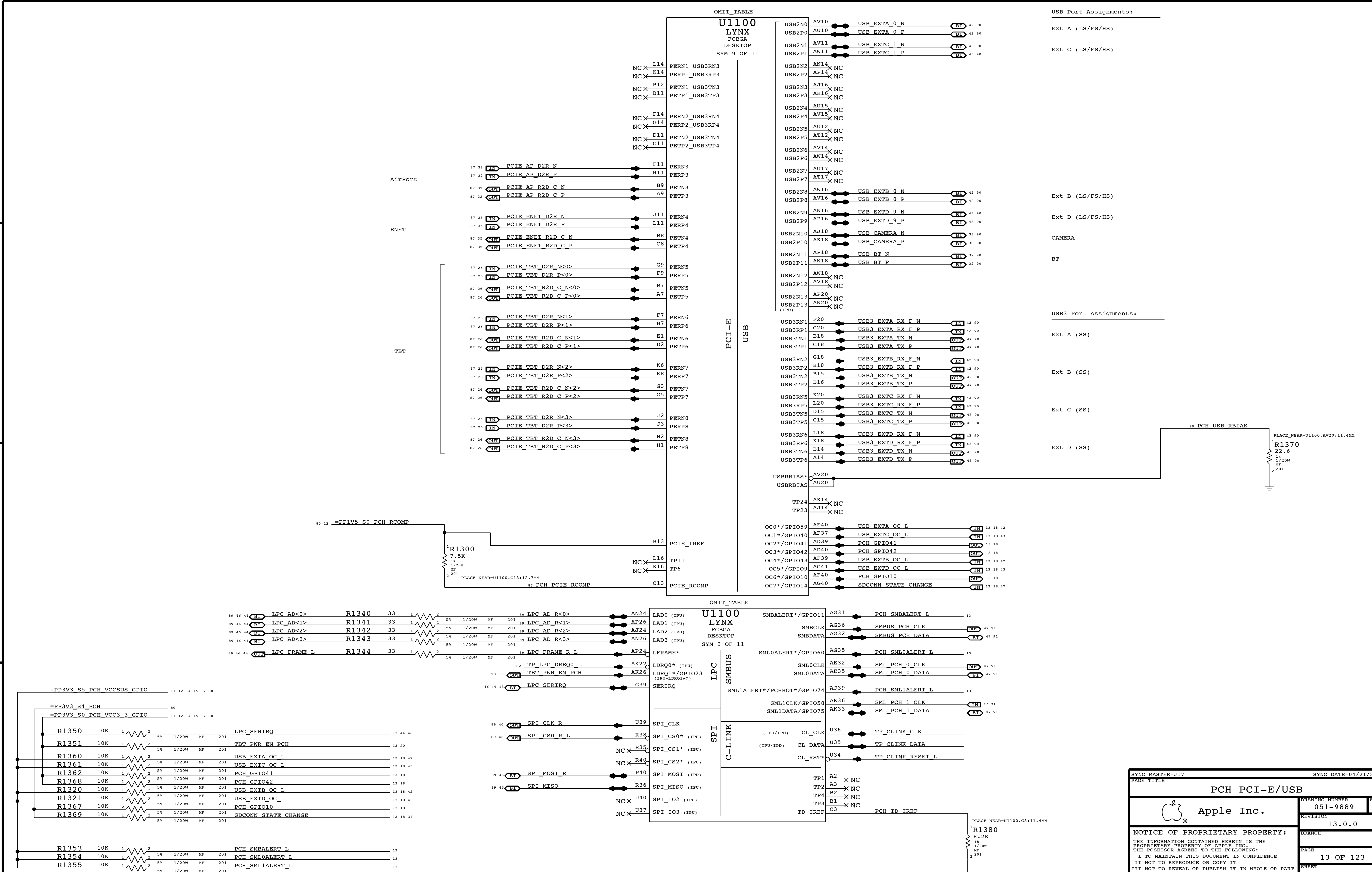
PCH DMI/FDI/PM/GFX/PCI

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

BRANCH



SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

PCH PCI-E/USB

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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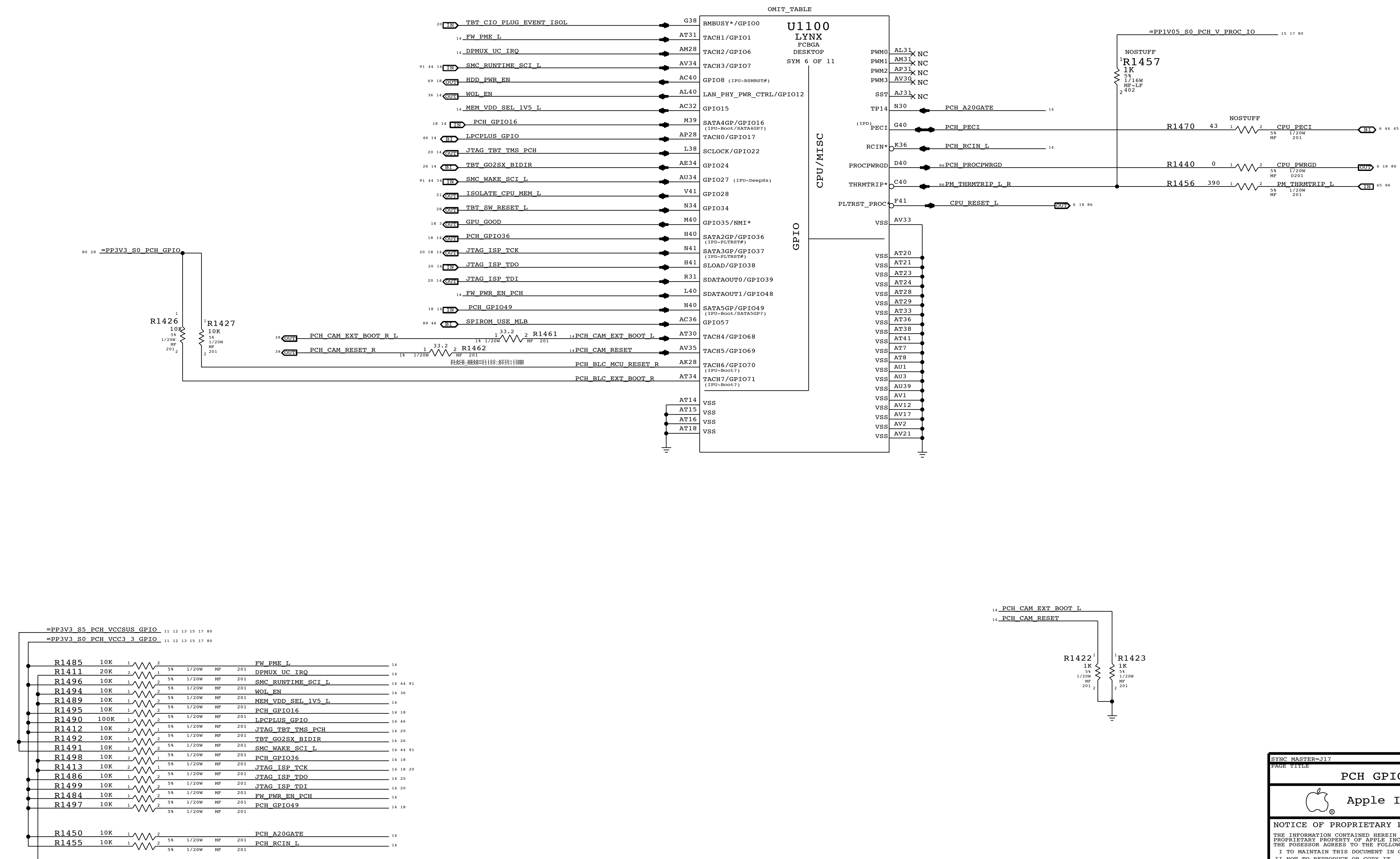

D C B A

SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

PCH GPIO/MISC/NCTF

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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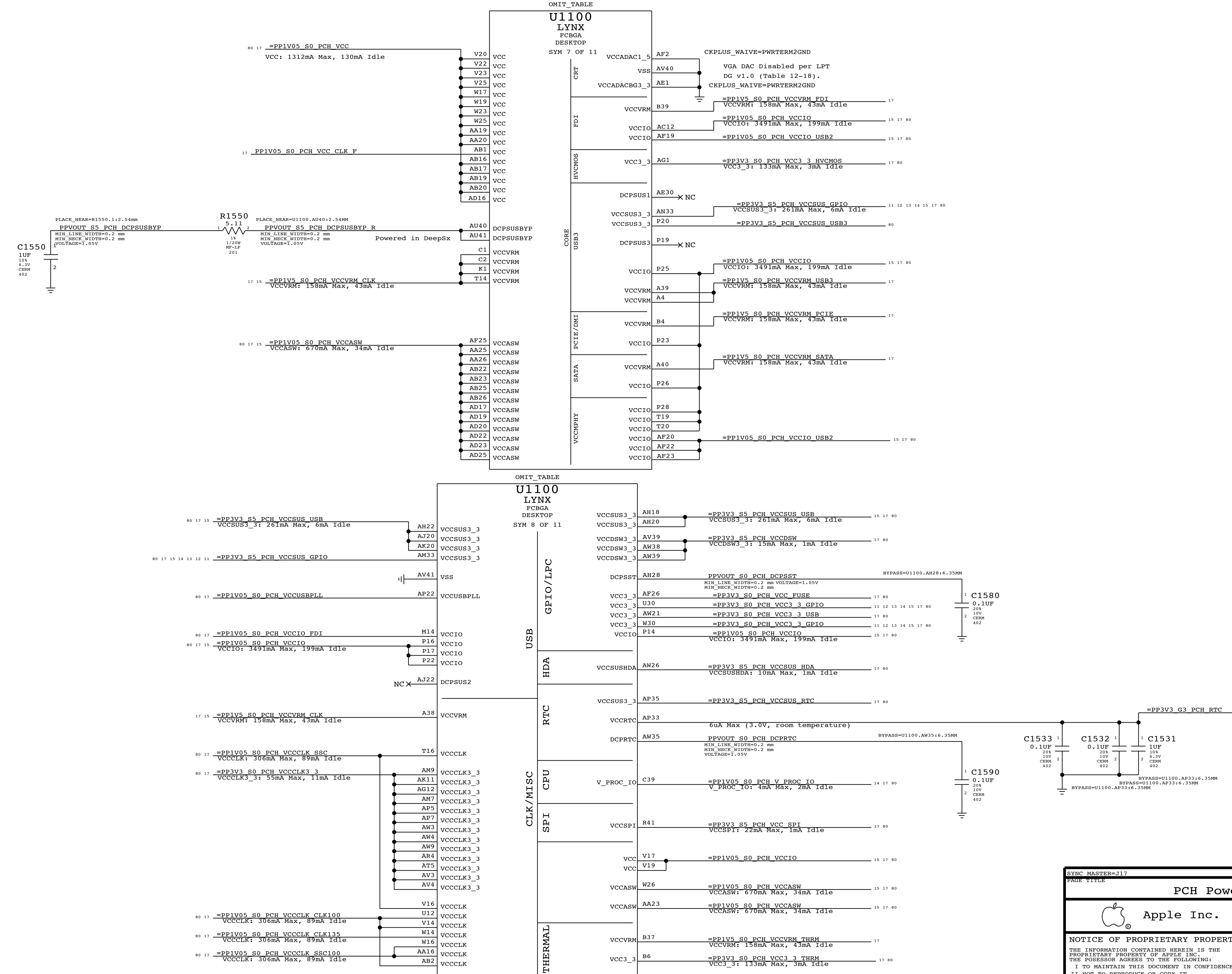

Current data from LPT EDS (doc #486708, Rev 1.0).

SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

PCH Power

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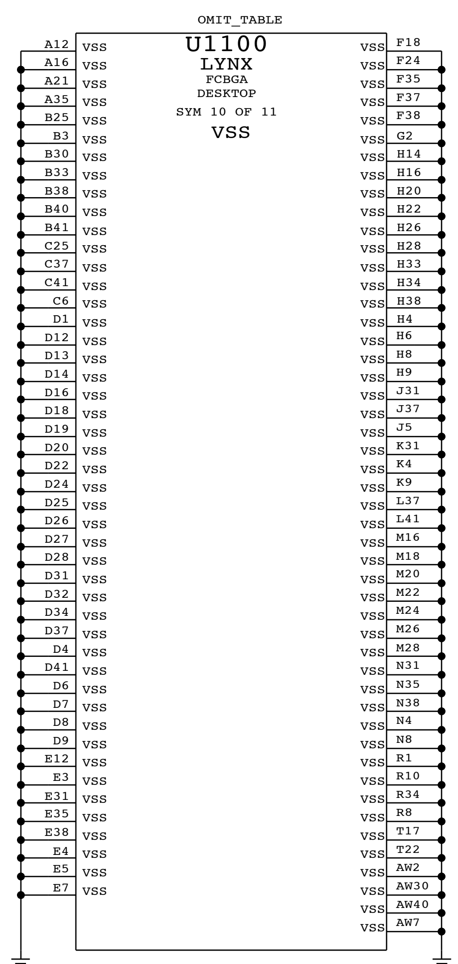

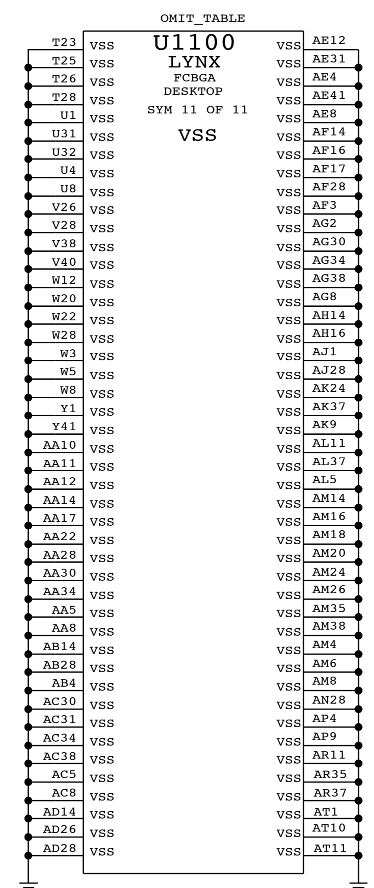

SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE

PCH Grounds

Apple Inc.

DRAWING NUMBER 051-9889

SIZE D

REVISION 13.0.0

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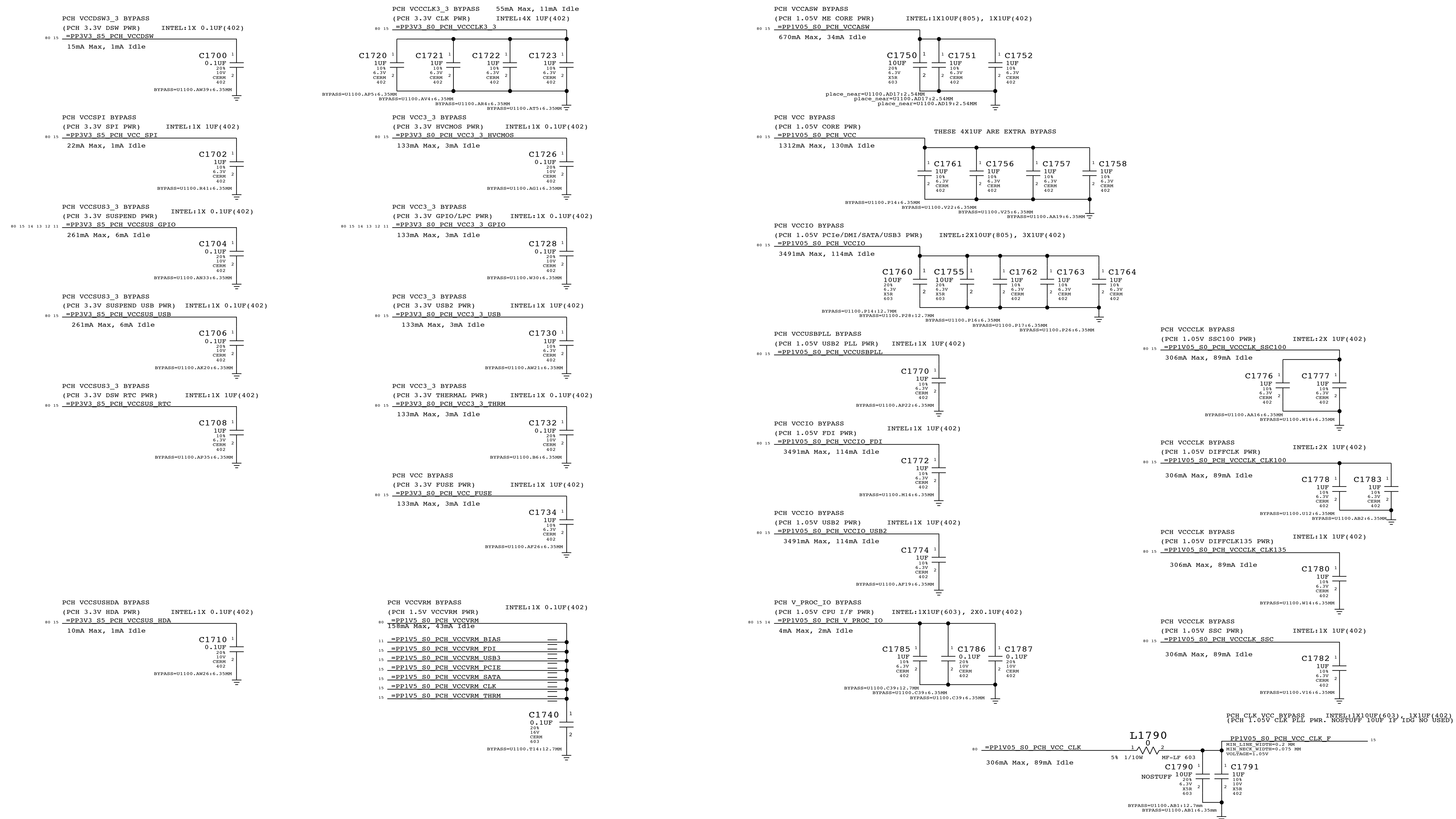
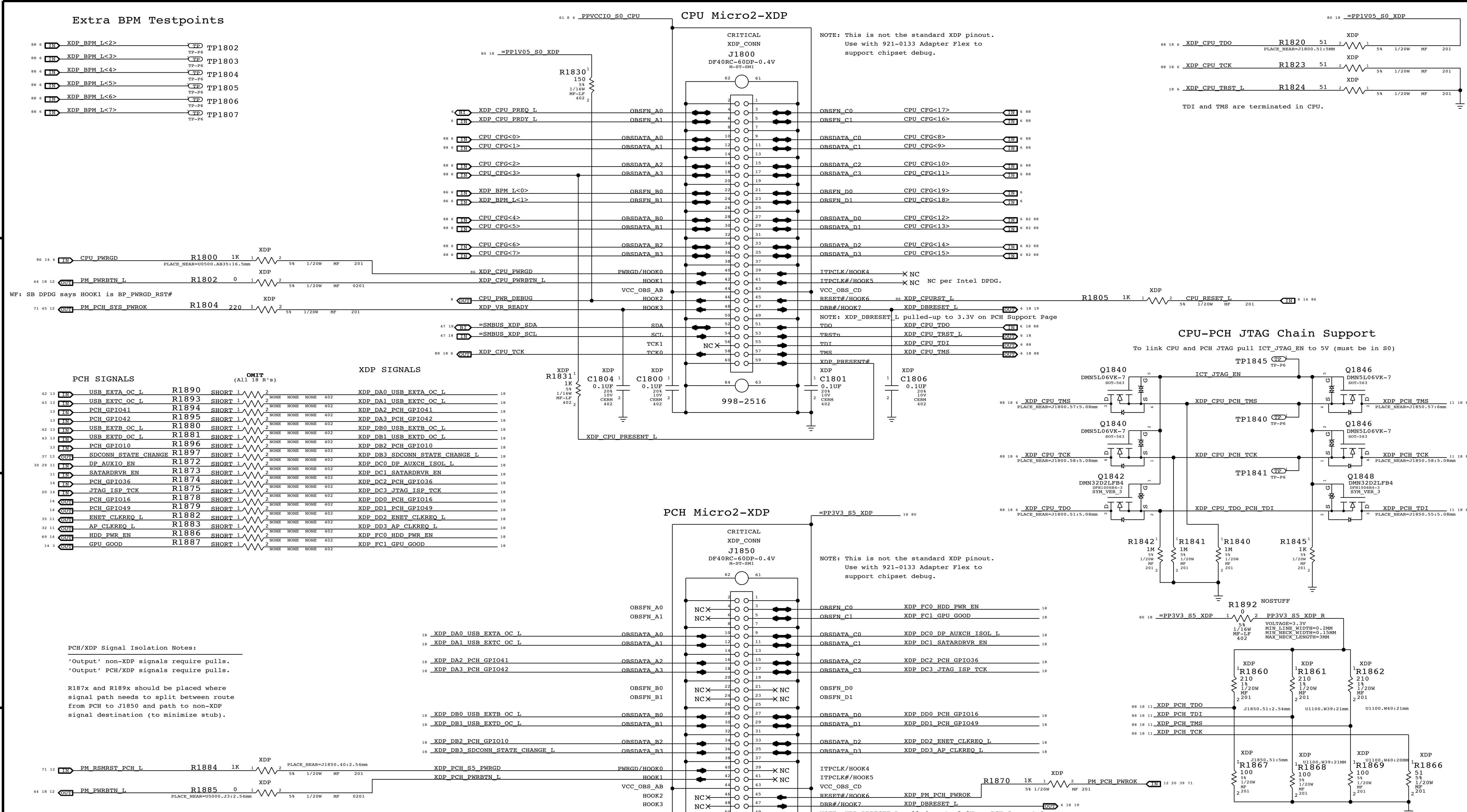


SYNC_MASTER=J17	SYNC_DATE=04/21/2013
PAGE TITLE	PCH DECOUPLING
Apple Inc.	DRAWING NUMBER 051-9889 SIZE D
	REVISION 13.0.0
	BRANCH
	PAGE 17 OF 123
	SHEET 17 OF 97

Current data from LPT EDS (doc #486708, Rev 1.0).



SYNC_MASTER=MASTER SYNC_DATE=04/21/2013

PAGE TITLE

CPU & PCH XDP

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DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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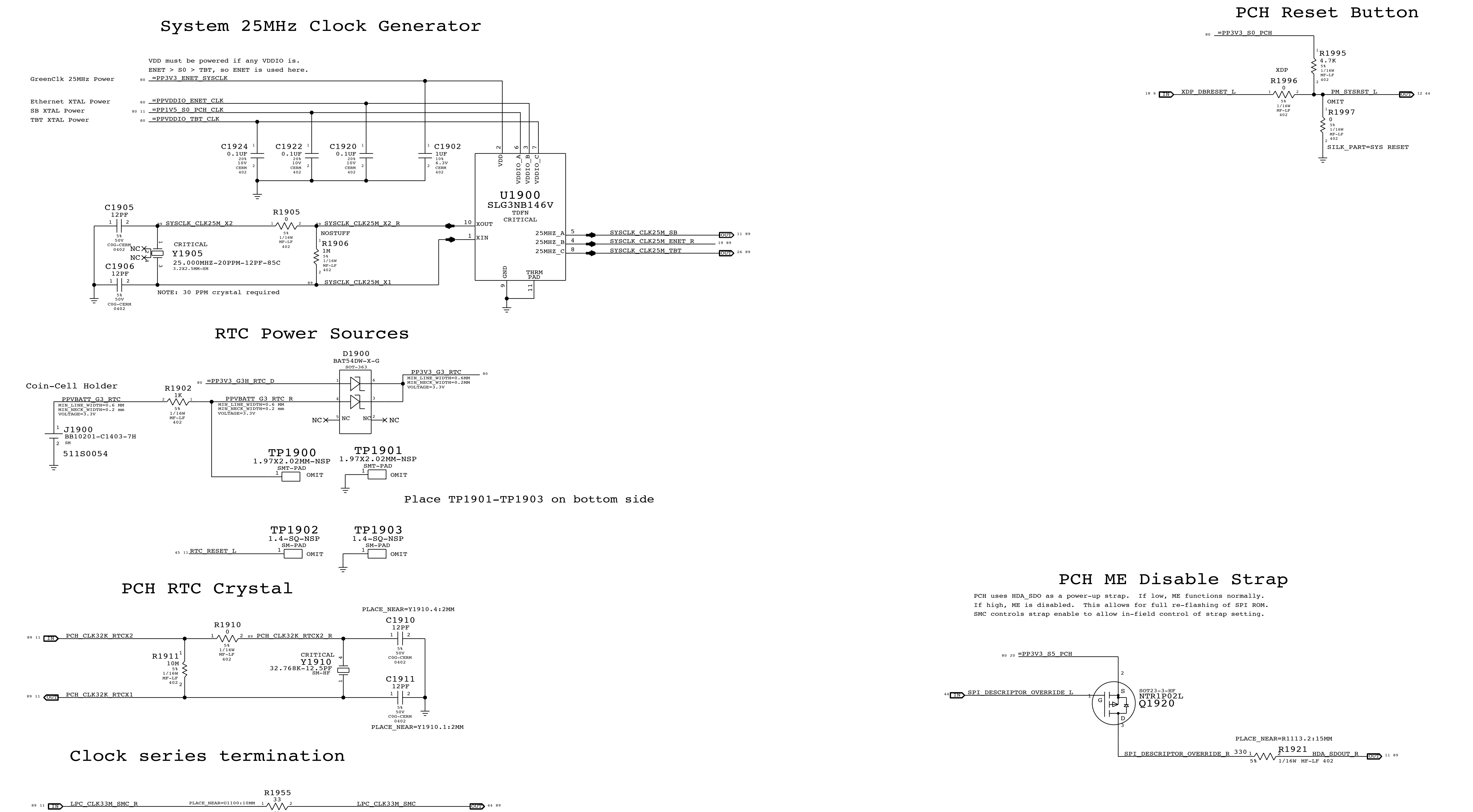


SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE: Chipset Support

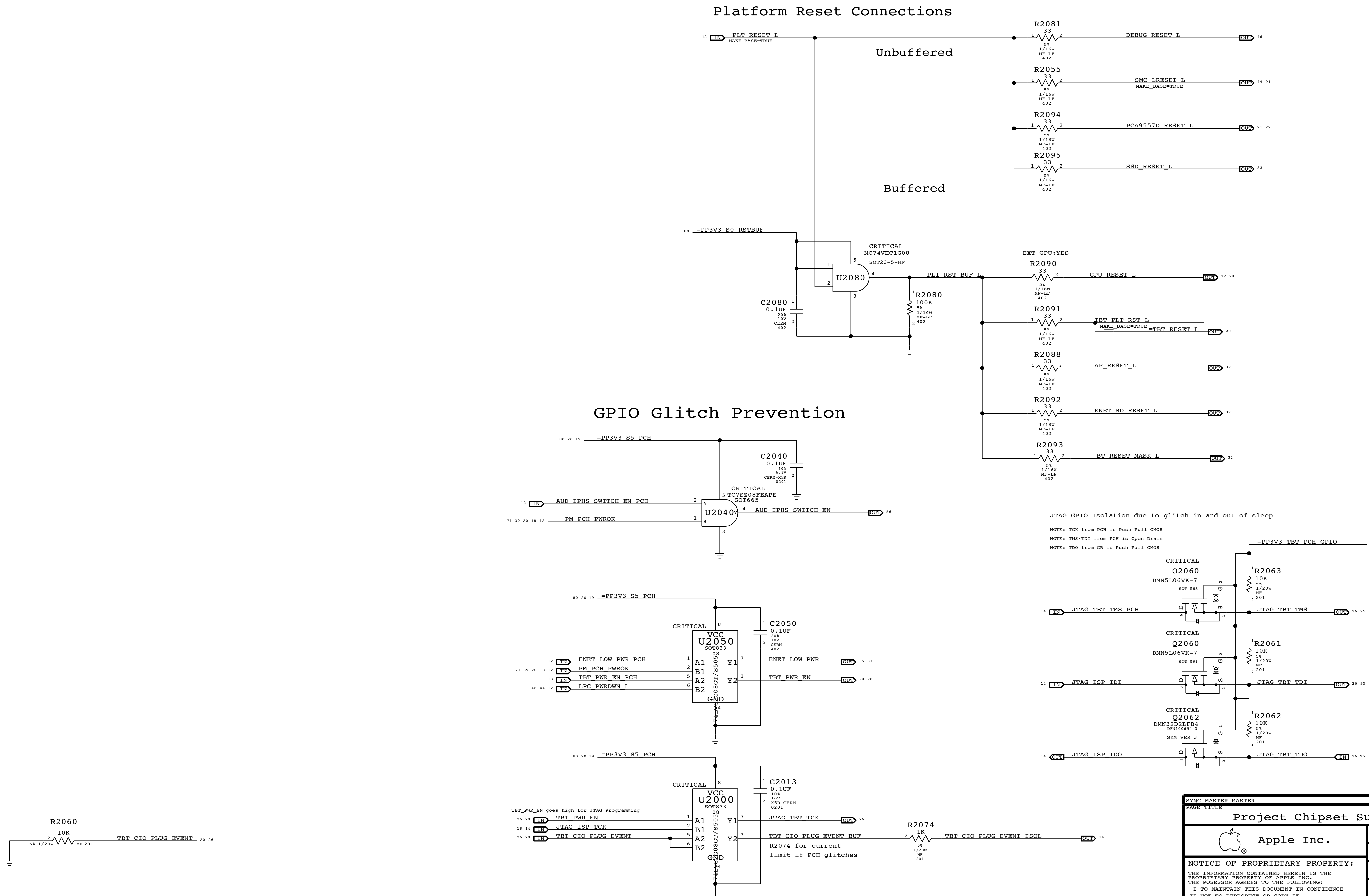
Apple Inc.

DRAWING NUMBER: 051-9889 SIZE: D

REVISION: 13.0.0

BRANCH

PAGE: 19 OF 123



SYNC_MASTER=MASTER SYNC_DATE=04/21/2013

PAGE TITLE

Project Chipset Support

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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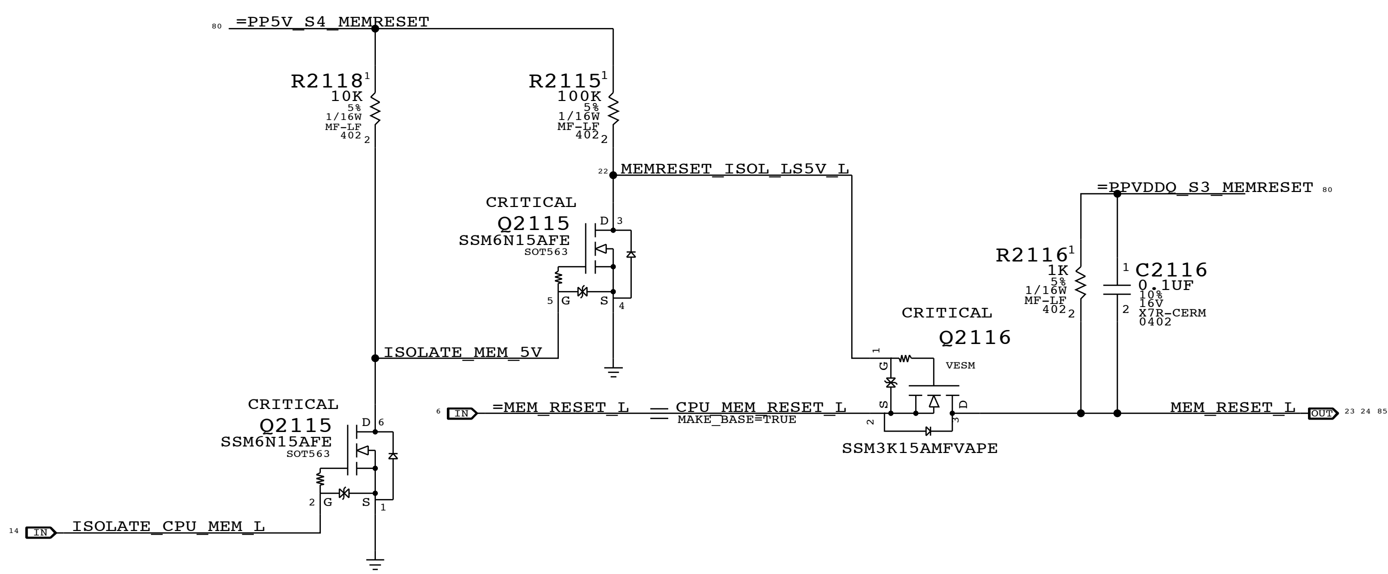
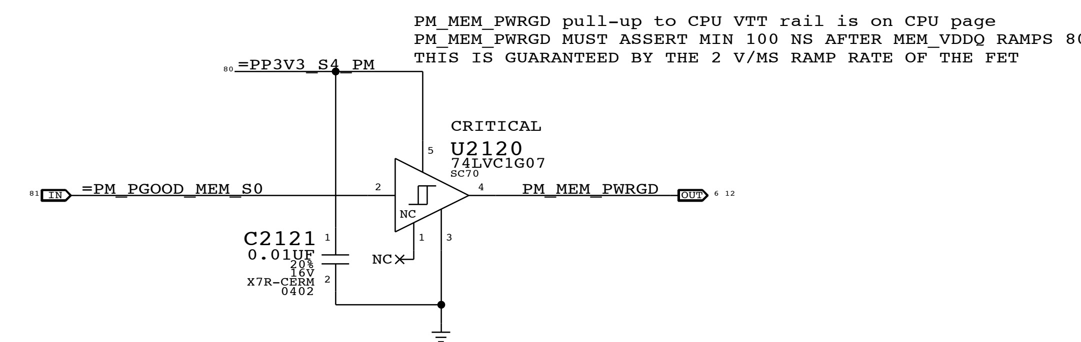
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

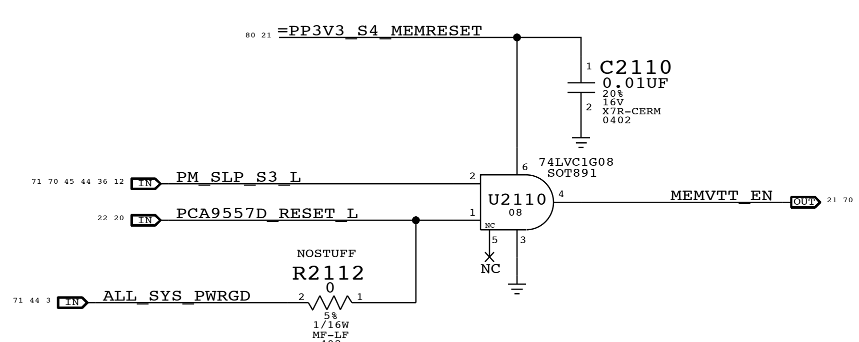
ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

MEMVTT_EN = PLT_RESET_L * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L



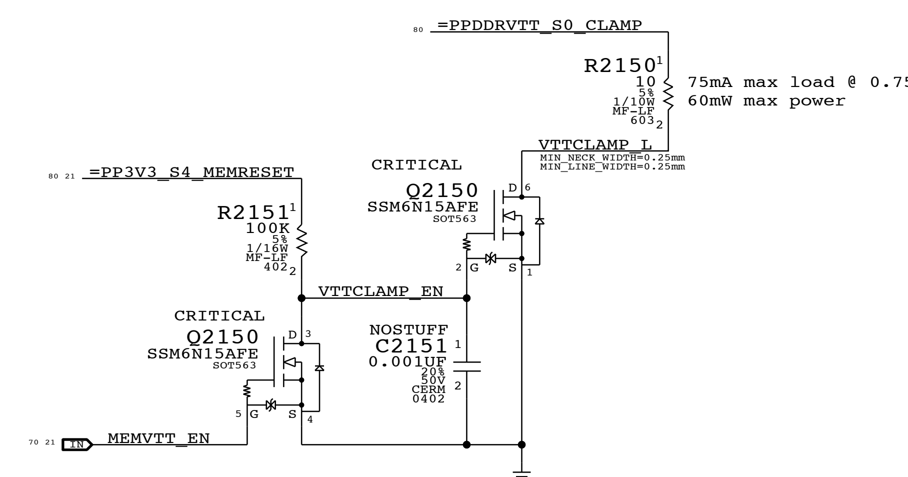
MEM S0 "PGOOD" FOR CPU





MEMVTT Clamp

Ensures CKE signals are held low in S3



	Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1
to	1	0	1	1	1	1	1
	2	0	0	1	1	1	0
S3	3	0	0	0	X	1	0
	4	0	0	1	X	1	0
to	5	0	1	1	0 (*)	1	1
	6	0	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC_MASTER=J17 SYNC_DATE=04/21/2013
PAGE TITLE
CPU Memory S3 Support
Apple Inc.
DRAWING NUMBER 051-9889 SIZE D
REVISION 13.0.0
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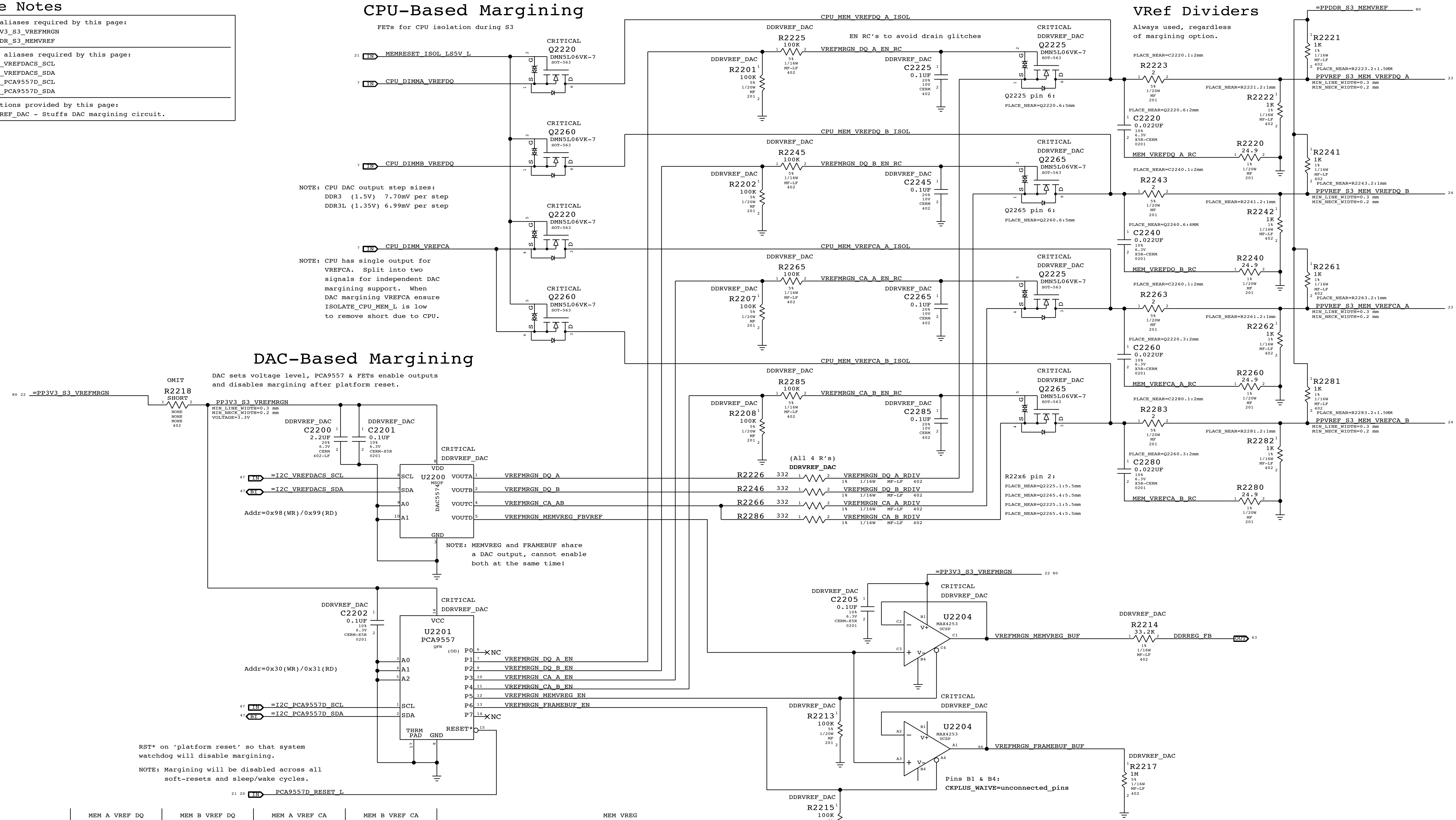
Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
- DDRVREF_DAC - Stuffs DAC margining circuit.



	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	
DAC Channel:	A	B	C	C	D	
PCA9557D Pin:	1	2	3	4	5	
	DDR3 (1.5V)		DDR3L (1.35V)		DDR3 (1.5V)	DDR3L (1.35V)
Nominal value	0.750V (DAC: 0x3A = 0.747mV)		0.675V (DAC: 0x34 = 0.670mV)		1.500V (DAC: 0x74 = 1.495V)	1.343V (DAC: 0x68 = 1.341V)
Margined target:	0.300V - 1.200V (+/- 450mV)		0.275V - 1.075V (+/- 400mV)		1.200V - 1.800V (+/- 300mV)	0.950V - 1.750V (+/- 400mV)
DAC range:	0.000V - 1.508V (0x00 - 0x75)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 3.004V (0x00 - 0xE9)	0.000V - 2.707V (0x00 - 0xD2)
Margined range:	0.299V - 1.206V (+/- 453mV)		0.269V - 1.083V (+/- 406mV)		1.199V - 1.801V (+/- 301mV)	0.932V - 1.760V (+/- 414mV)
VRef current:	+901uA - -911uA (- = sourced)		+811uA - -816uA (- = sourced)		+36uA - -36uA (- = sourced)	+28uA - -29uA (- = sourced)
DAC step size:	7.68mV / step @ output		7.67mV / step @ output		2.575mV / step @ output	3.923mV / step @ output

NOTE: DDR3 assumes TPS51916 supply with 10.0k/49.9k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

SYNC_MASTER=J17 SYNC_DATE=04/21/2013

PAGE TITLE: DDR3 VREF MARGINING

Apple Inc.

DRAWING NUMBER: 051-9889 SIZE: D

REVISION: 13.0.0

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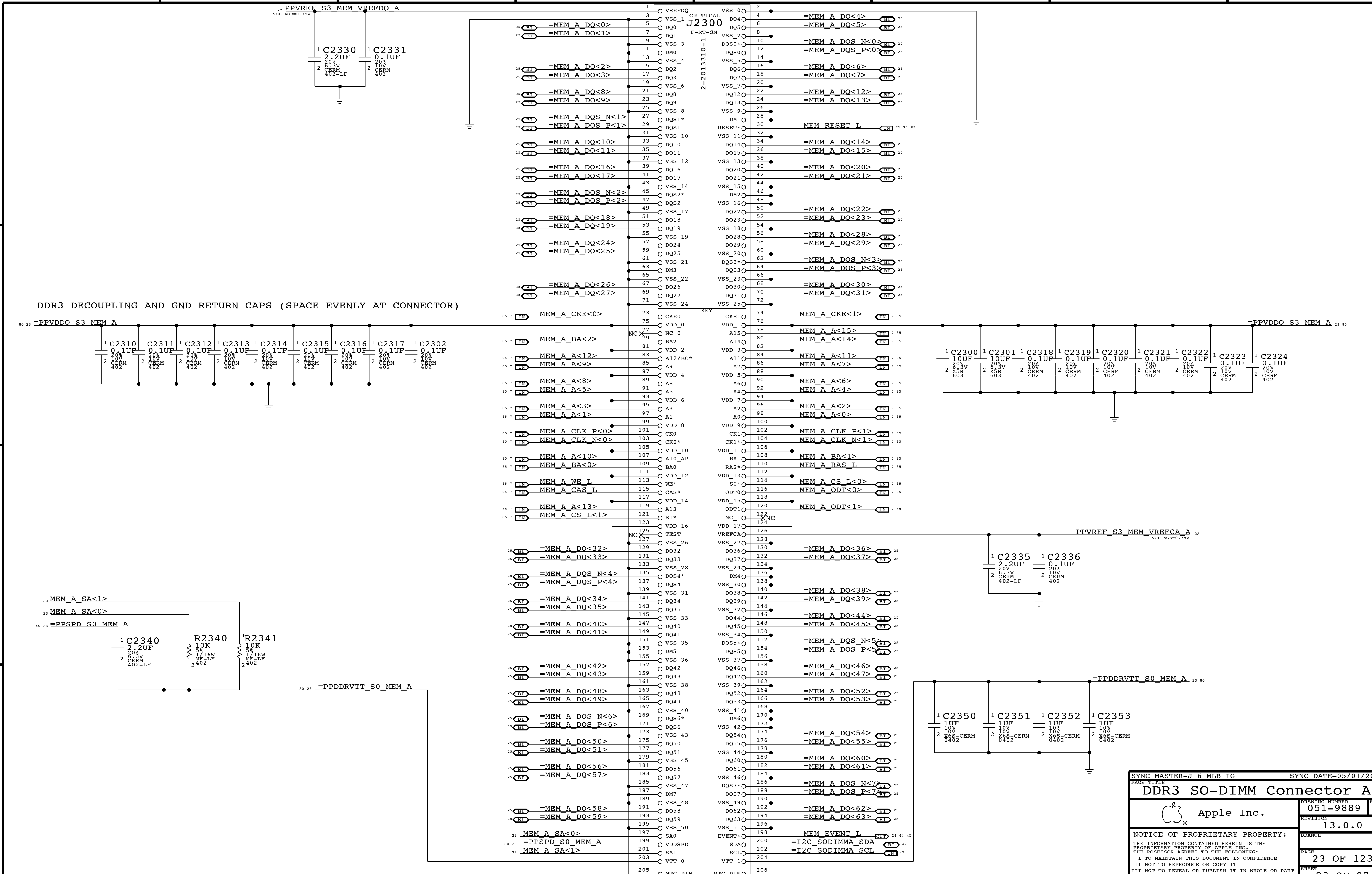

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

DDR3 SO-DIMM Connector A

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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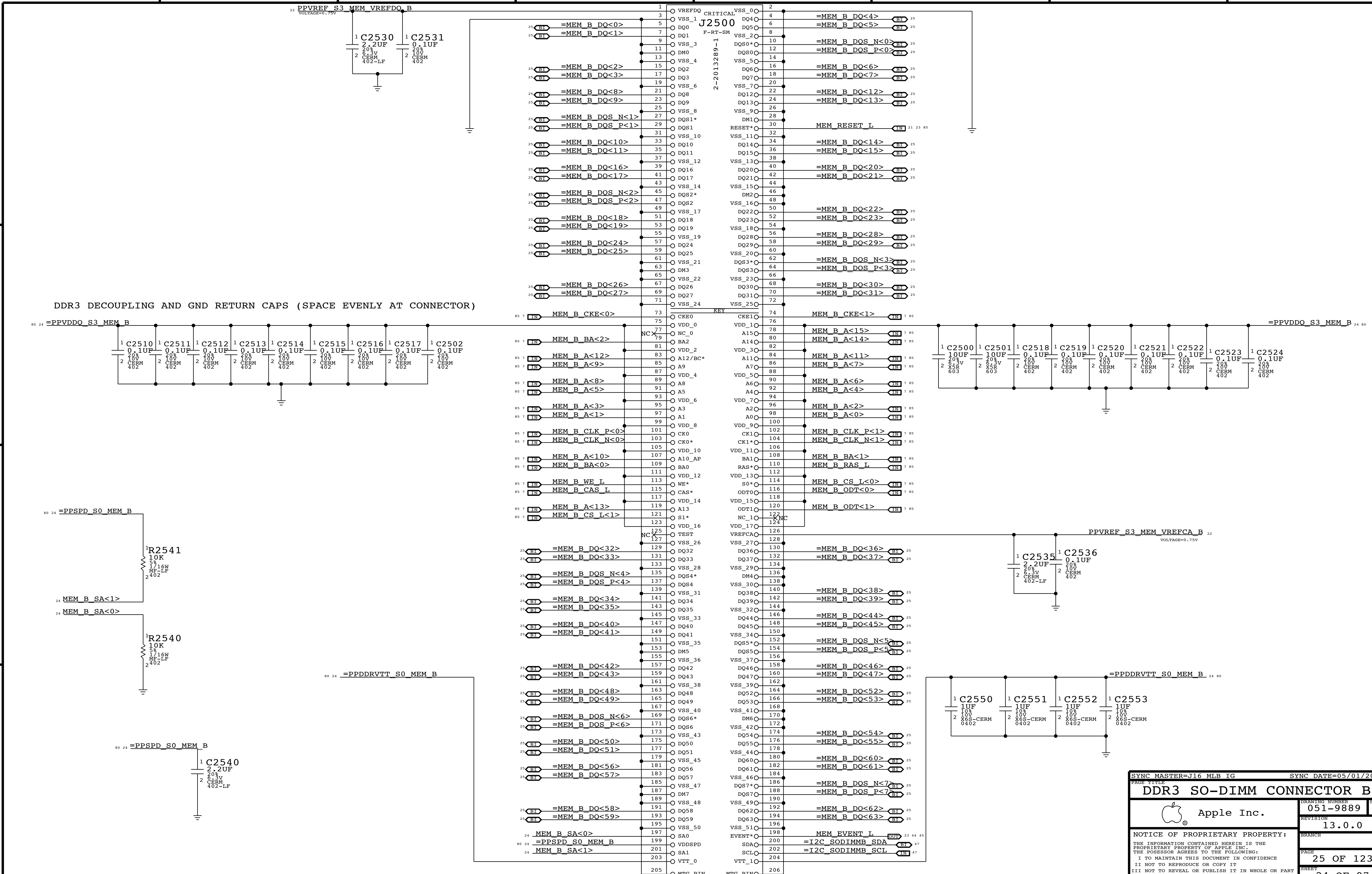

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

DDR3 SO-DIMM CONNECTOR B

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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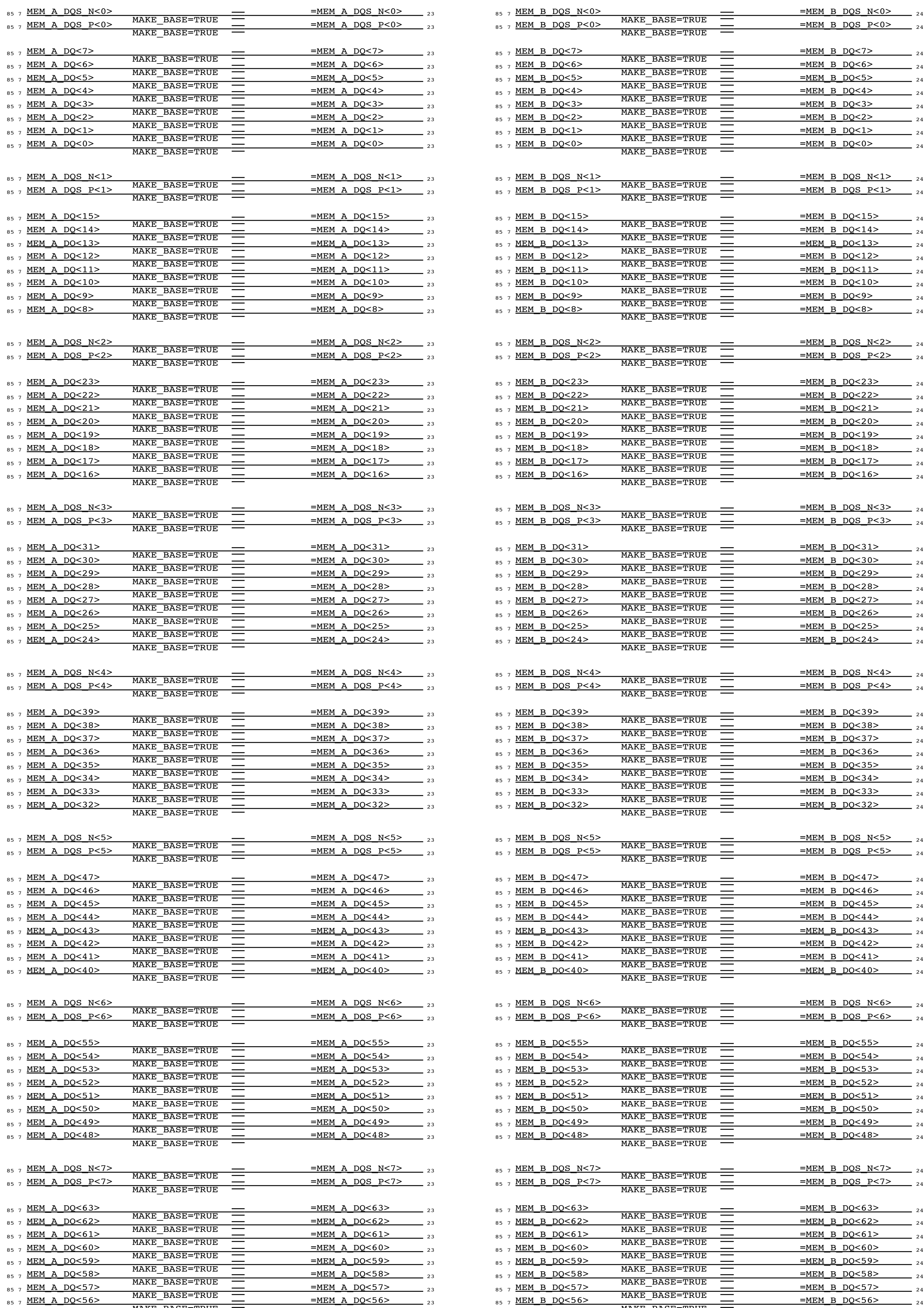
BRANCH

THERE ARE NO PIN SWAPS



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

DDR3 ALIASES AND BITSWAPS

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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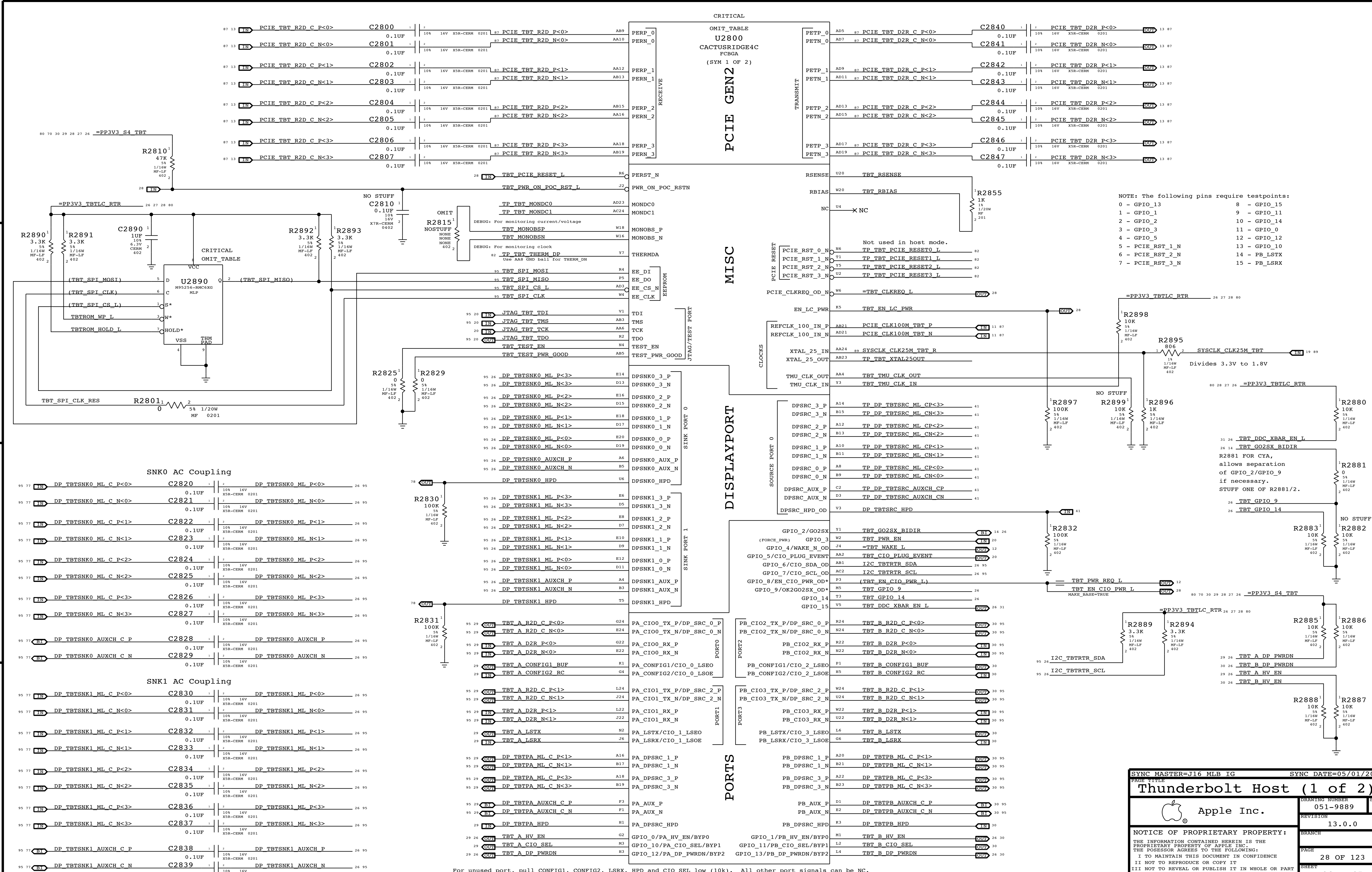
PAGE 27 OF 123

SHEET 25 OF 97



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

Thunderbolt Host (1 of 2)

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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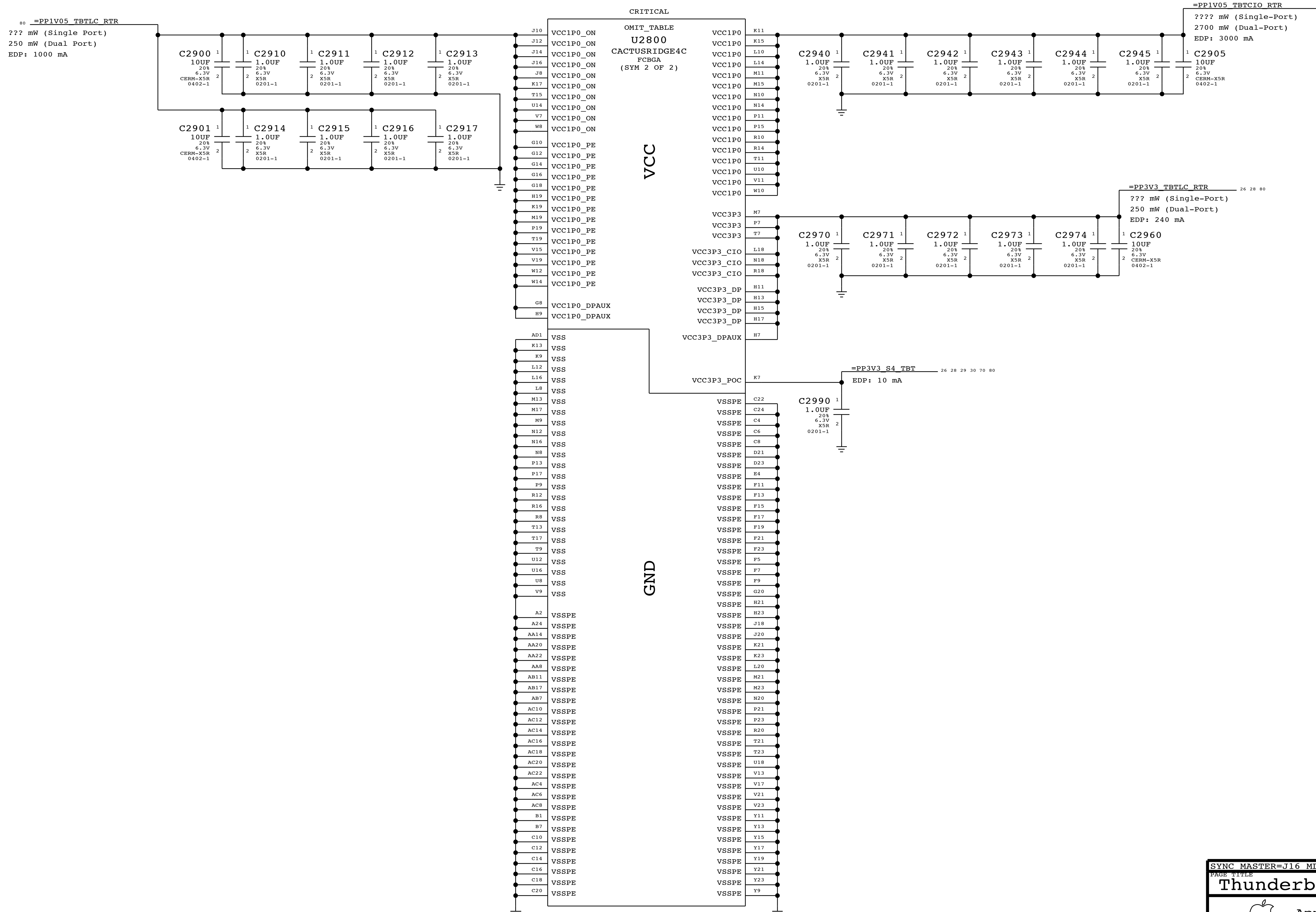
PAGE 28 OF 123

SHEET 26 OF 97



EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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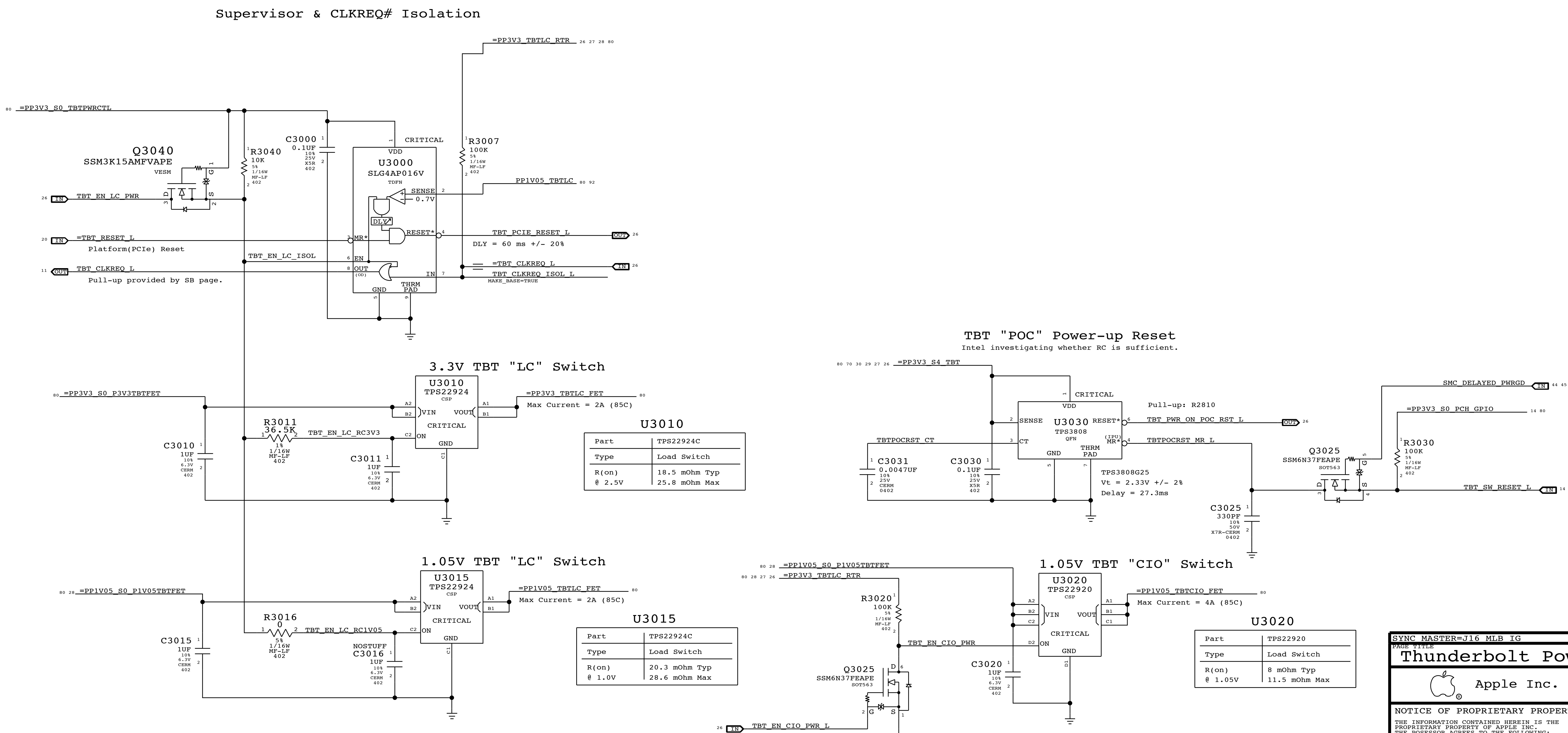
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTPWRCTL
- =PP1V05_TBT_P1V05TBTFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

Signal aliases required by this page:
- =TBT_CLKREQ_L
- =TBT_RESET_L

BOM options provided by this page:
TBTBST:Y - Stuffs 15V boost circuitry.



U3010

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

U3015

Part	TPS22924C
Type	Load Switch
R(on) @ 1.0V	20.3 mOhm Typ 28.6 mOhm Max

U3020

Part	TPS22920
Type	Load Switch
R(on) @ 1.05V	8 mOhm Typ 11.5 mOhm Max

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE
Thunderbolt Power Support

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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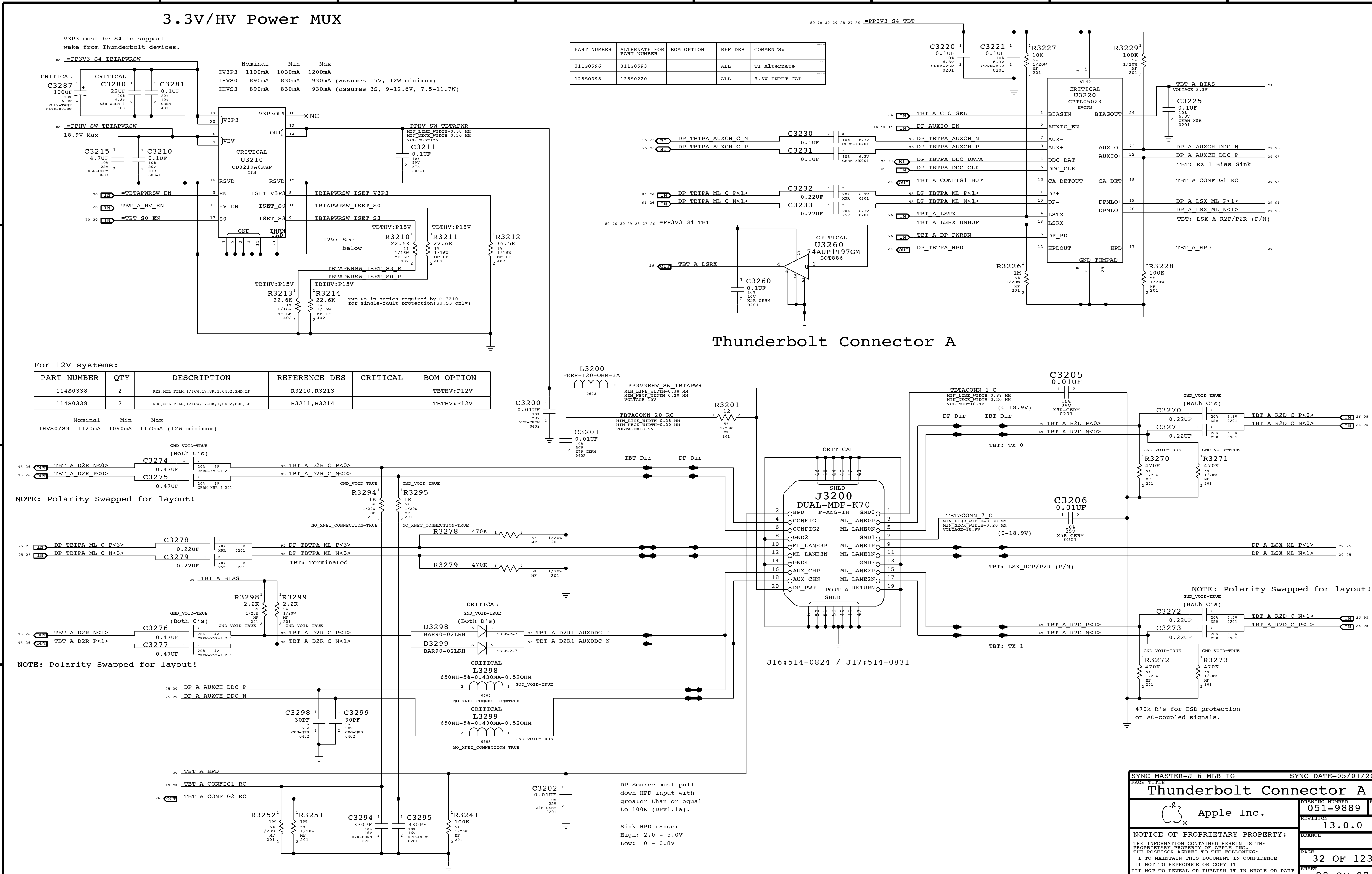
3.3V/HV Power MUX



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S0596	311S0593		ALL	TI Alternate
128S0398	128S0220		ALL	3.3V INPUT CAP

For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R3210,R3213		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R3211,R3214		TBTHV:P12V

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE: Thunderbolt Connector A

Apple Inc.

DRAWING NUMBER: 051-9889 SIZE: D

REVISION: 13.0.0

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3.3V/HV Power MUX

V3P3 must be S4 to support
wake from Thunderbolt devices.

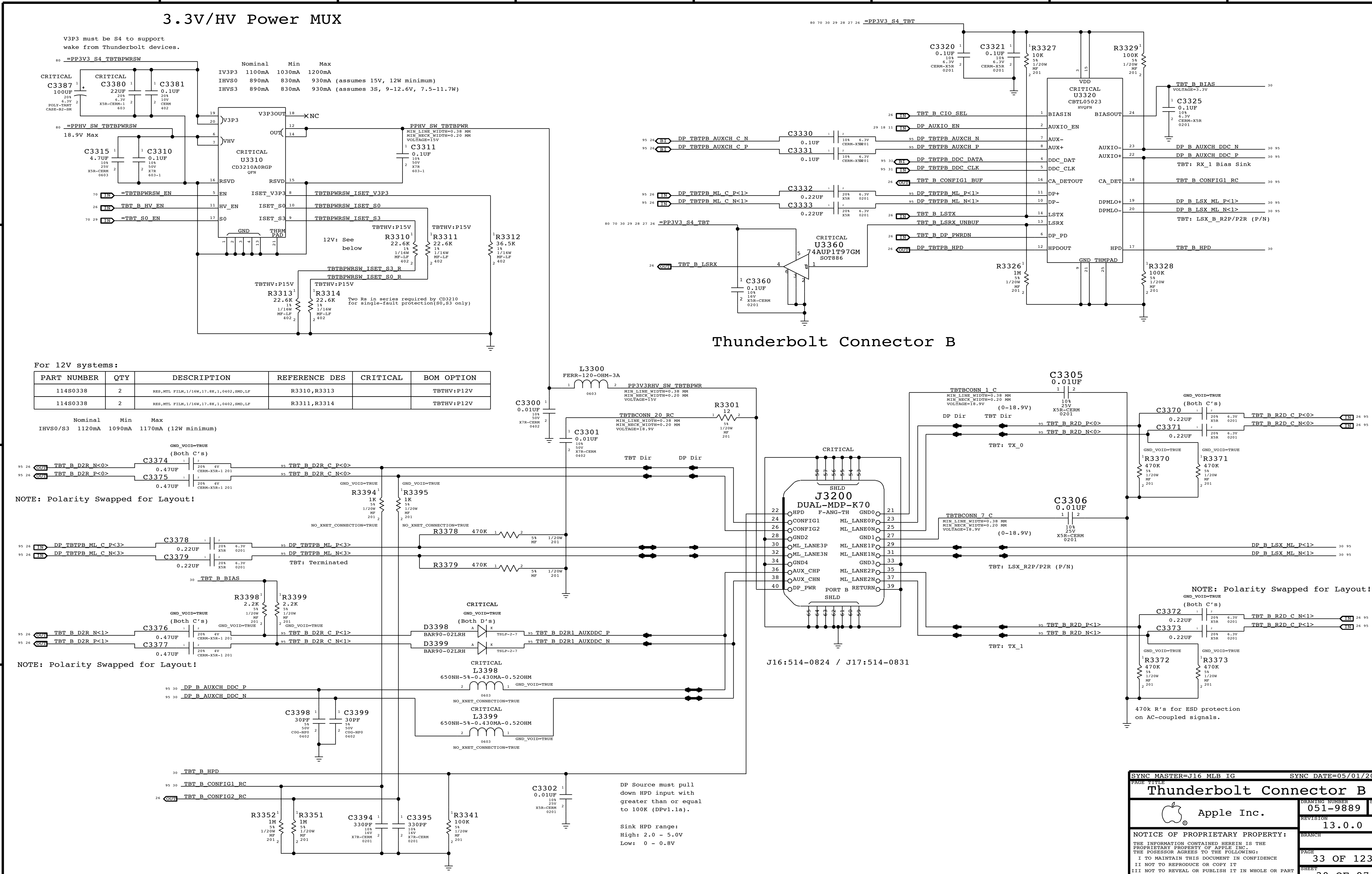
	Nominal	Min	Max
IV3P3	1100mA	1030mA	1200mA
IHVS0	890mA	830mA	930mA (assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA (assumes 3S, 9-12.6V, 7.5-11.7W)

For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R3310,R3313		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R3311,R3314		TBTHV:P12V

Nominal Min Max
IHVS0/S3 1120mA 1090mA 1170mA (12W minimum)

Thunderbolt Connector B



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

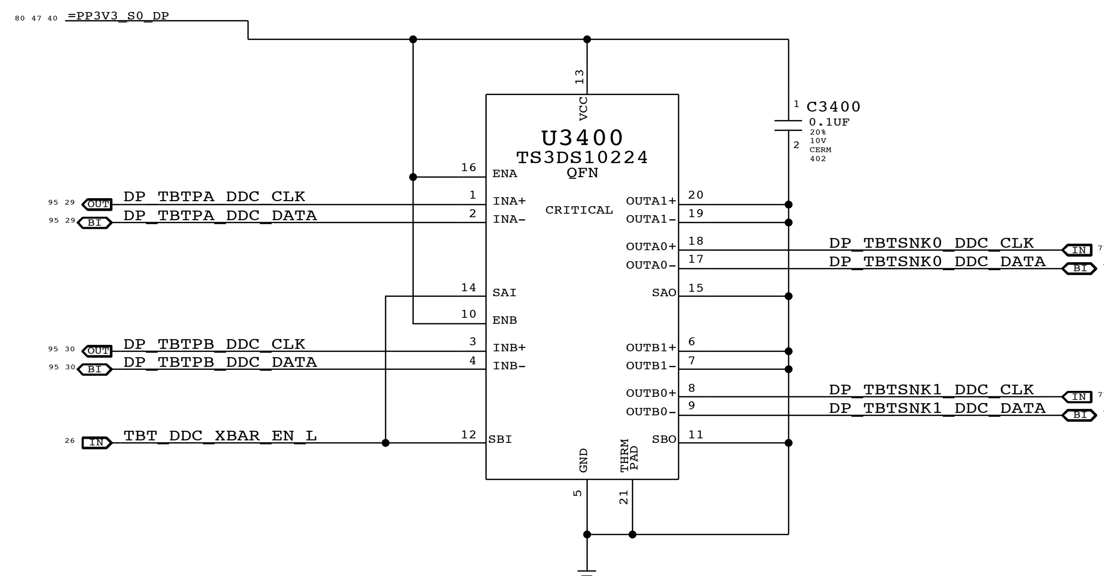
PAGE TITLE: Thunderbolt Connector B

Apple Inc.

DRAWING NUMBER: 051-9889 SIZE: D

REVISION: 13.0.0

Dual-Port Host DDC Crossbar



SYNC MASTER=J16 DG | SYNC DATE=04/21/2013

PAGE TITLE

TBT DDC Crossbar

Apple Inc.

DRAWING NUMBER 051-9889 | SIZE D

REVISION 13.0.0

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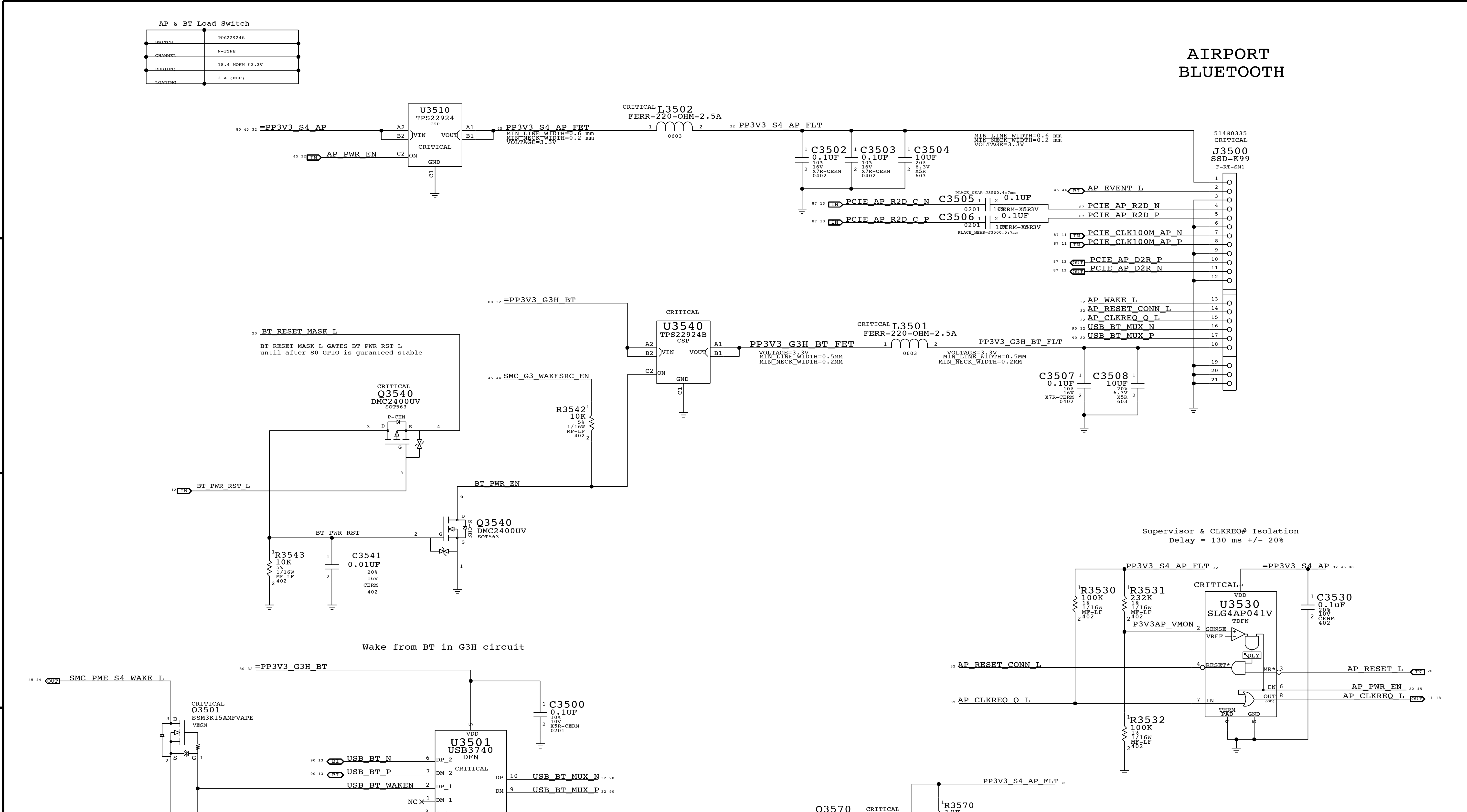

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE: AIRPORT/BT

Apple Inc.

DRAWING NUMBER: 051-9889 SIZE: D

REVISION: 13.0.0

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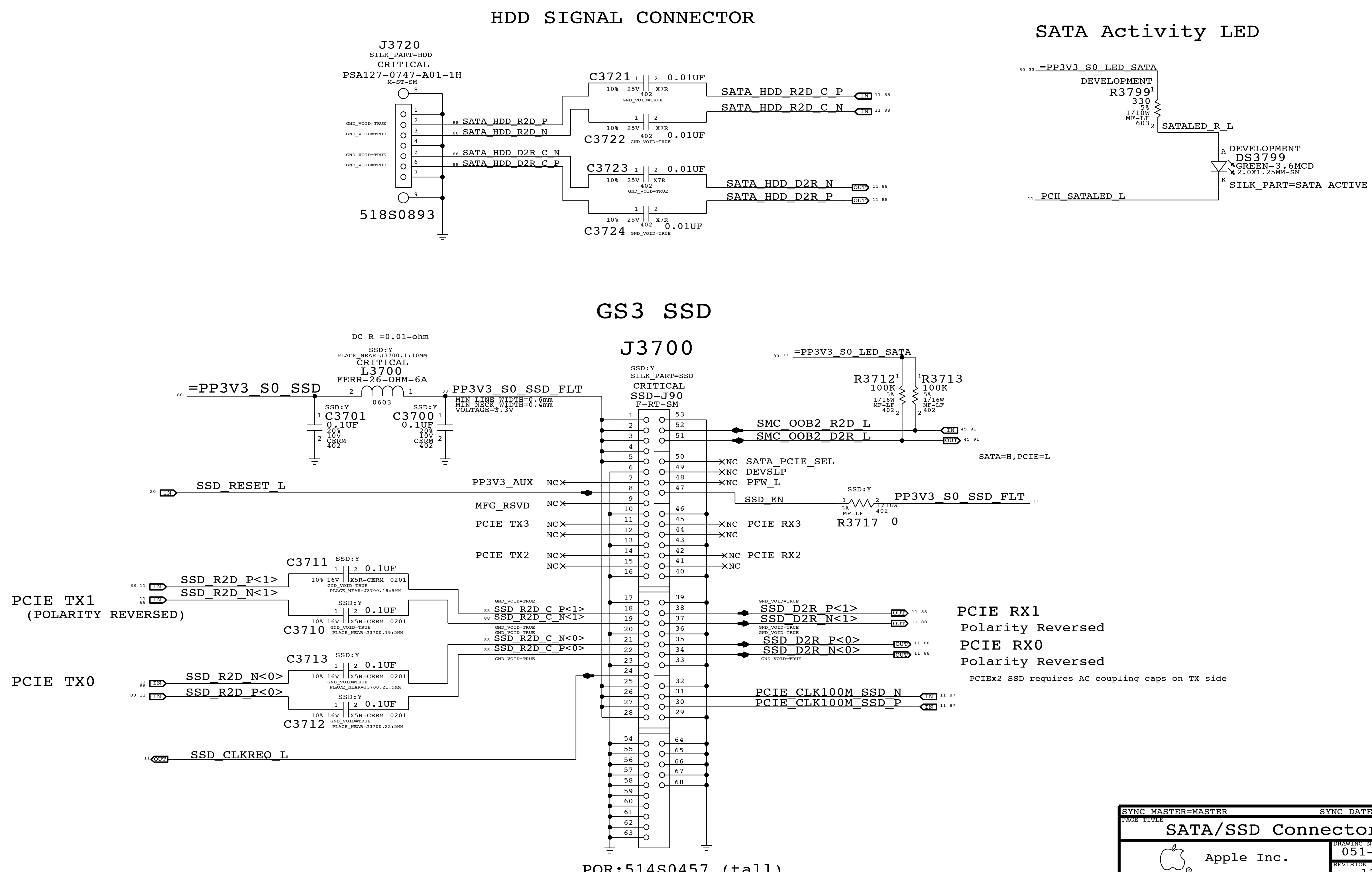

SYNC_MASTER=MASTER SYNC_DATE=05/01/2013

PAGE TITLE

SATA/SSD Connectors

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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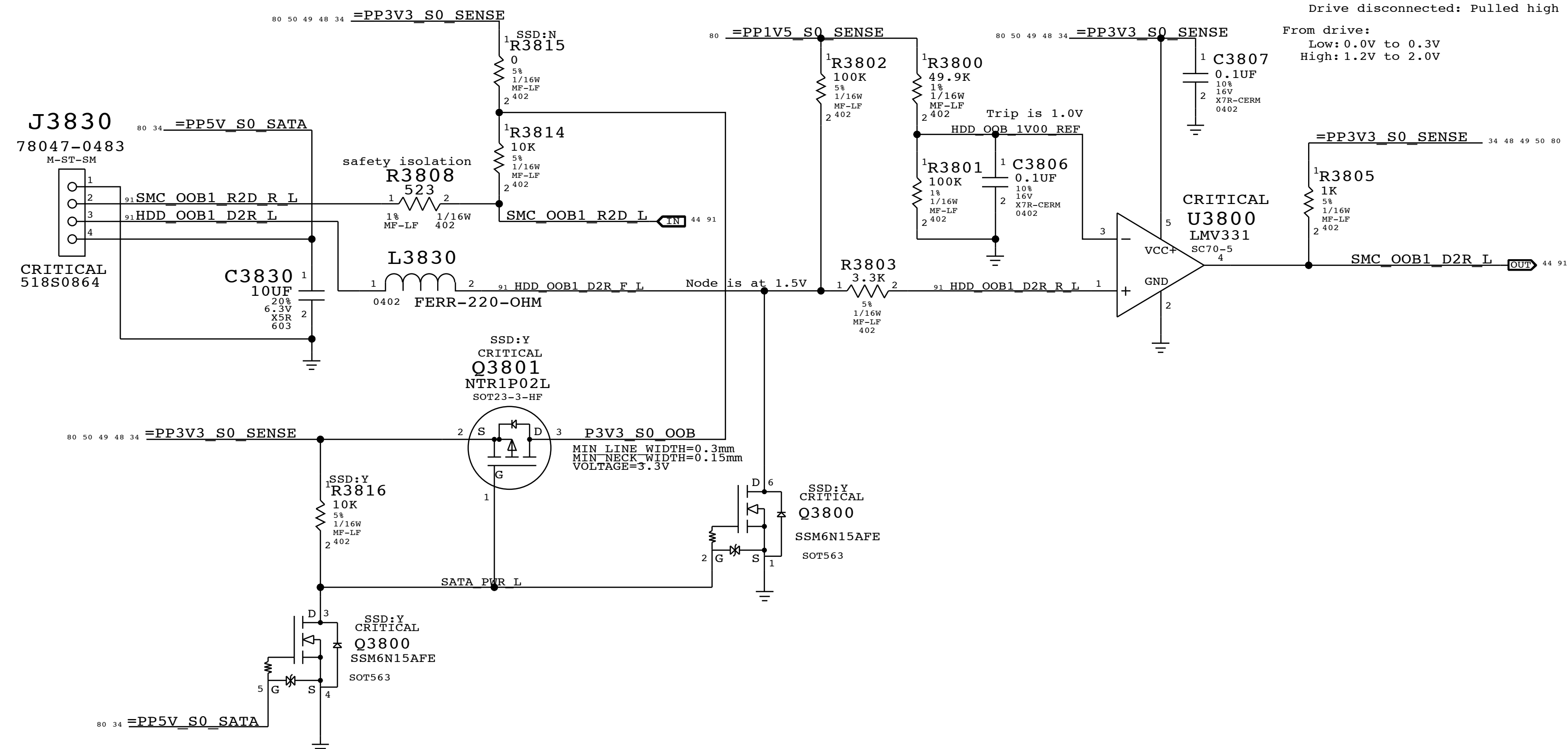
HDD POWER/OOB CONNECTOR

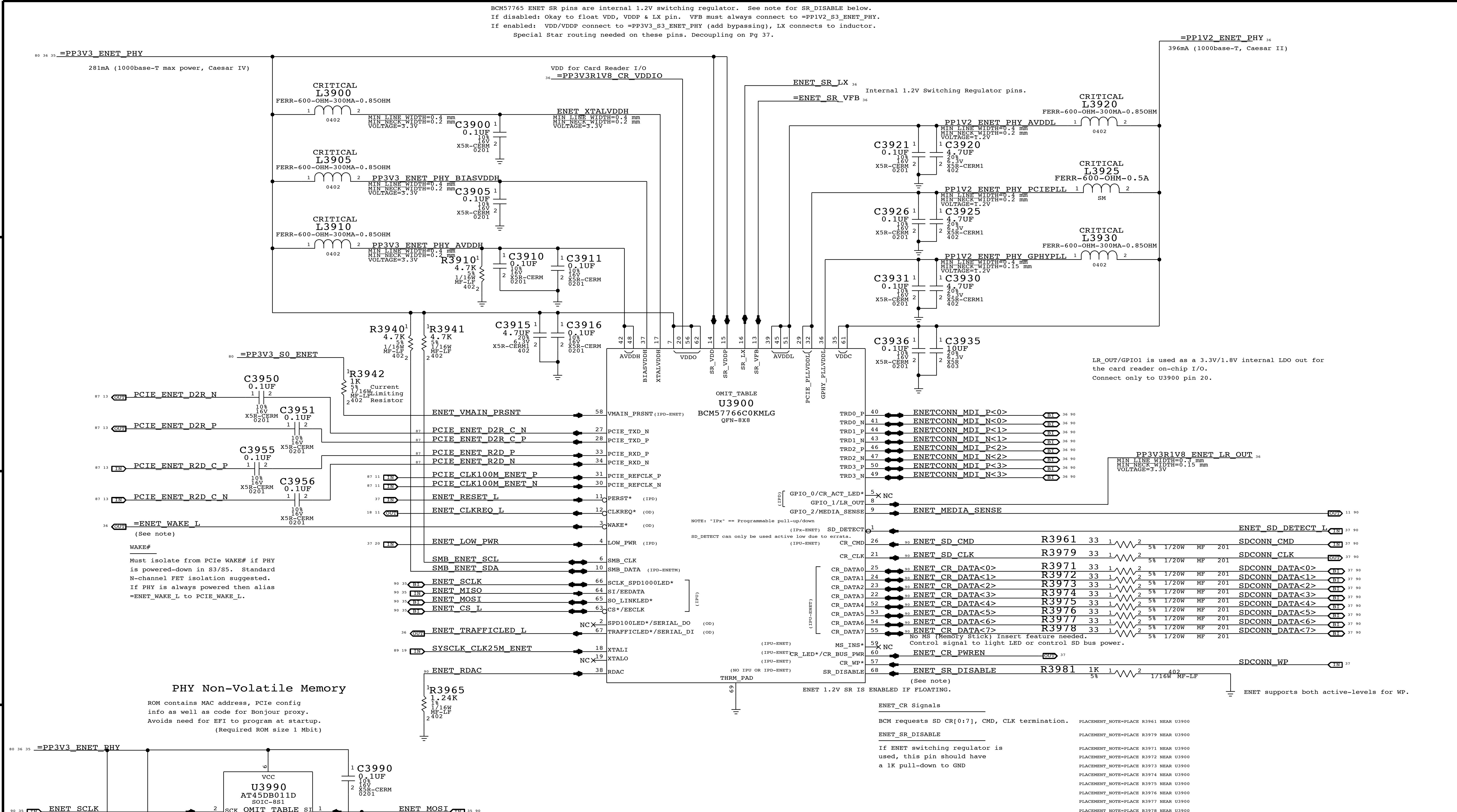
HDD Out-of-Band Temperature Sensing

Notes:
Drive active: Valid signal protocol
Drive asleep: HDD drives HDD_OOB_TEMP low
Drive disconnected: Pulled high

From drive:
Low: 0.0V to 0.3V
High: 1.2V to 2.0V



SYNC_MASTER=J16_MLB_IG		SYNC_DATE=05/01/2013
PAGE TITLE: HDD Connector		
Apple Inc.	DRAWING NUMBER: 051-9889	SIZE: D
	REVISION: 13.0.0	
	BRANCH	
	PAGE: 38 OF 123	
	SHEET: 34 OF 97	



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE
ETHERNET PHY (CAESAR IV)

DRAWING NUMBER 051-9889 SIZE D

Apple Inc.

REVISION 13.0.0

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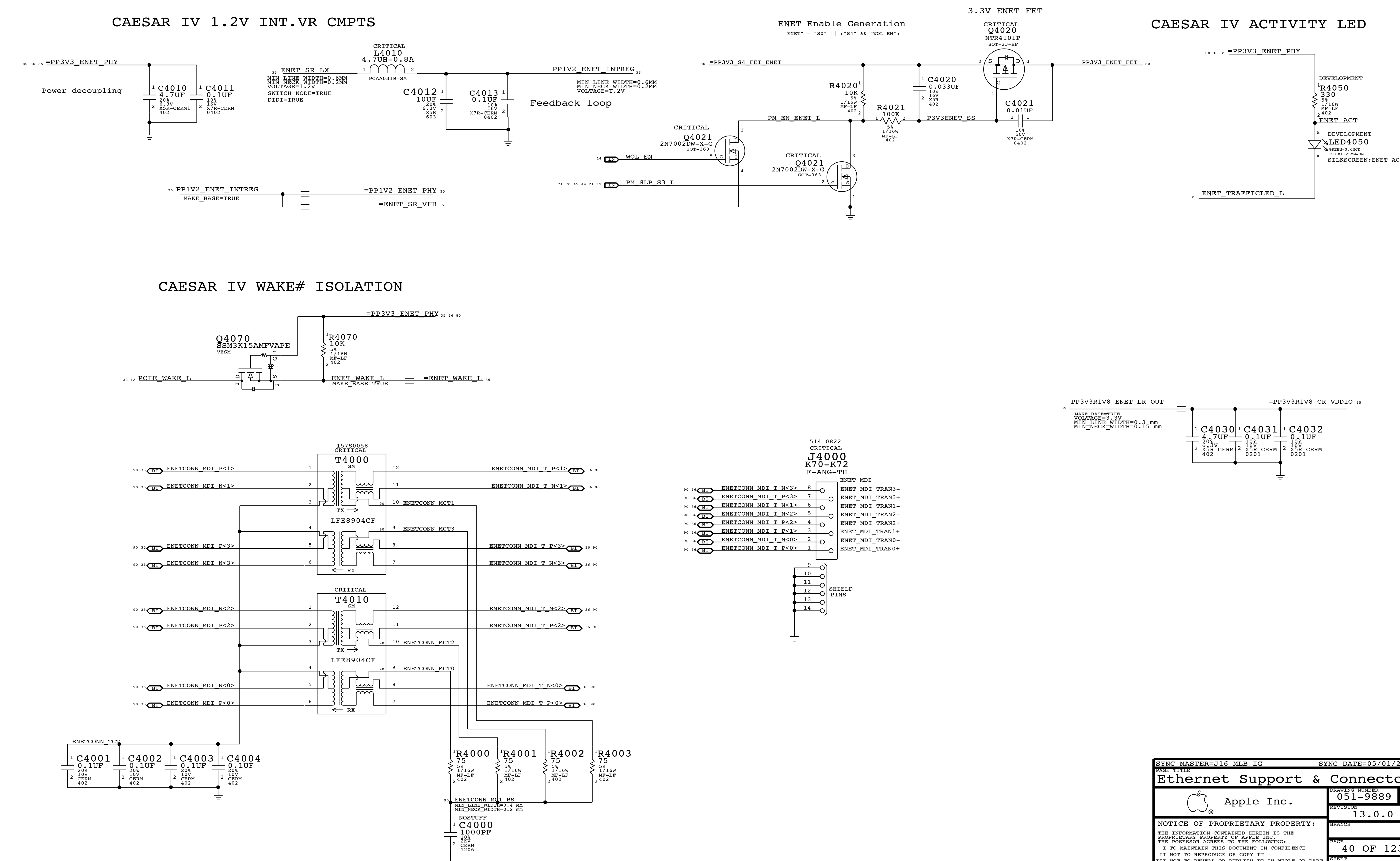

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

Ethernet Support & Connector

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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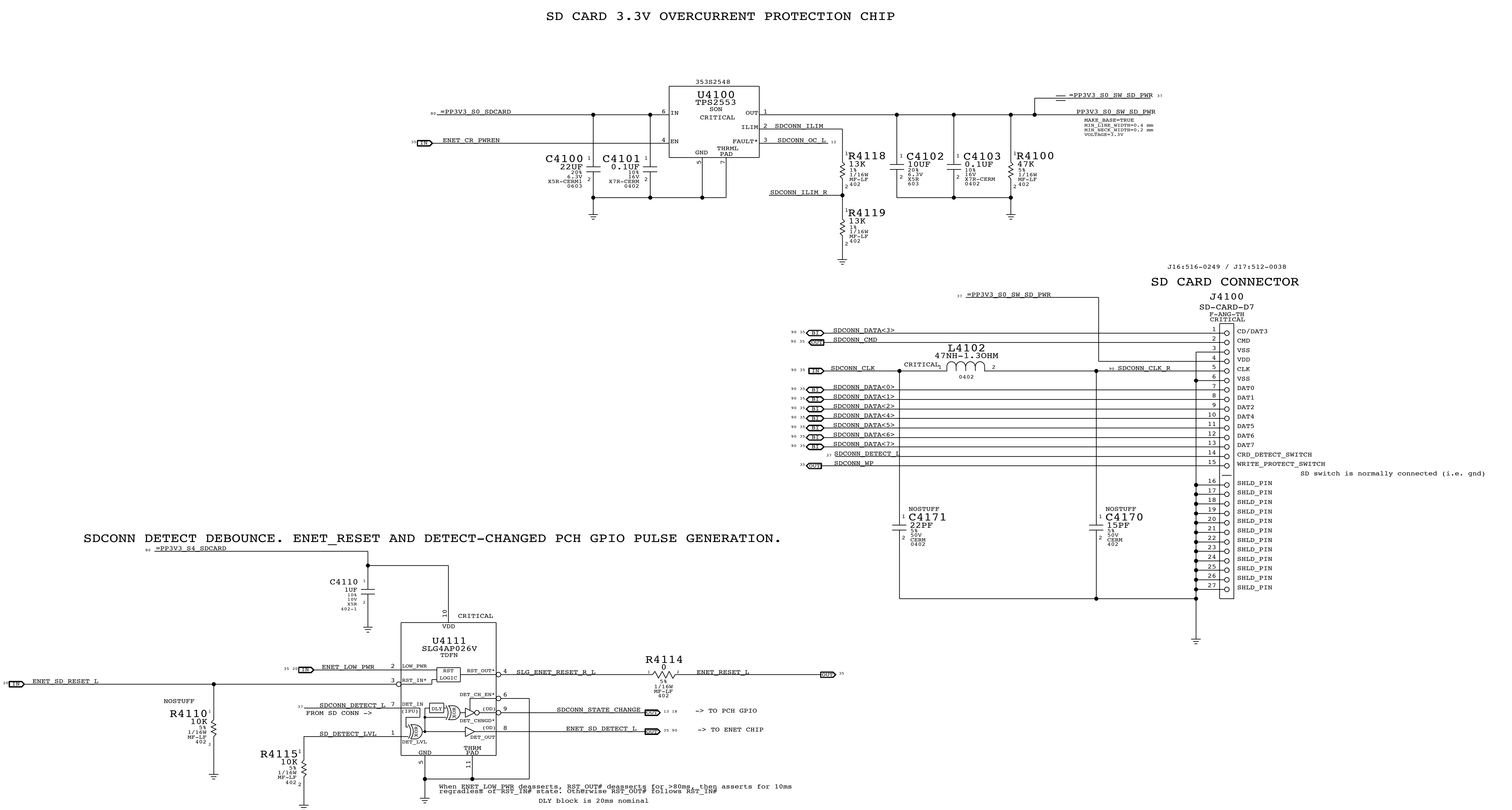
SHEET 36 OF 97



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

SD READER CONNECTOR

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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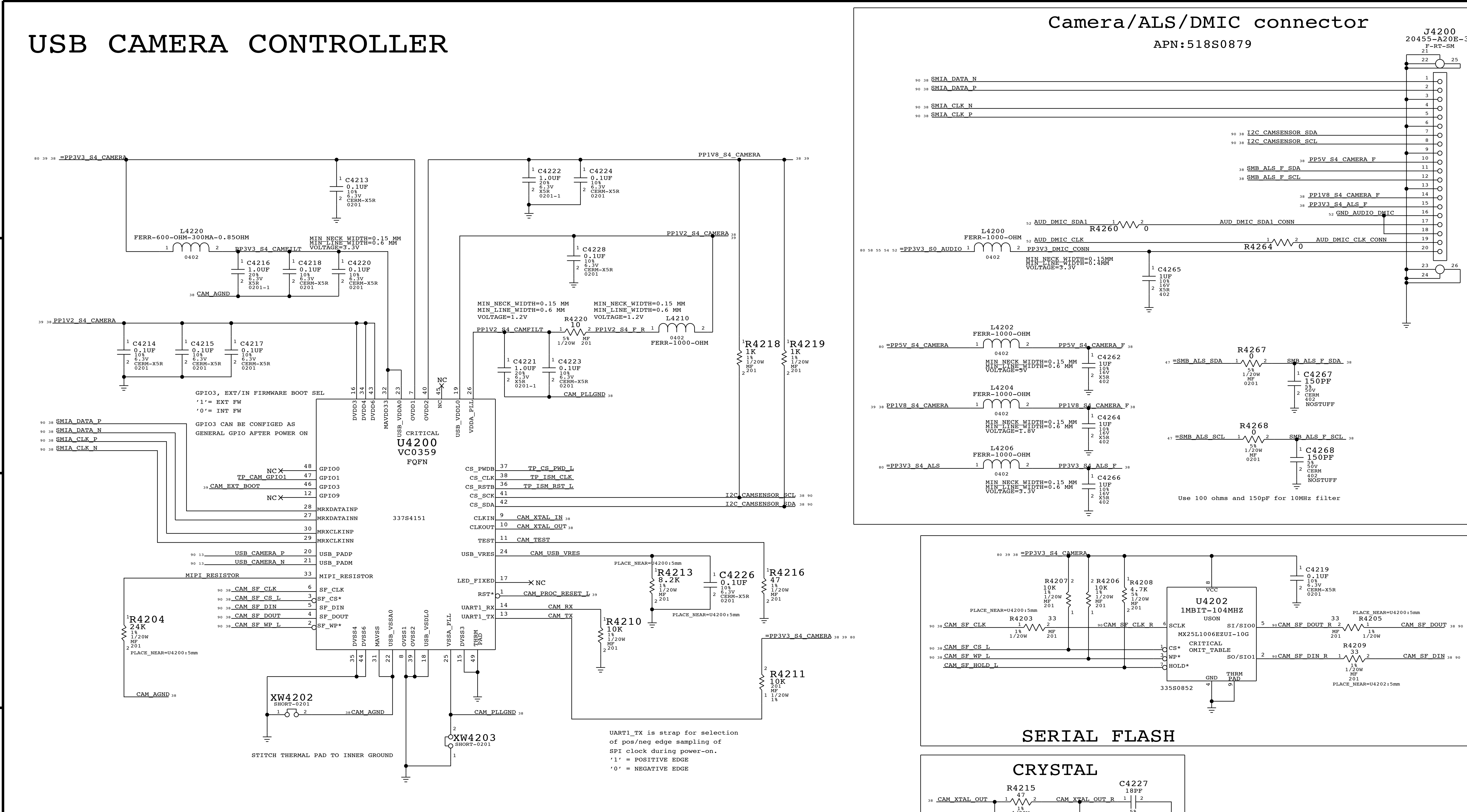

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE: Camera Controller

Apple Inc.

DRAWING NUMBER: 051-9889 SIZE: D

REVISION: 13.0.0

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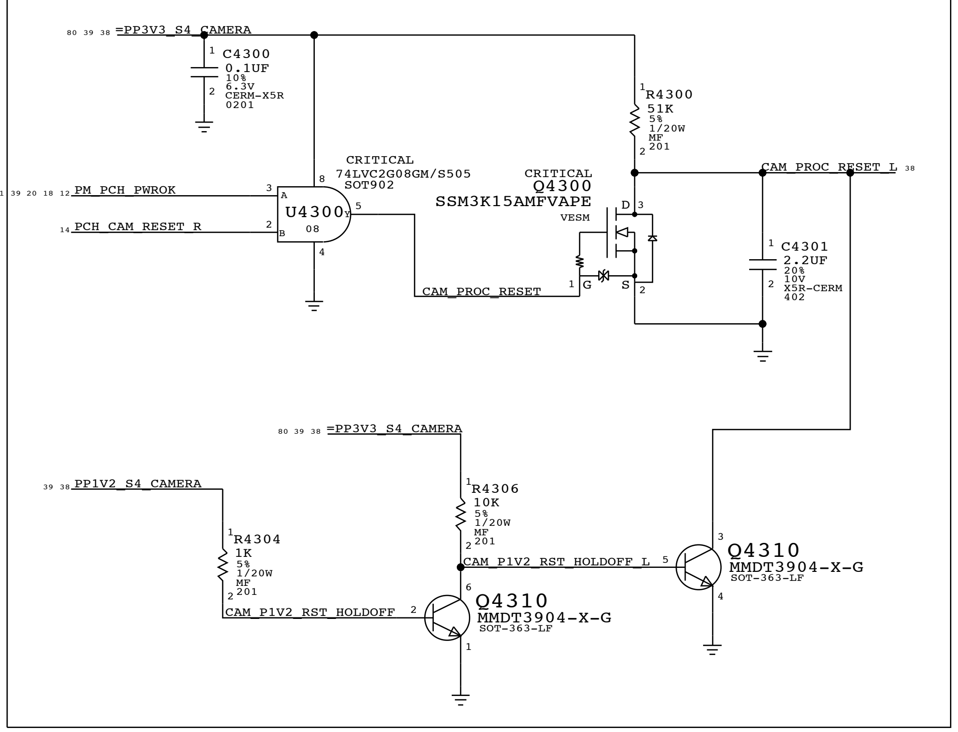
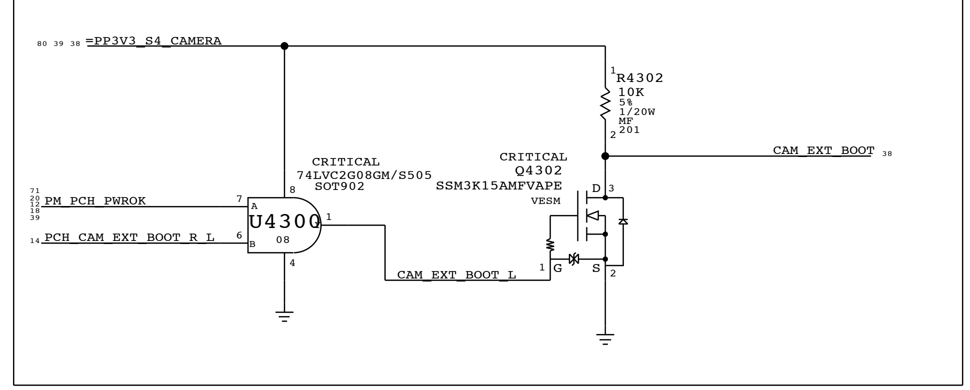
PAGE: 42 OF 123

SHEET: 38 OF 97

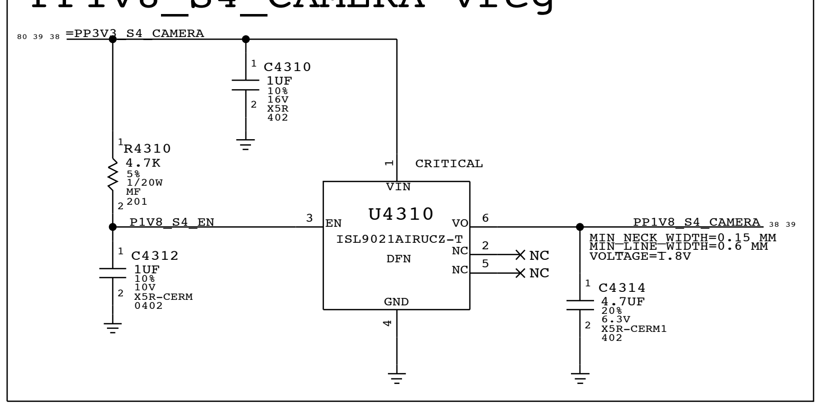
Camera Processor Reset



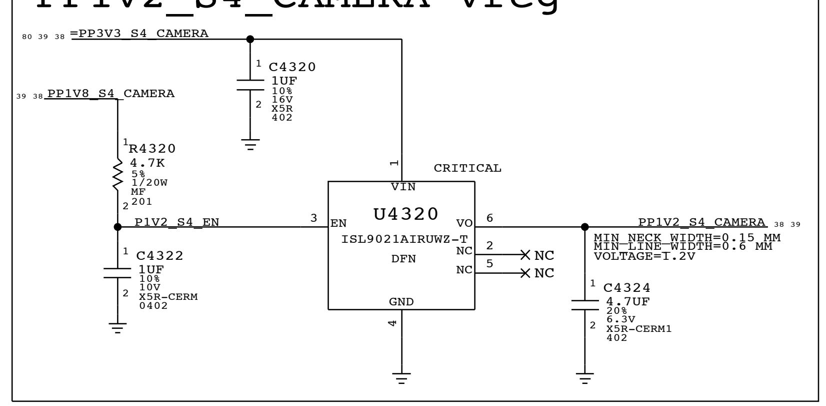
Camera Processor ExtBoot Cntl



PP1V8_S4_CAMERA Vreg



PP1V2_S4_CAMERA Vreg



SYNC MASTER=J16 MLB IG SYNC DATE=05/01/2013

PAGE TITLE

Camera Controller Support

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

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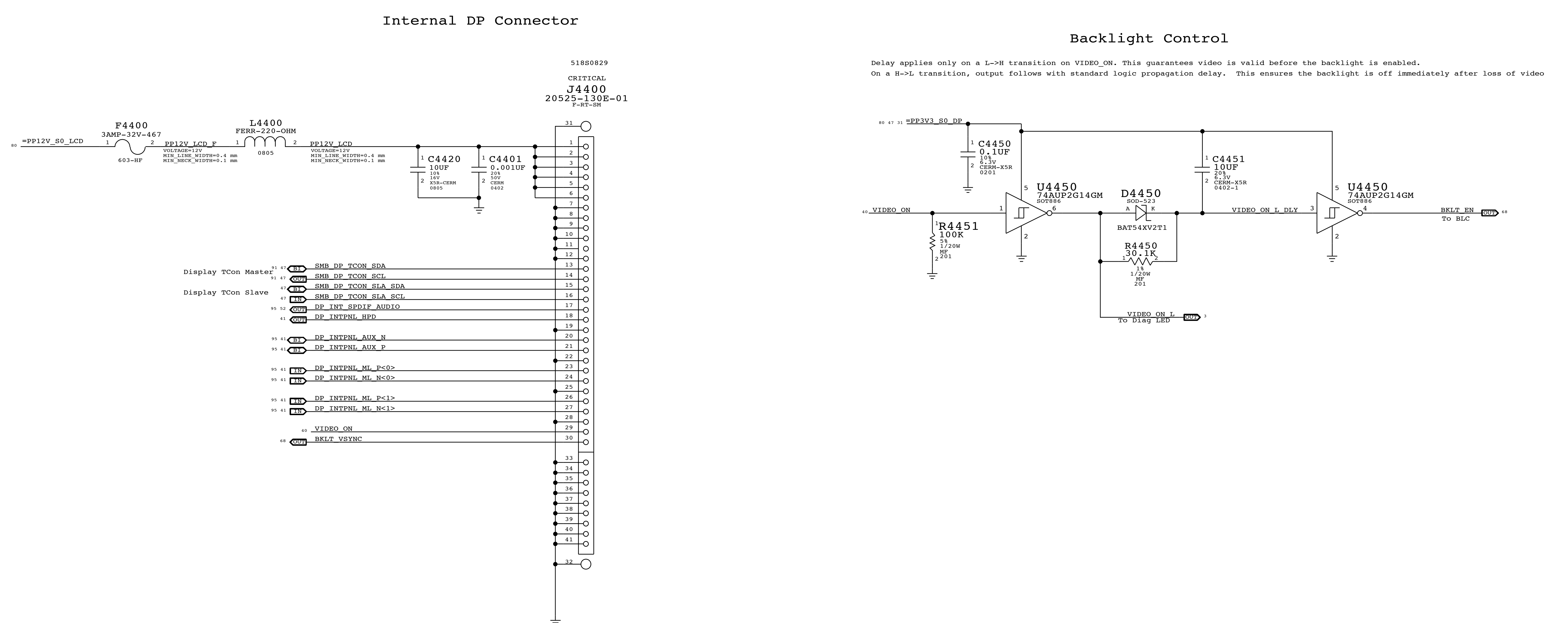
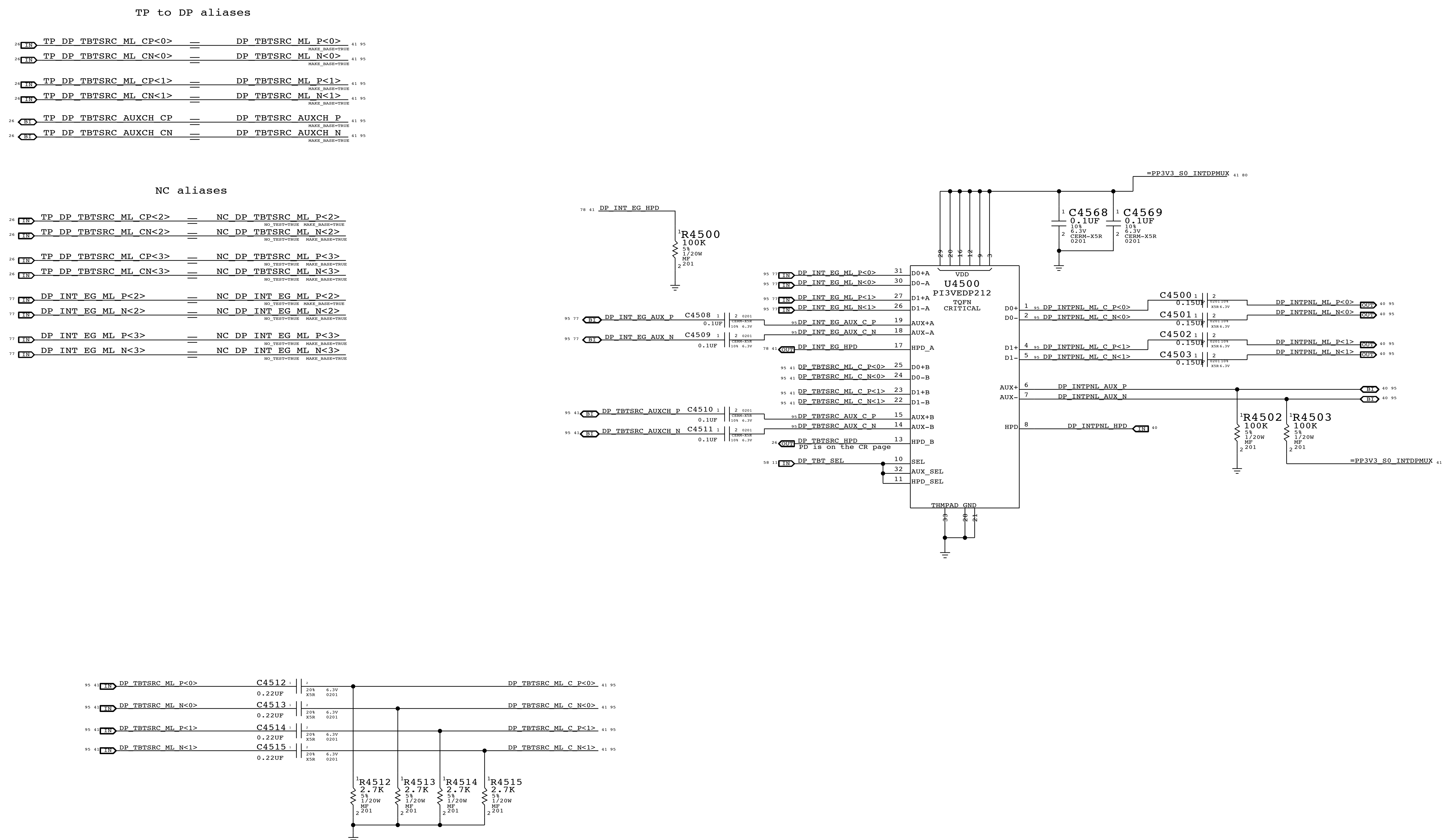


SYNC_MASTER=J16_MLB_IG	SYNC_DATE=05/01/2013
PAGE TITLE	Internal DP Support
DRAWING NUMBER	051-9889
SIZE	D
REVISION	13.0.0
BRANCH	
PAGE	44 OF 123
SHEET	40 OF 97

Apple Inc.



SYNC_MASTER=J16_DG SYNC_DATE=04/21/2013

PAGE TITLE

Internal DP MUXing

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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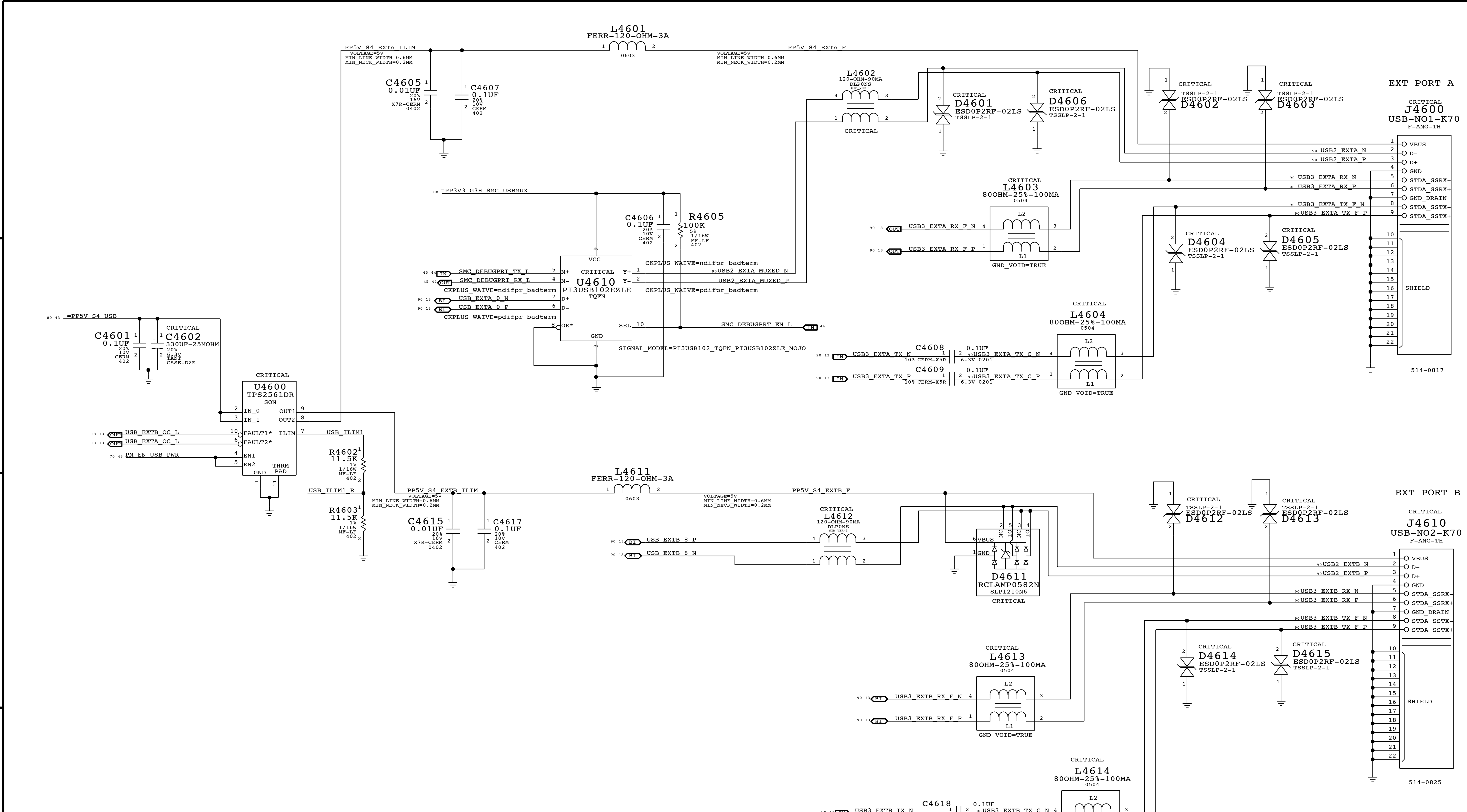

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

EXTERNAL USB PORTS A & B

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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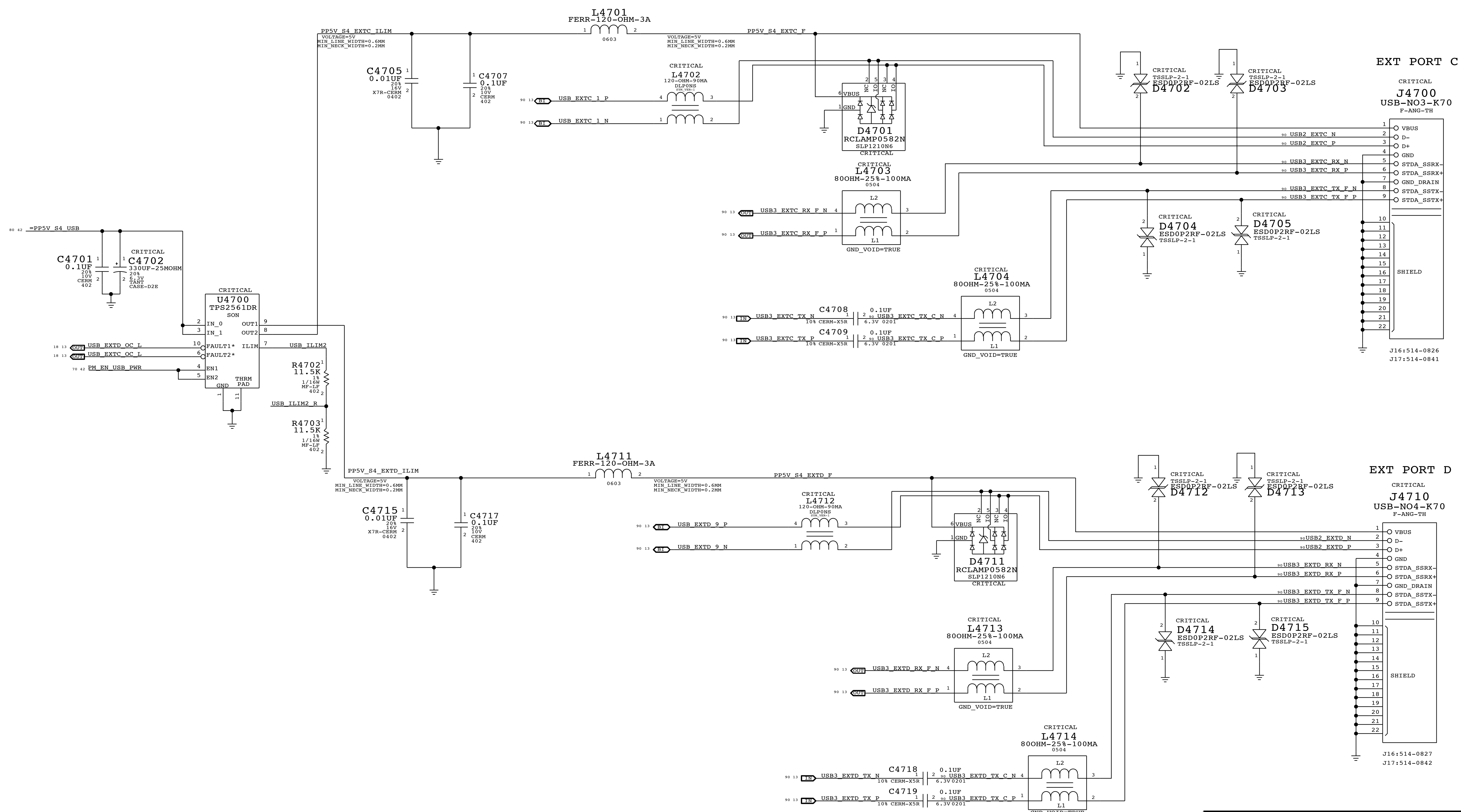

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

EXTERNAL USB PORTS C & D

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

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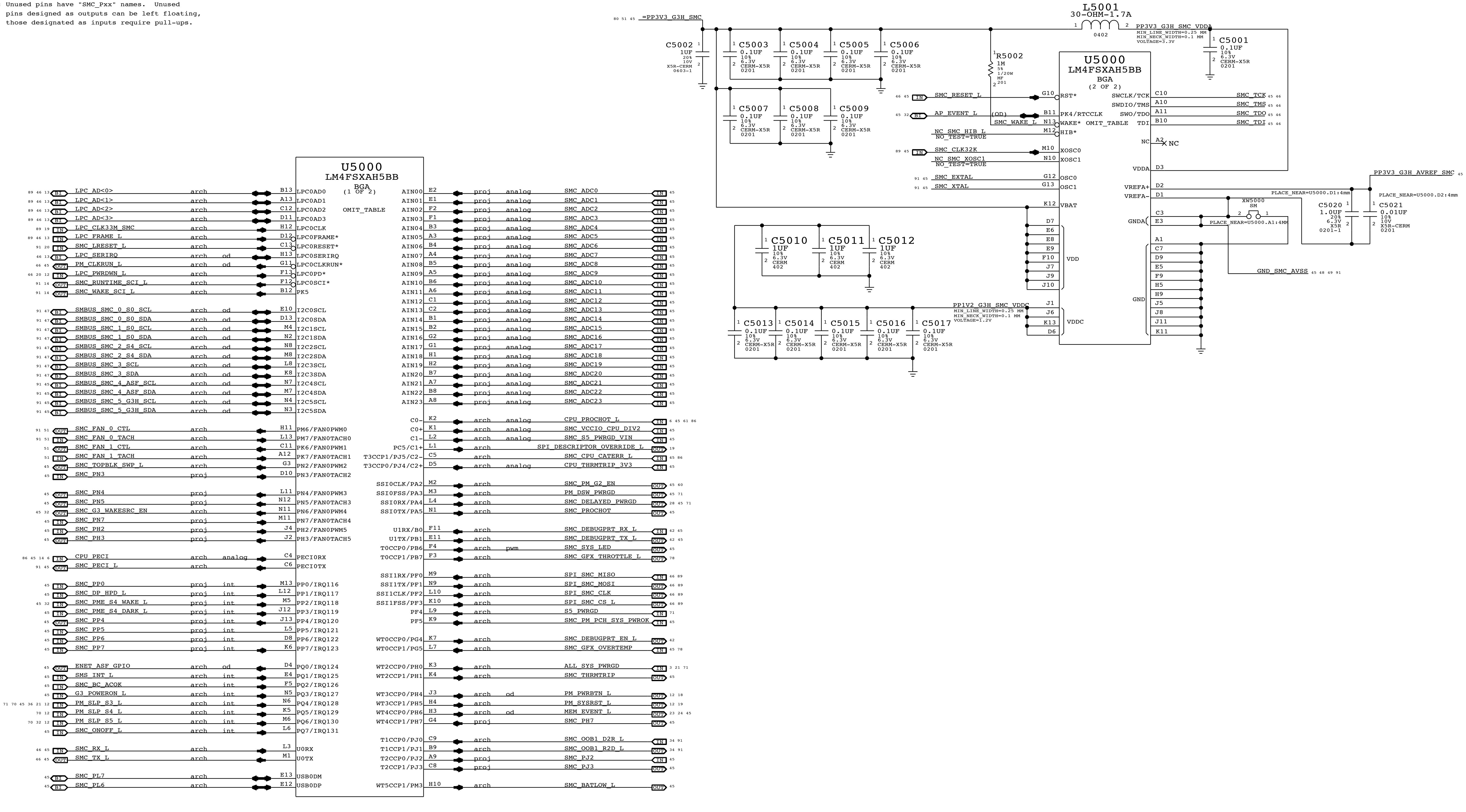
NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.



SYNC_MASTER=J16_MLB_IG	SYNC_DATE=05/01/2013
PAGE TITLE	SMC
DRAWING NUMBER	051-9889
SIZE	D
REVISION	13.0.0
BRANCH	
PAGE	50 OF 123
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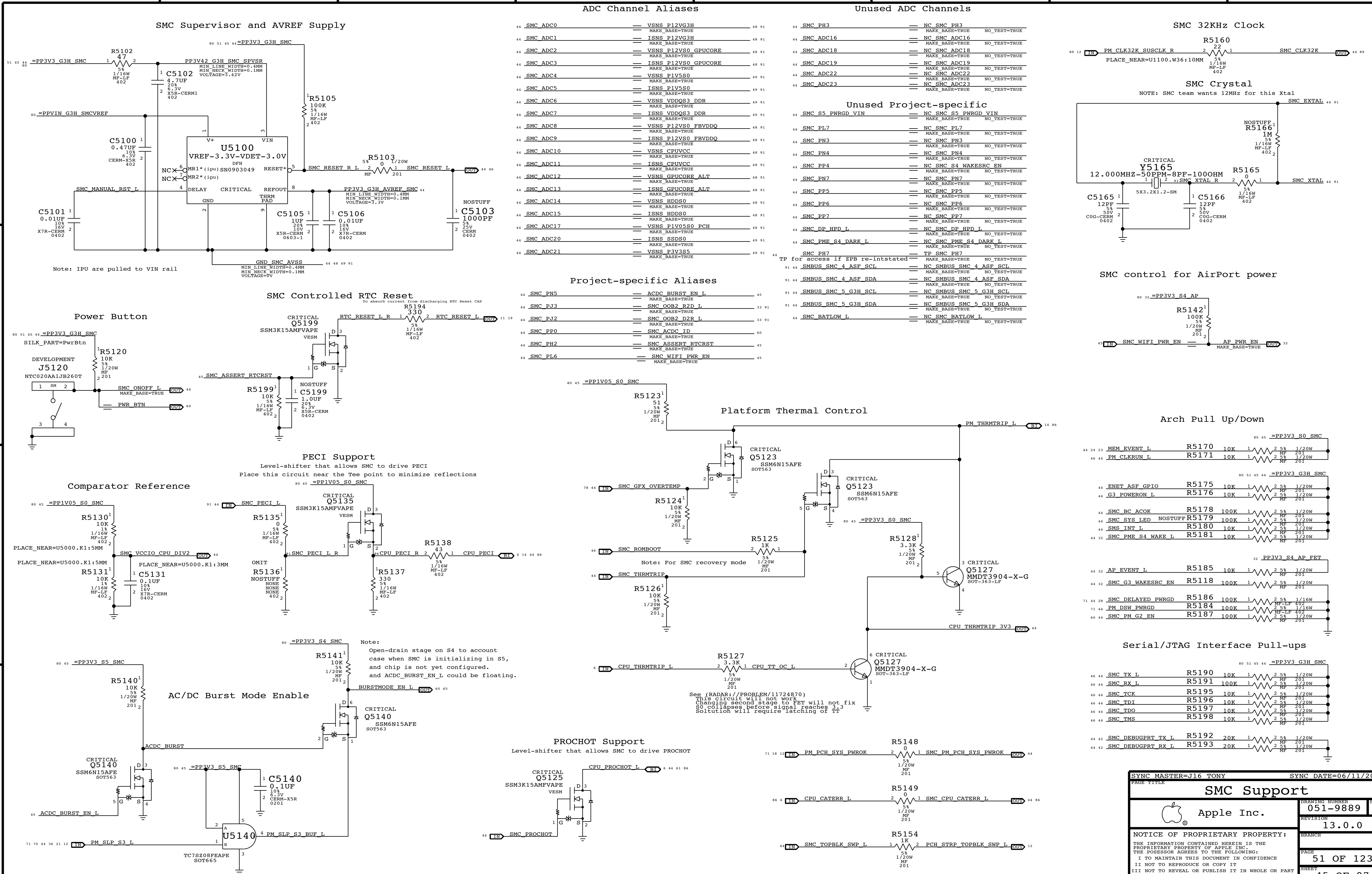
Apple Inc.



SYNC_MASTER=J16_TONY SYNC_DATE=06/11/2013

PAGE TITLE

SMC Support

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

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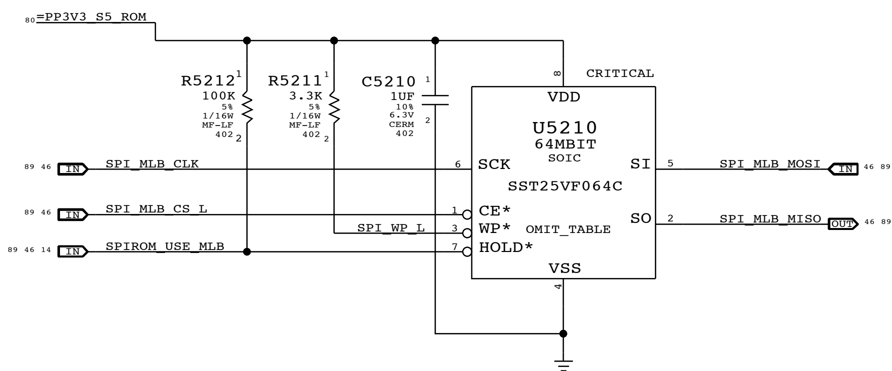
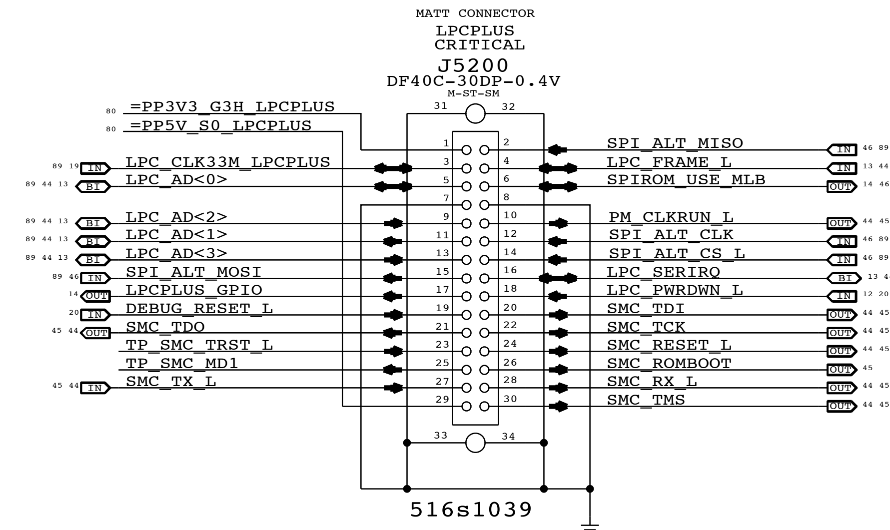
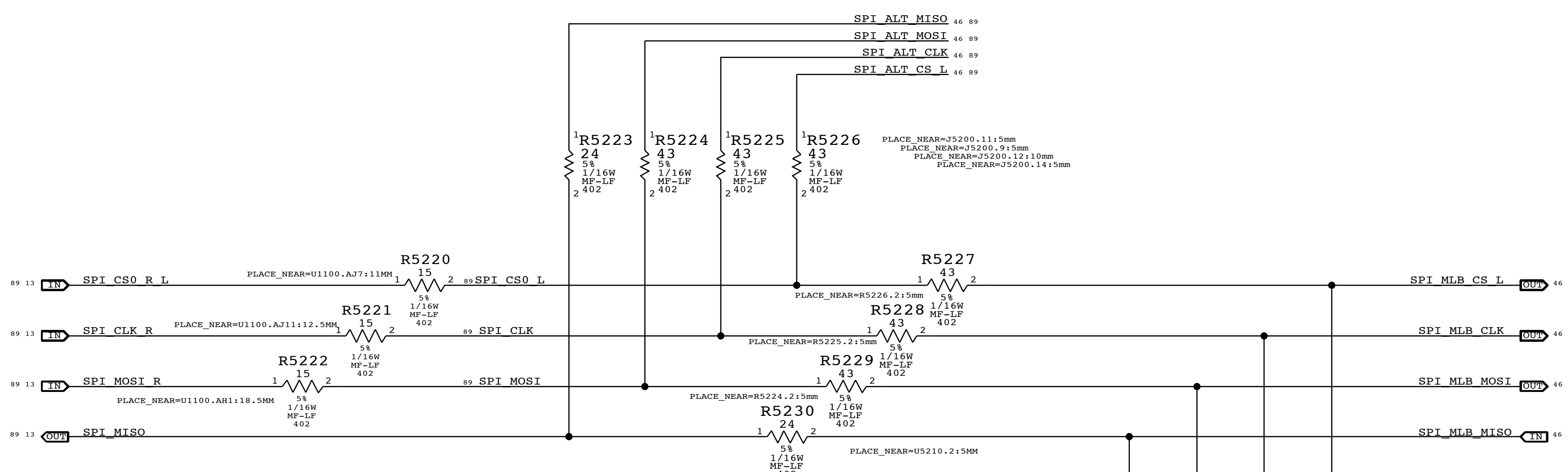
SPI BootROM



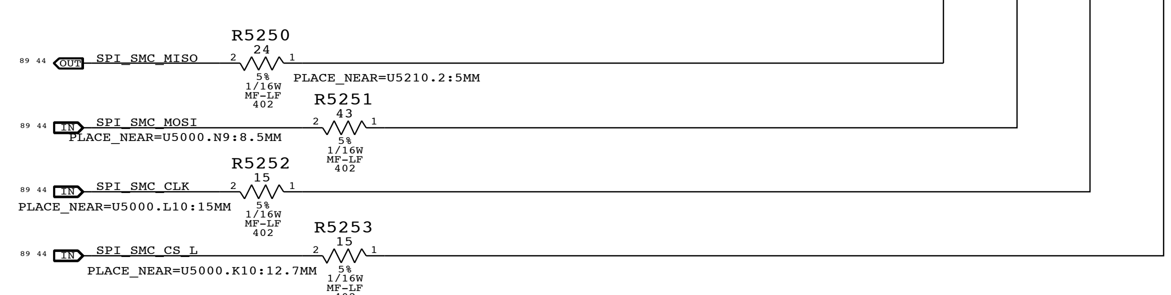
LPC+SPI Connector



SPI Series Termination



SMC SPI Support



SYNC_MASTER=J16_TONY SYNC_DATE=06/17/2013

PAGE TITLE
SPI and Debug Connector

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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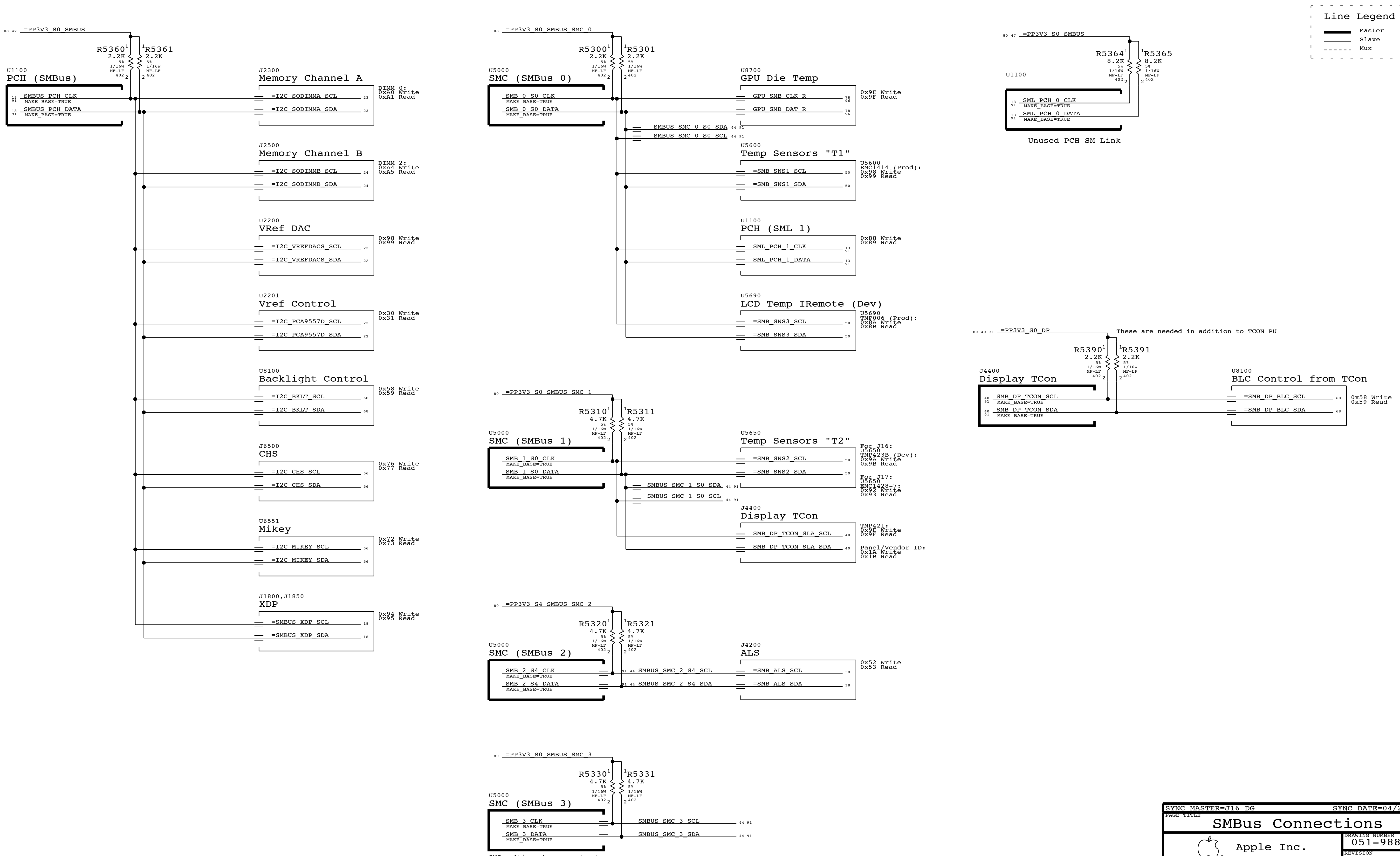

SYNC_MASTER=J16_DG SYNC_DATE=04/21/2013

PAGE TITLE

SMBus Connections

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

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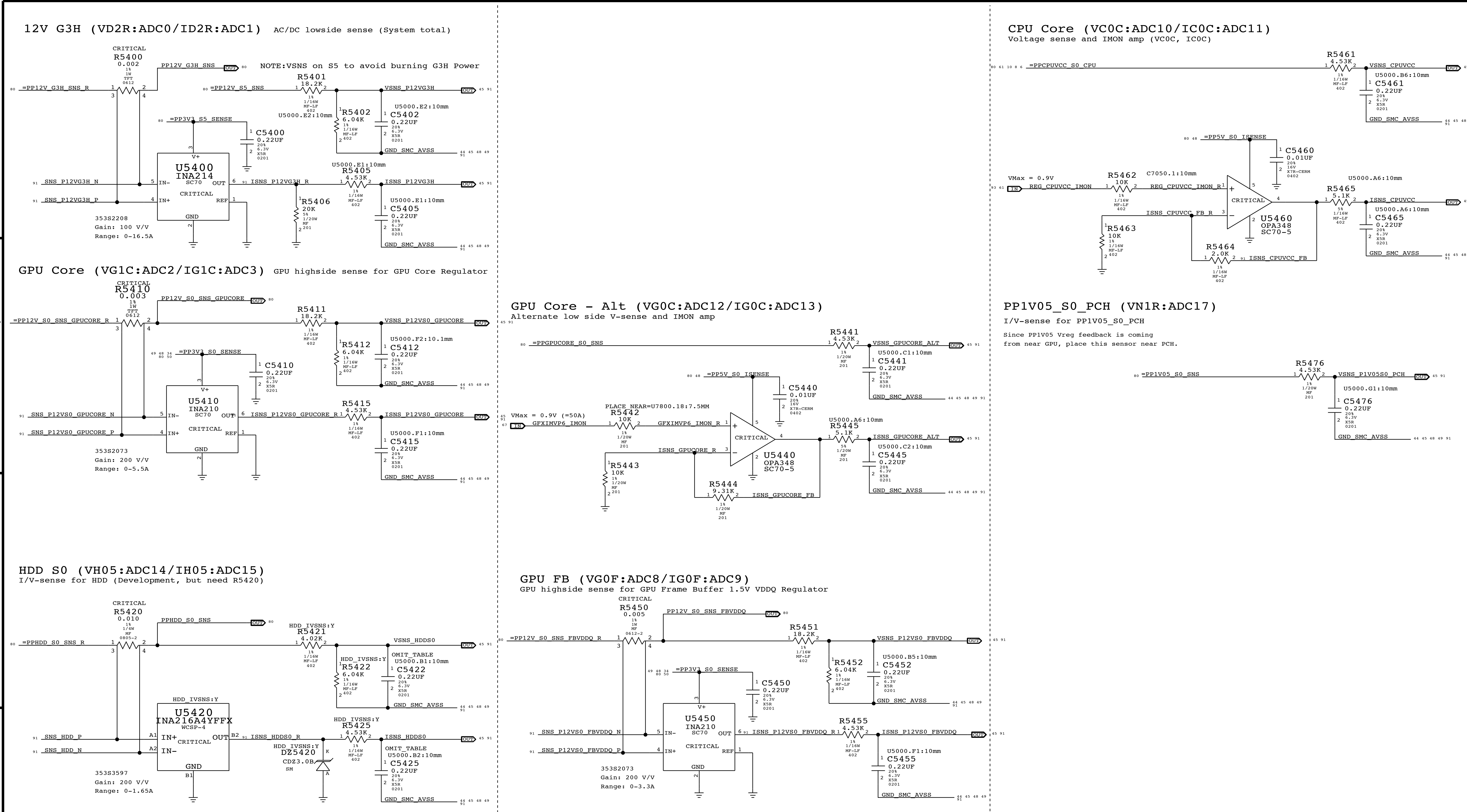

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP,0.22UF,201	C5422,C5425	HDD_IVSNS:Y
117S0002	2	RES,0 OHM,201	C5422,C5425	HDD_IVSNS:N

SYNC_MASTER=J16_TONY SYNC_DATE=06/11/2013

PAGE TITLE

I and V Sense

Apple Inc.

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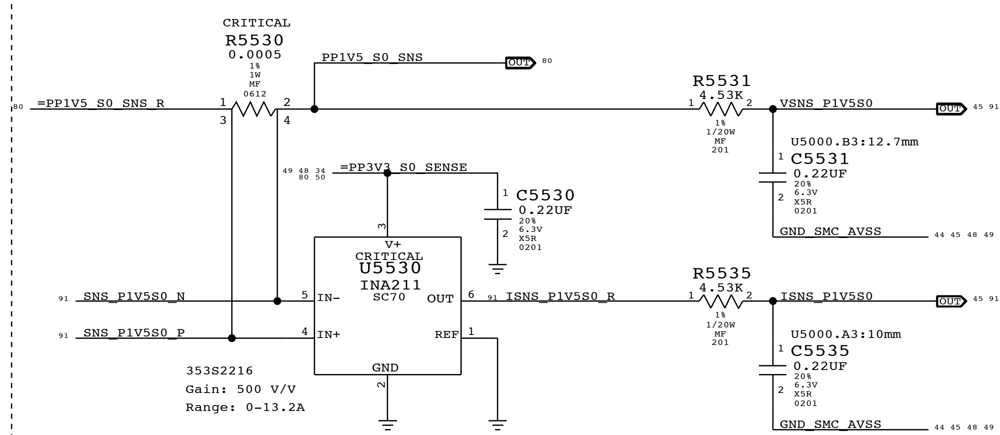
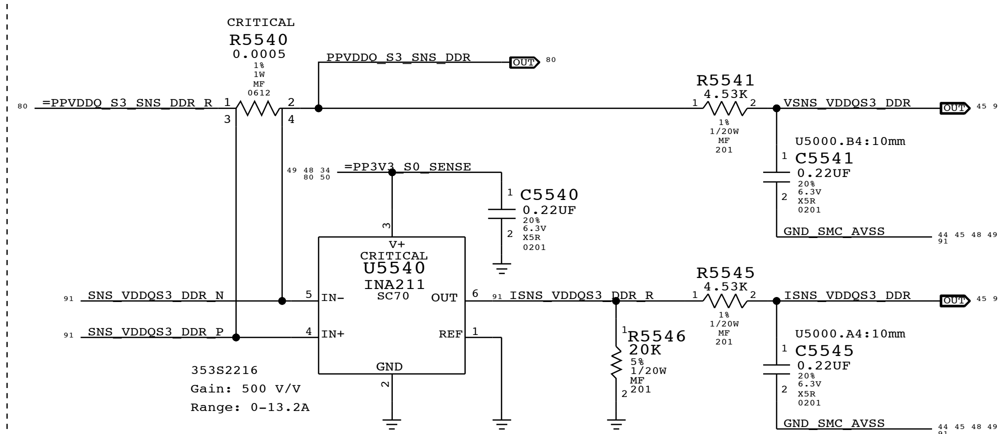
PP1V5_S0 (VC0M:ADC4/IC0M:ADC5)

lowside sense for VCCVRM,PCH,CPU mem, audio



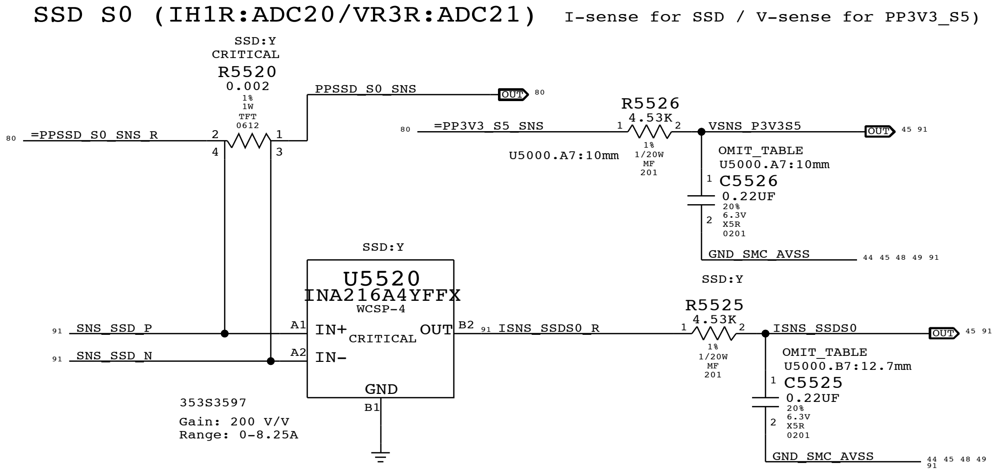
VDDQ S3 (VM0R:ADC6/IM0R:ADC7)

VDDQ lowside sense for SO-DIMM modules



SSD S0 (IH1R:ADC20/VR3R:ADC21)

I-sense for SSD / V-sense for PP3V3_S5)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP,0.22UF,201	C5525,C5526	SSD:Y
117S0002	2	RES,0 OHM,201	C5525,C5526	SSD:N

SYNC_MASTER=J16_TONY SYNC_DATE=06/03/2013

PAGE TITLE

I and V Sense(Continued)

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DRAWING NUMBER 051-9889 SIZE D

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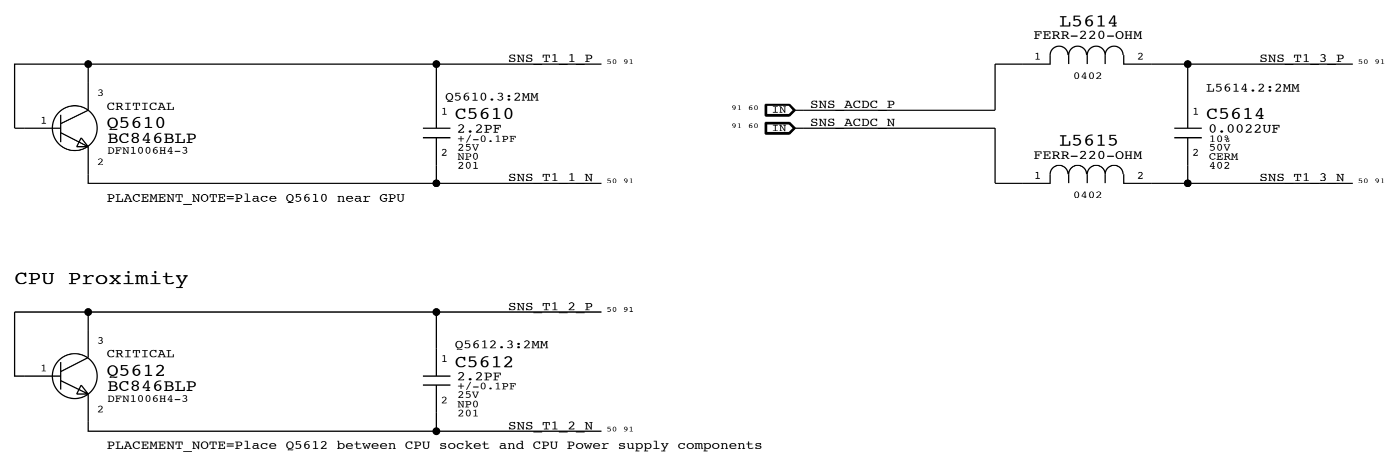
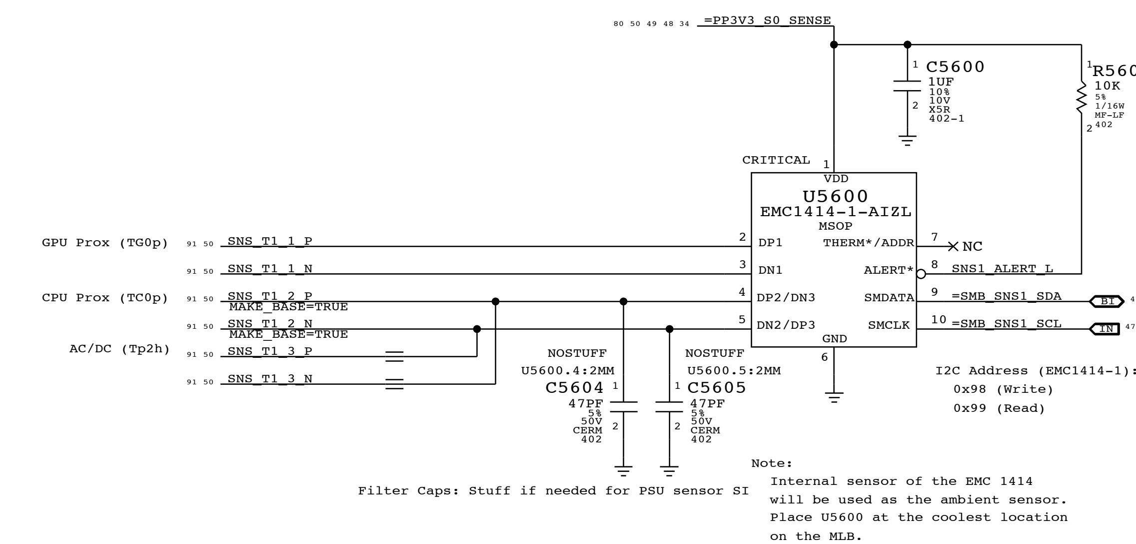
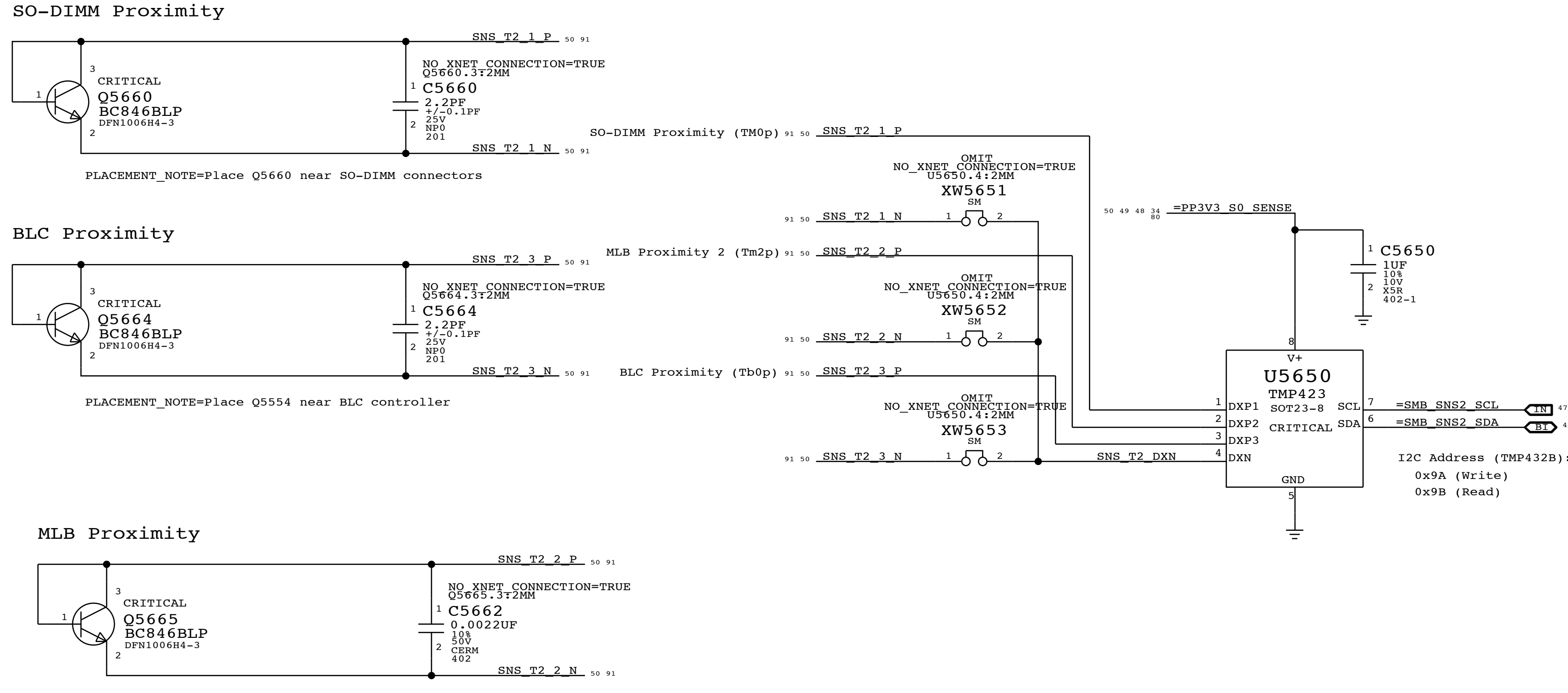
Temperature Sensor T1



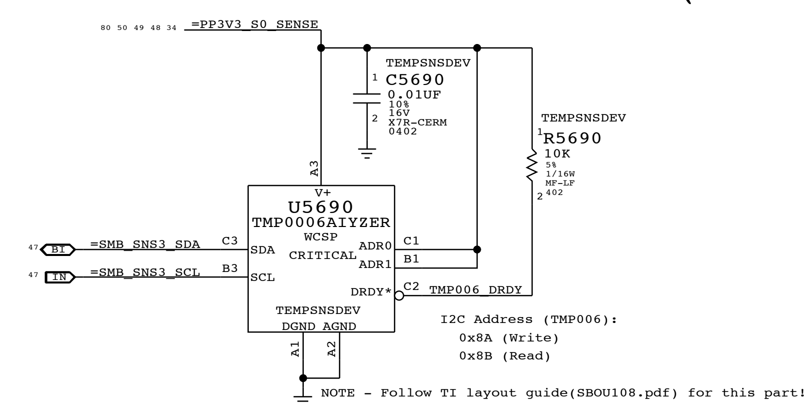


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

Temperature Sensor T2



Temperature Sensor T3: LCD Remote Sensor(Dev Only)



This PD part is a rubber bumper to protect TMP006
Added to board BOM (DEV only) to clean up PD BOM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
875-6433	1	BUMPER,U5690,D7	BUMPER_U5690	TEMPSNSDEV

SYNC_MASTER=J16_FIYIN SYNC_DATE=06/11/2013

PAGE TITLE

Temperature Sensors

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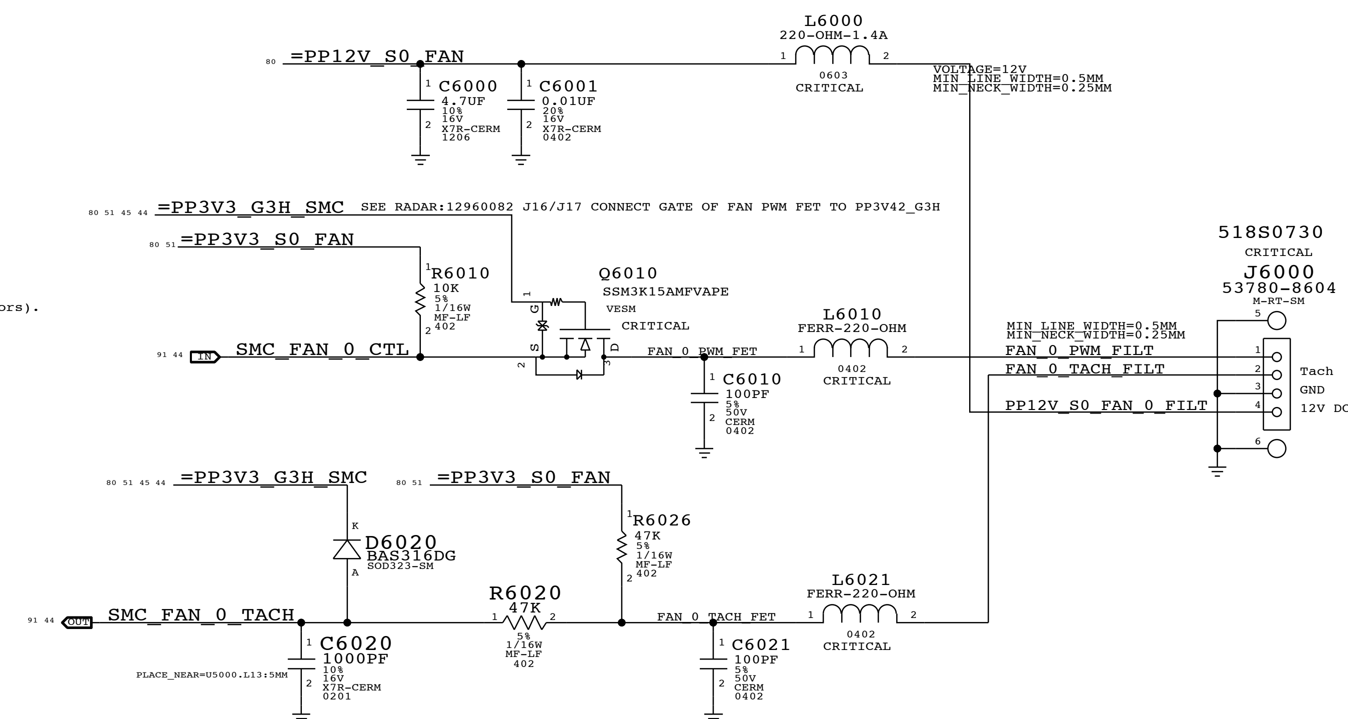
Note:
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

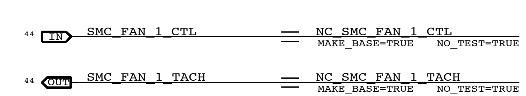
Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need scematic and PCB file of fan(All Vendors).

SMC Fan 0 (System)



Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.

SMC Fan 1 (Unused)



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE: System Fan

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DRAWING NUMBER: 051-9889 SIZE: D

REVISION: 13.0.0

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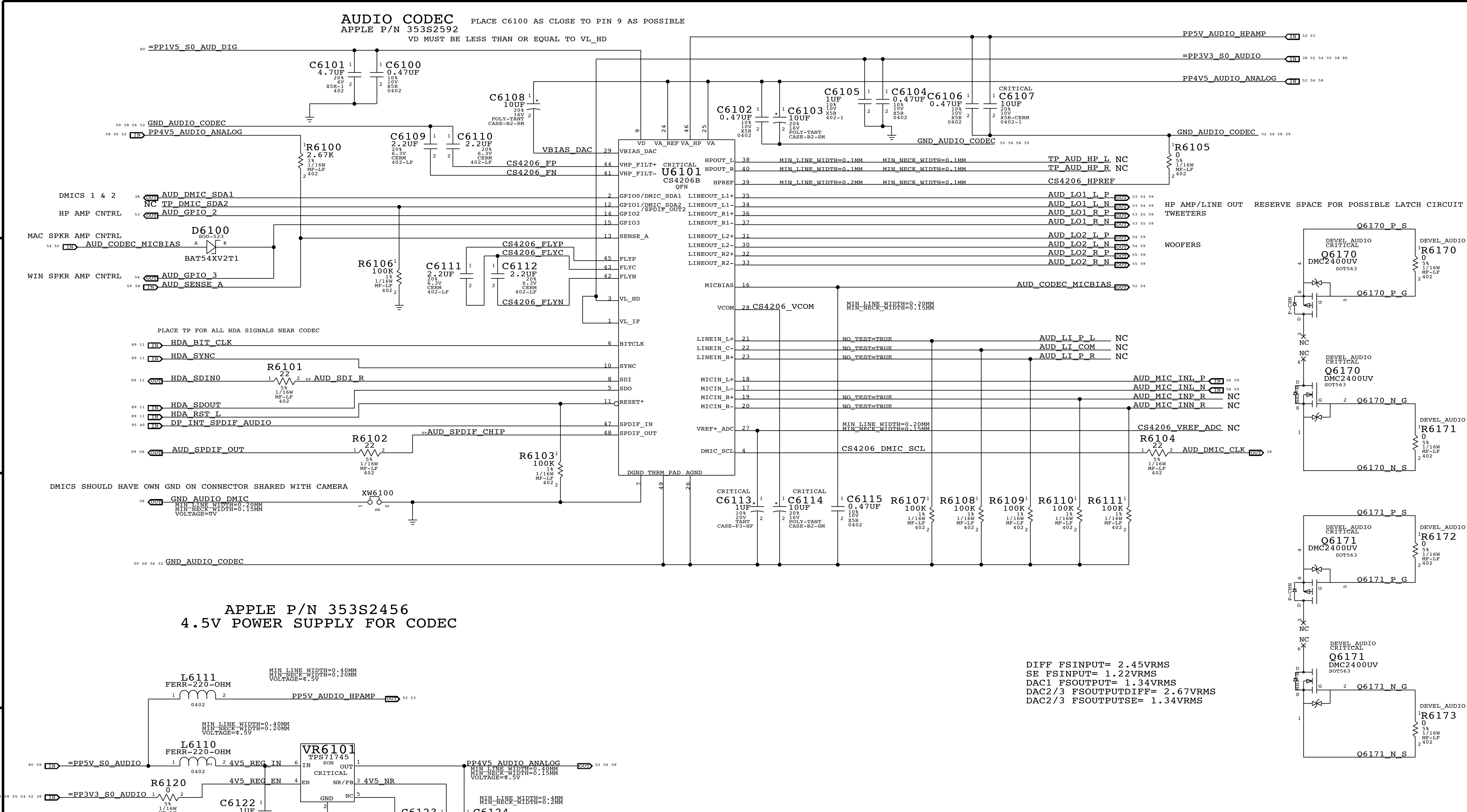

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

AUDIO: CODEC/REGULATORS

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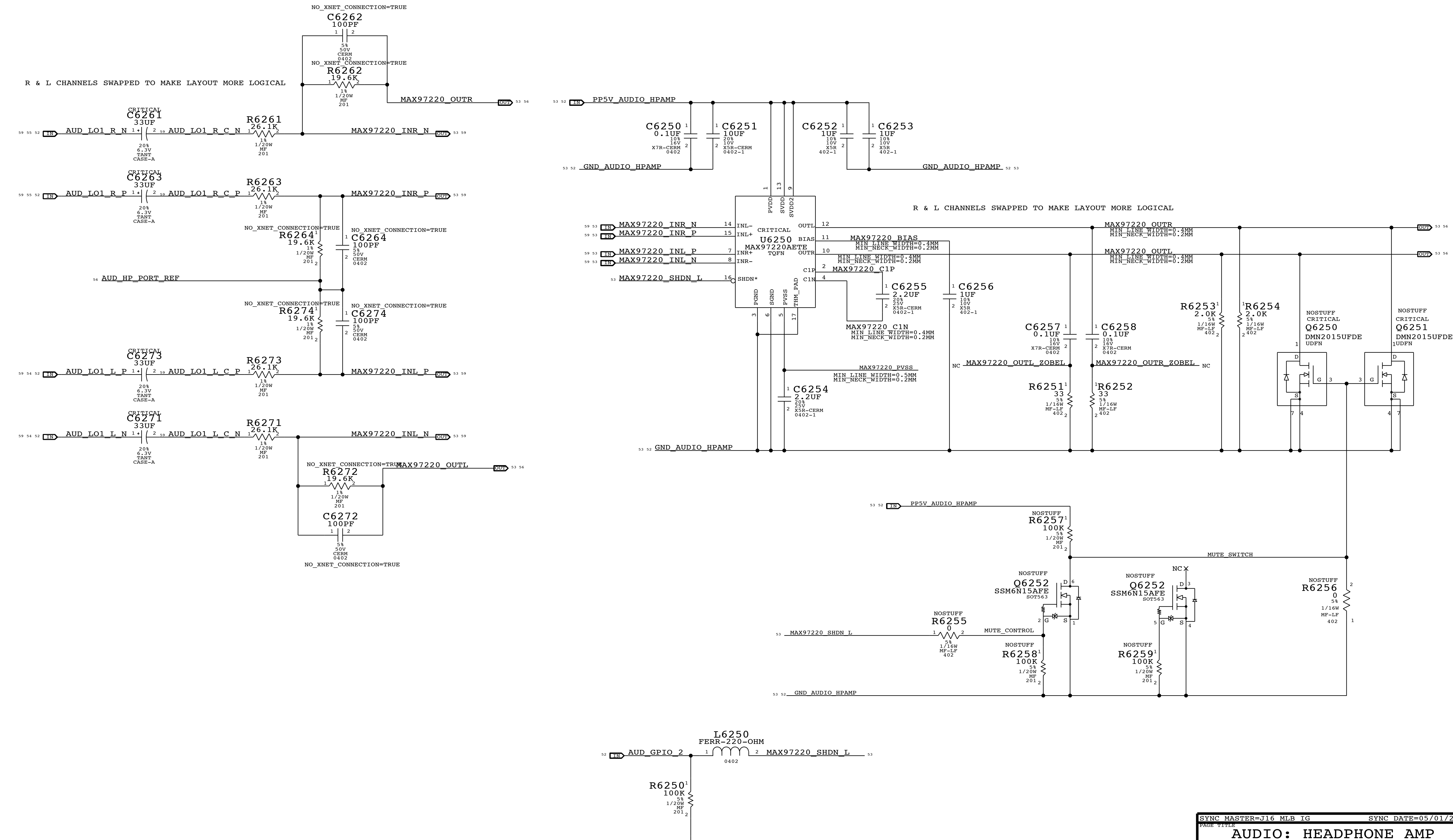

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

AUDIO: HEADPHONE AMP

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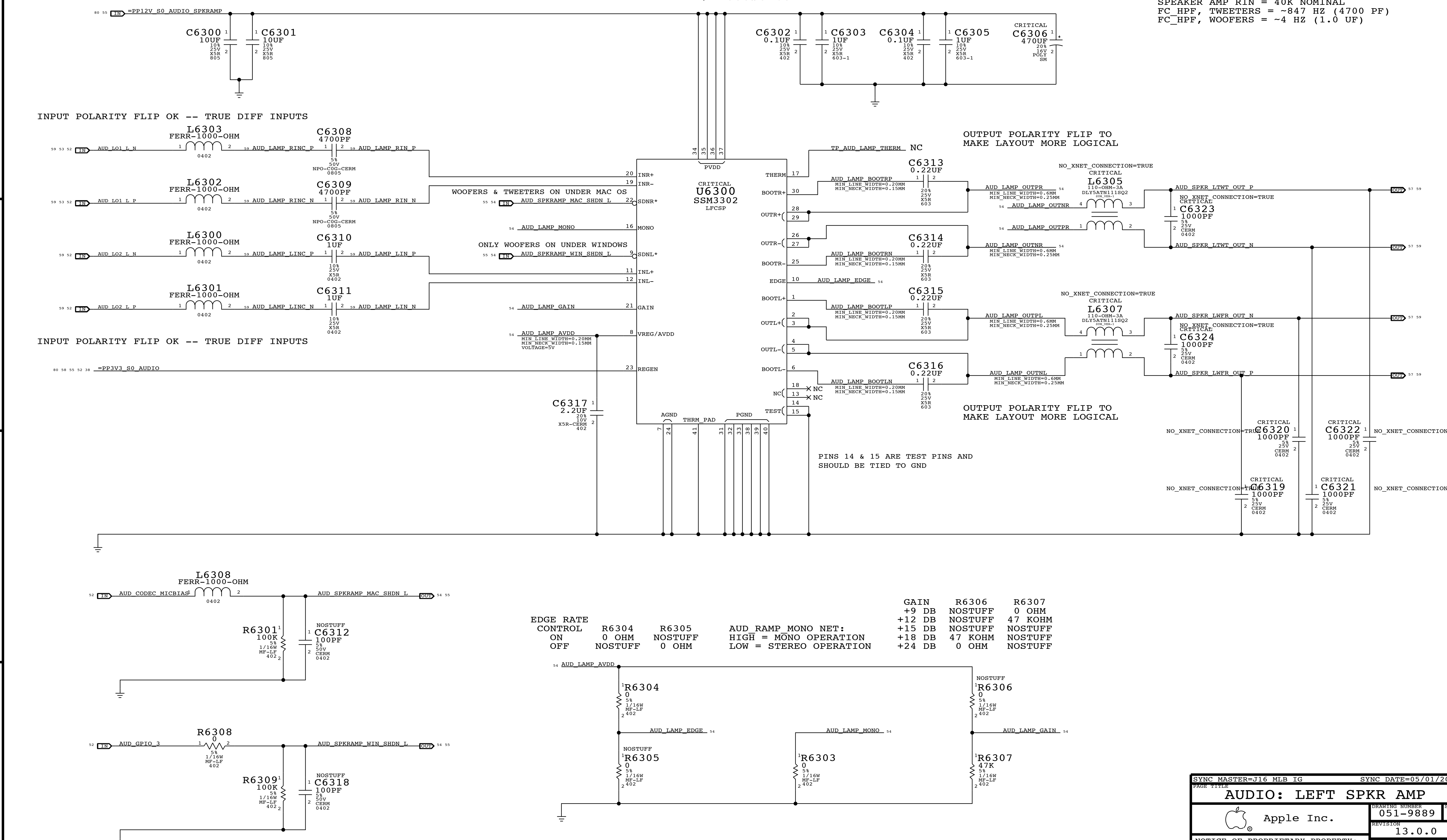
LEFT CH SPEAKER AMP

APPLE P/N 353S3163



EDGE RATE CONTROL	R6304	R6305
ON	0 OHM	NOSTUFF
OFF	NOSTUFF	0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN	R6306	R6307
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	47 KOHM
+15 DB	NOSTUFF	NOSTUFF
+18 DB	47 KOHM	NOSTUFF
+24 DB	0 OHM	NOSTUFF

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE: AUDIO: LEFT SPKR AMP

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DRAWING NUMBER: 051-9889 SIZE: D

REVISION: 13.0.0

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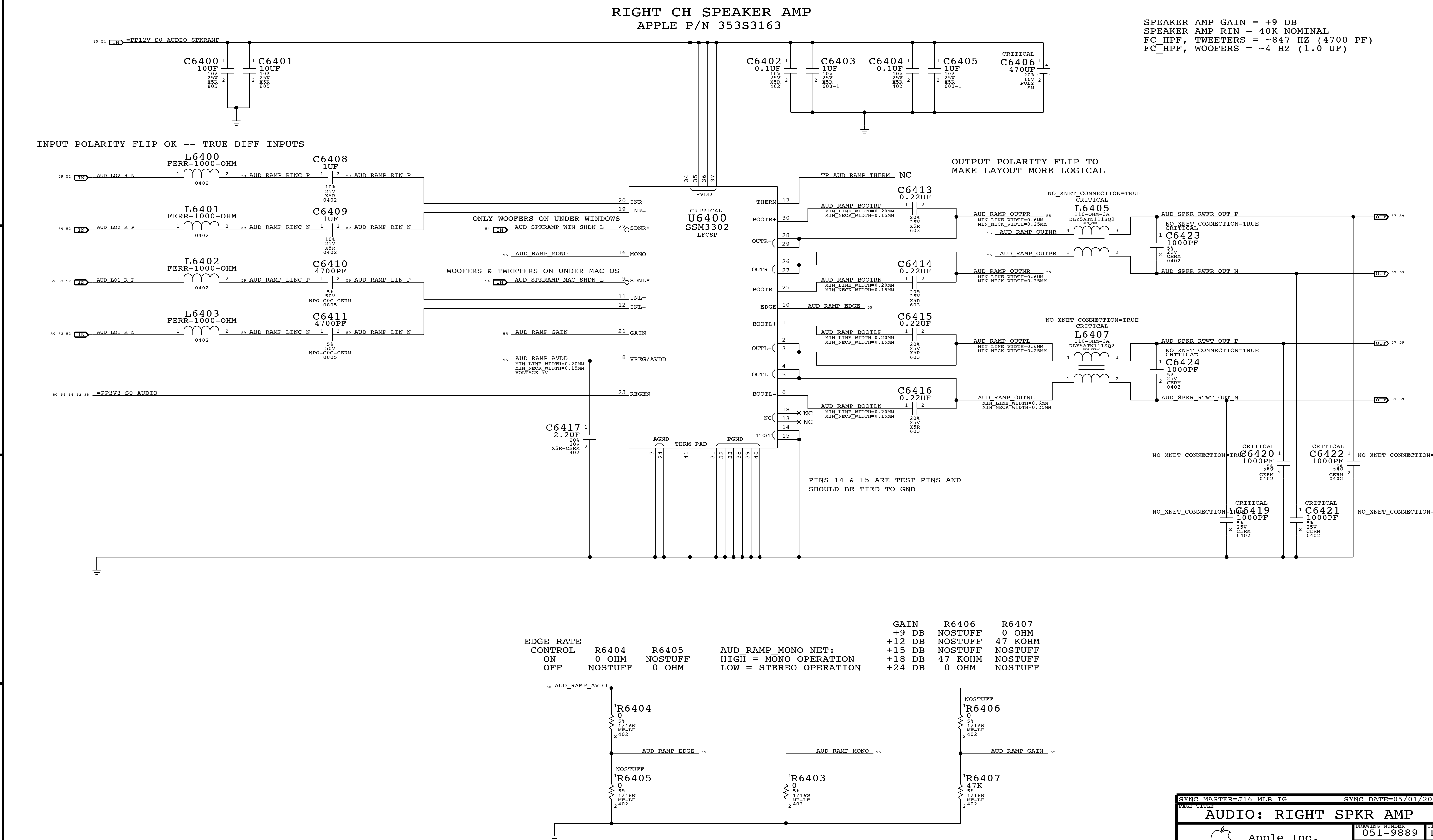

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE
AUDIO: RIGHT SPKR AMP

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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2640

PORT B LEFT(HEADSET MIC)
HP=80HZ, LP=10.63KHZ

MIKEY 1A
APN:353S2640
MIKEY ADDRESS: WRITE=72H, READ=73H

I2C ADDRESSES

MIKEY	READ	0111 0011	0X73
MIKEY	WRITE	0111 0010	0X72
CHS	READ	0111 0111	0X77
CHS	WRITE	0111 0110	0X76

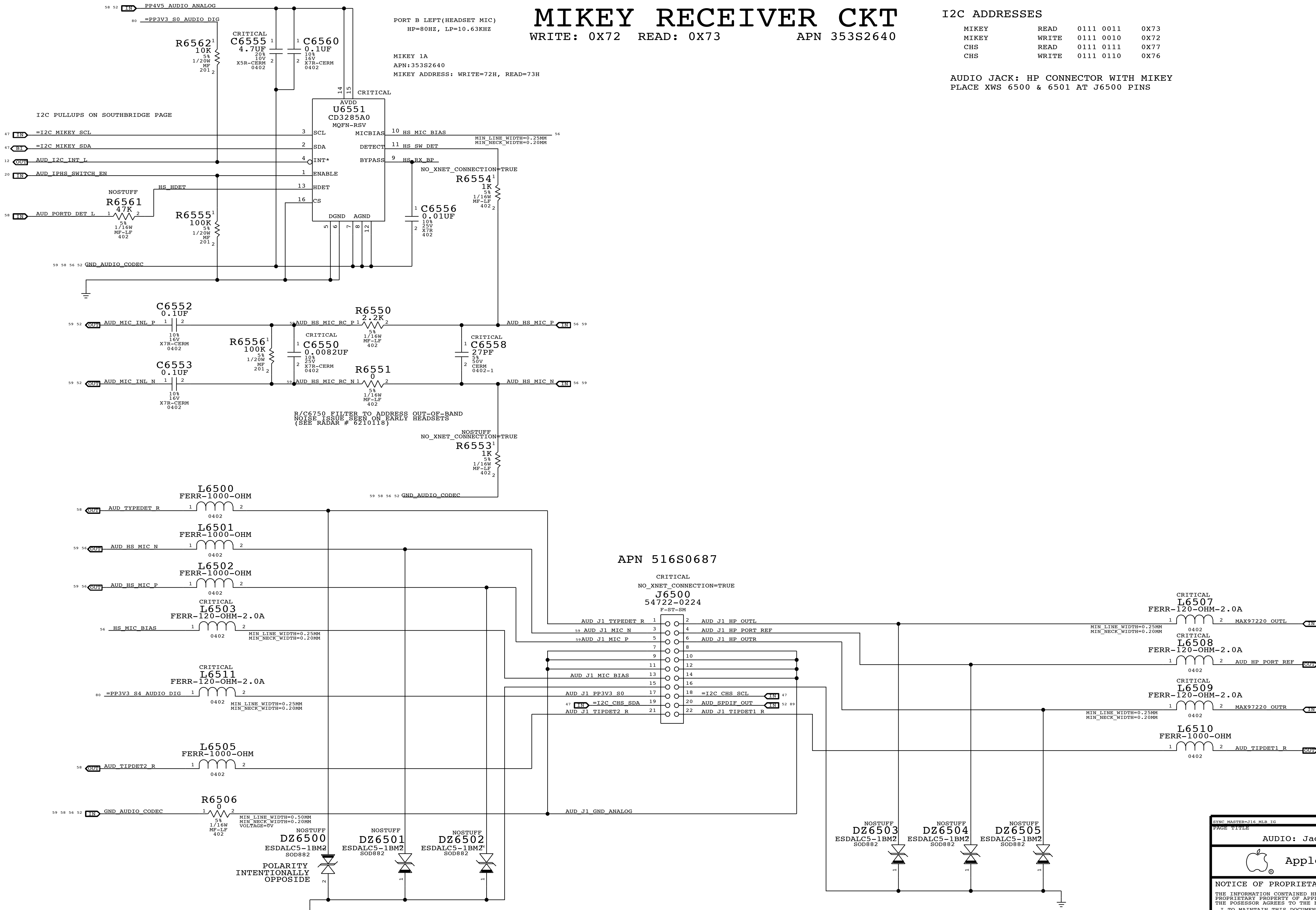
AUDIO JACK: HP CONNECTOR WITH MIKEY
PLACE XWS 6500 & 6501 AT J6500 PINS

I2C PULLUPS ON SOUTHBRIDGE PAGE

R/C6750 FILTER TO ADDRESS OUT-OF-BAND
NOISE ISSUE SEEN ON EARLY HEADSETS
(SEE RADAR # 6210118)

APN 516S0687

POLARITY
INTENTIONALLY
OPPOSIDE



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2012

PAGE TITLE
AUDIO: Jack, Mikey, CHS Switch

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DRAWING NUMBER 051-9889 SIZE D

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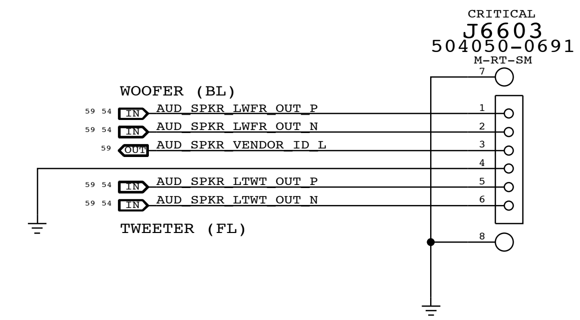
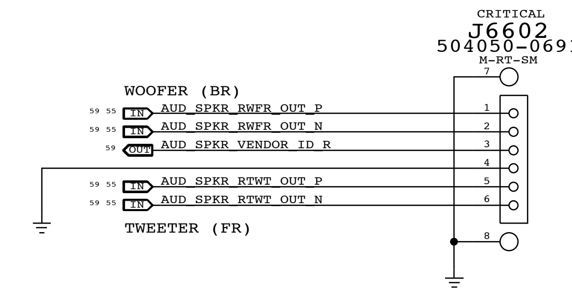
SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862





SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

Audio: Spkr/Mic Conn.

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DRAWING NUMBER 051-9889 SIZE D

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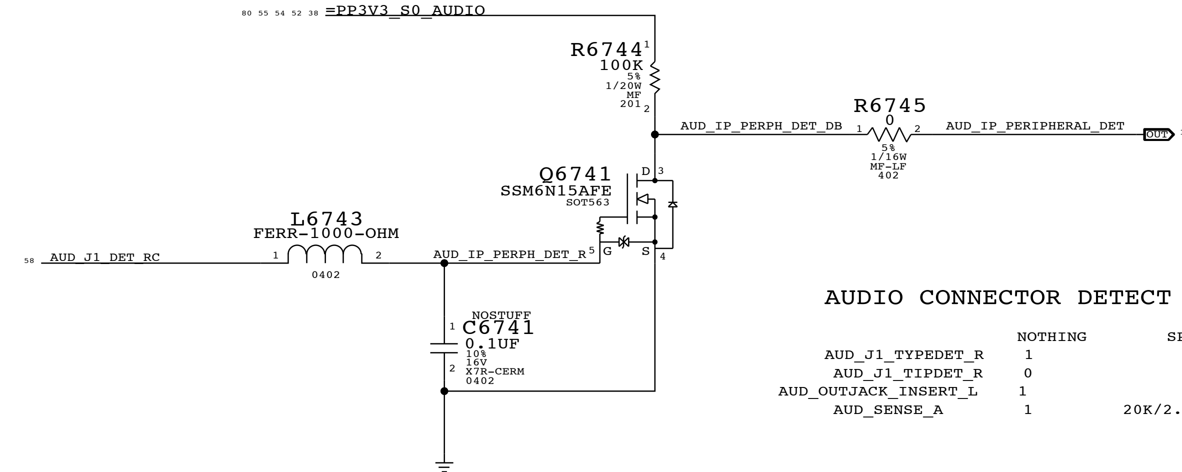
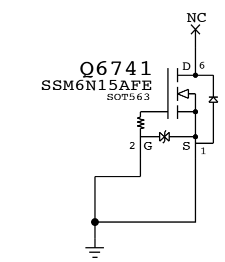
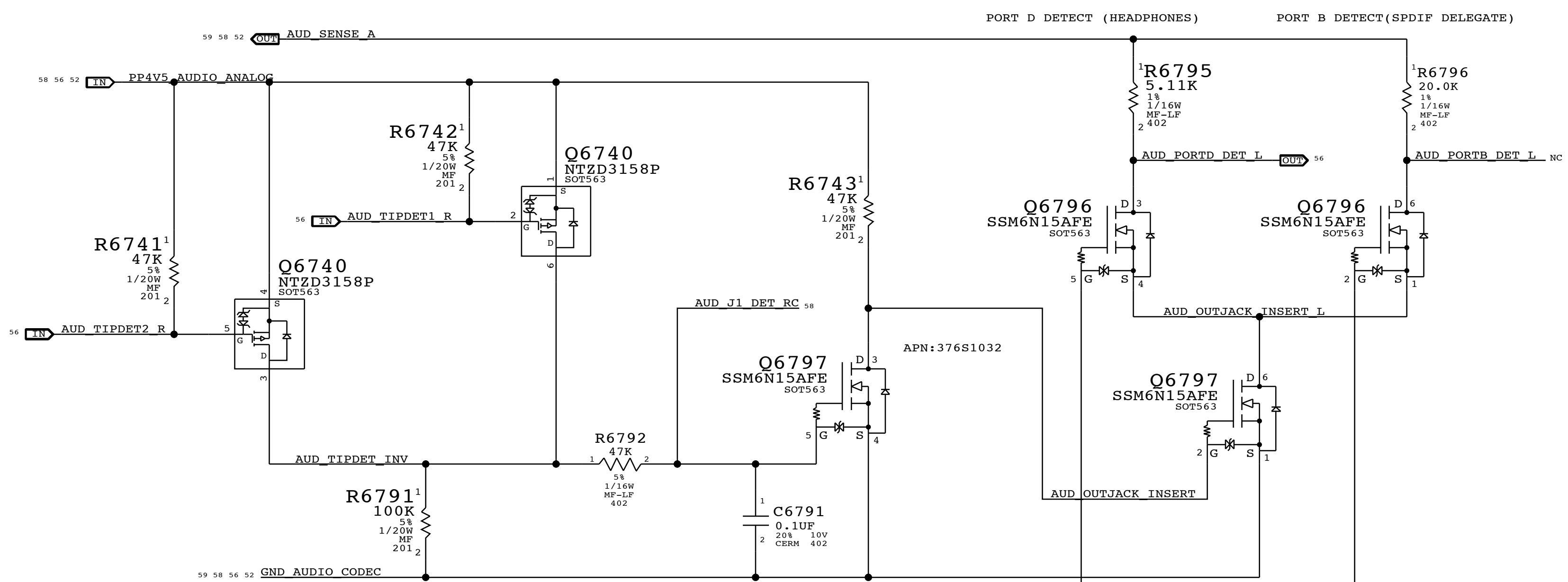
IPHS HS Detect Debounce CKT

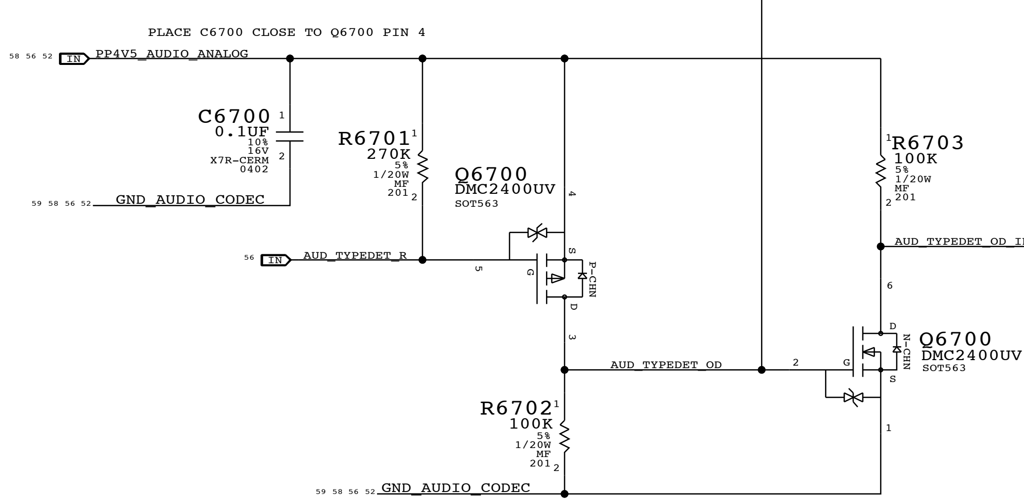


AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_J1_TYPEDET_R	1	1	0
AUD_J1_TIPDET_R	0	1	1
AUD_OUTJACK_INSERT_L	1	0	0
AUD_SENSE_A	1	20K/2.67K RDIV	39.2K/2.67K RDIV

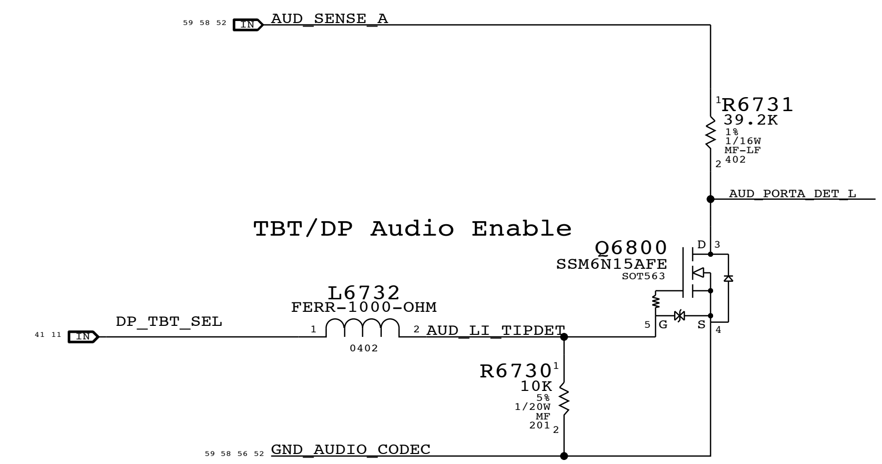






Target Display Mode Detect

TBT/DP Audio Enable



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE
AUDIO: Detects/Grounding

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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	MICBIAS	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
SPDIF IN	0X07 (7)	0X0F (15)	N/A	0X09 (DET A)
INTERNAL MIC ARRAY	0X06 (6) 0X05 (5)	0X0E (14,LEFT & RIGHT) 0X12 (18,LEFT)	N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	Lynx POINT GPIO 16	Lynx POINT GPIO 5 (RCVR INT) Lynx POINT GPIO 3 (PERIPH DET)

OTHER DETECT

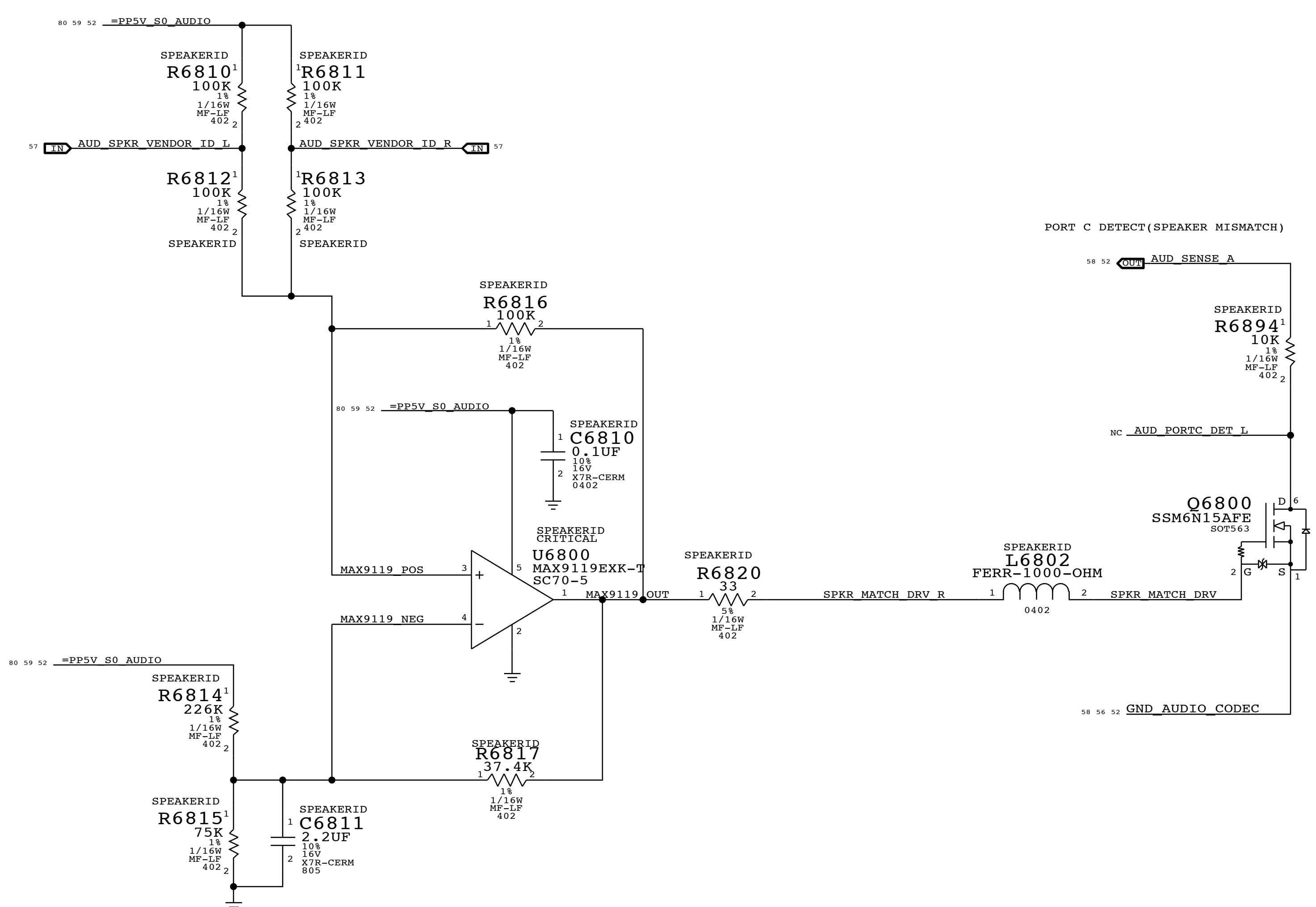
FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
MULTIPLE SPKR VENDORS	N/A	N/A	N/A	0X0C (DET C)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I215	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_P	52 53 54
I216	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_N	52 53 54
I217	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_P	53
I218	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_N	53
I211	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_P	52 53 55
I212	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_N	52 53 55
I210	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_P	53
I209	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_N	53
I208	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_P	52 54
I206	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_N	52 54
I207	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_P	52 55
I204	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_N	52 55
I205	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_P	55
I203	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_N	55
I220	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_P	55
I219	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_N	55
I222	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_P	55
I221	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_N	55
I224	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_P	55
I223	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_N	55
I226	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_P	54
I225	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_N	54
I227	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_P	54
I229	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_N	54
I228	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_P	54
I230	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_N	54
I264	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_P	54
I263	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_N	54
I236	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_P	53
I238	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_N	53
I237	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_P	53
I239	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_N	53
I240	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_P	55 57
I241	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_N	55 57
I242	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_P	55 57
I243	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_N	55 57
I244	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_P	54 57
I246	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_N	54 57
I245	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_P	54 57
I247	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_N	54 57
I326	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_P	52 56
I327	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_N	52 56
I328	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_P	56
I329	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_N	56
I254	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_P	56
I255	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_N	56
I324	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_P	56
I325	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_N	56



SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

AUDIO: Speaker ID

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3.425V "G3Hot" Regulator

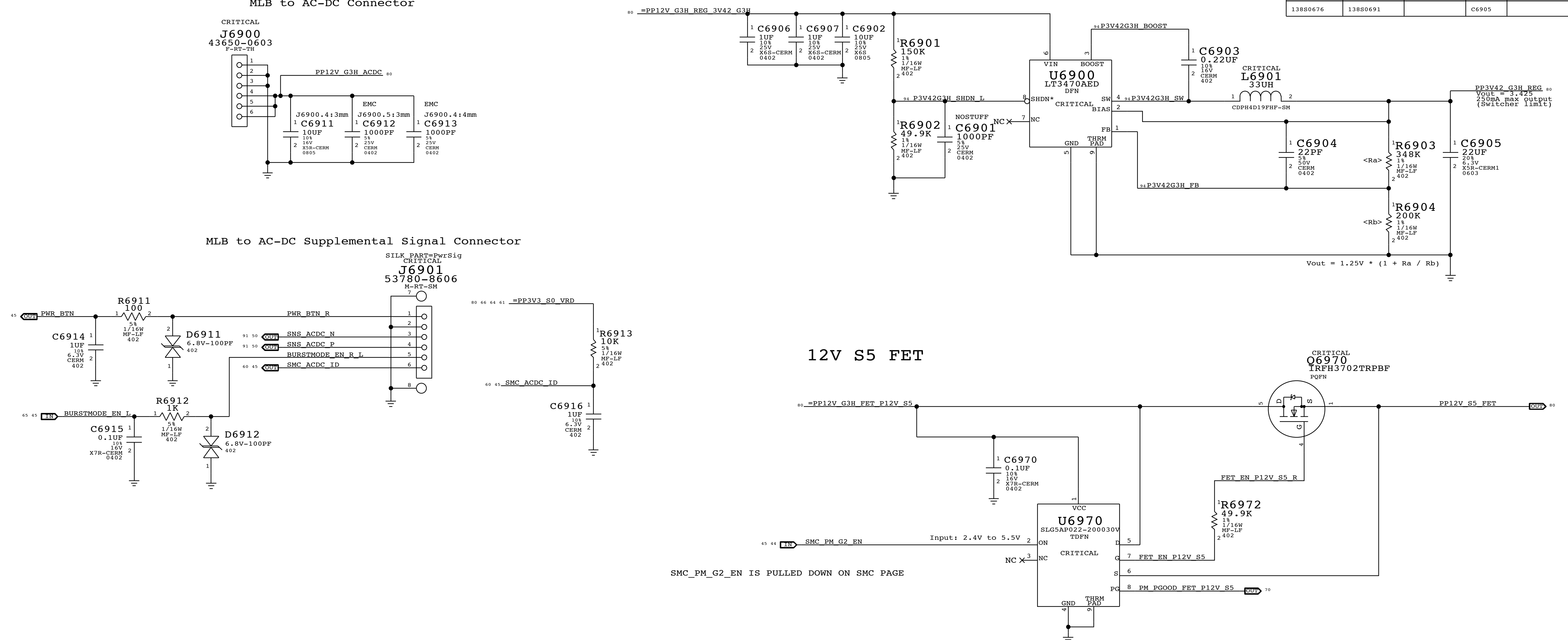
Switching freq: 409 kHz = $\frac{13.5}{L6901}$

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0676	138S0691		C6905	

MLB to AC-DC Connector

MLB to AC-DC Supplemental Signal Connector

12V S5 FET



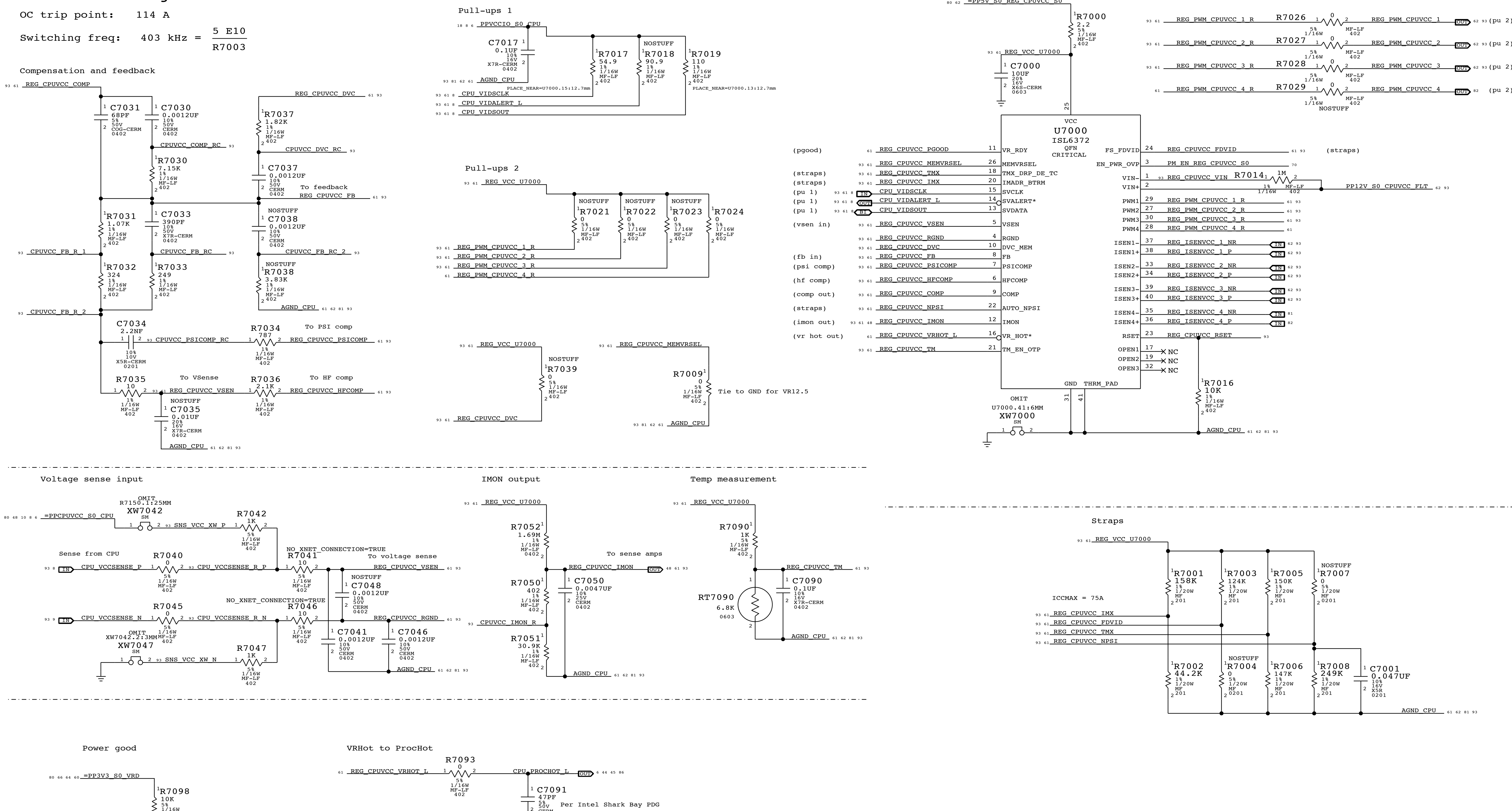
SMC_PM_G2_EN IS PULLED DOWN ON SMC PAGE

SYNC_MASTER=J16_MLB_IG	SYNC_DATE=05/01/2013
PAGE TITLE	Power Connectors / VReg G3Hot
Apple Inc.	DRAWING NUMBER 051-9889, SIZE D
	REVISION 13.0.0
	BRANCH
	PAGE 69 OF 123
	SHEET 60 OF 97

CPU VCC S0 Regulator

OC trip point: 114 A

Switching freq: 403 kHz = 5 E10 / R7003



SYNC_MASTER=J16_ROSSANA		SYNC_DATE=07/29/2013
PAGE TITLE: VReg CPU VCC Cntl		
Apple Inc.	DRAWING NUMBER: 051-9889	SIZE: D
	REVISION: 13.0.0	
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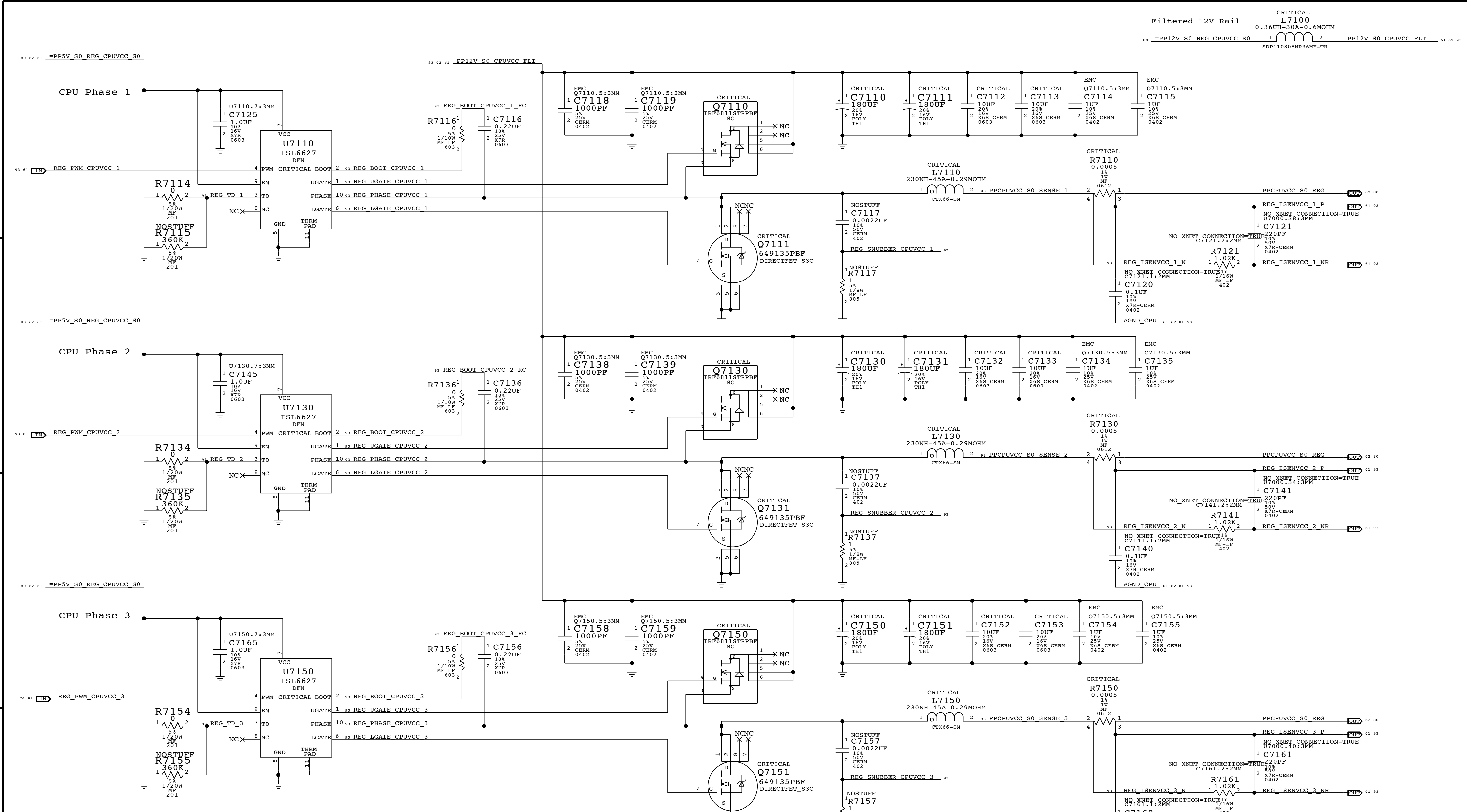
	SHEET: 61 OF 97	



SYNC_MASTER=J16_DG SYNC_DATE=04/21/2013

PAGE TITLE

VReg CPU VCC Phases

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DRAWING NUMBER 051-9889 SIZE D

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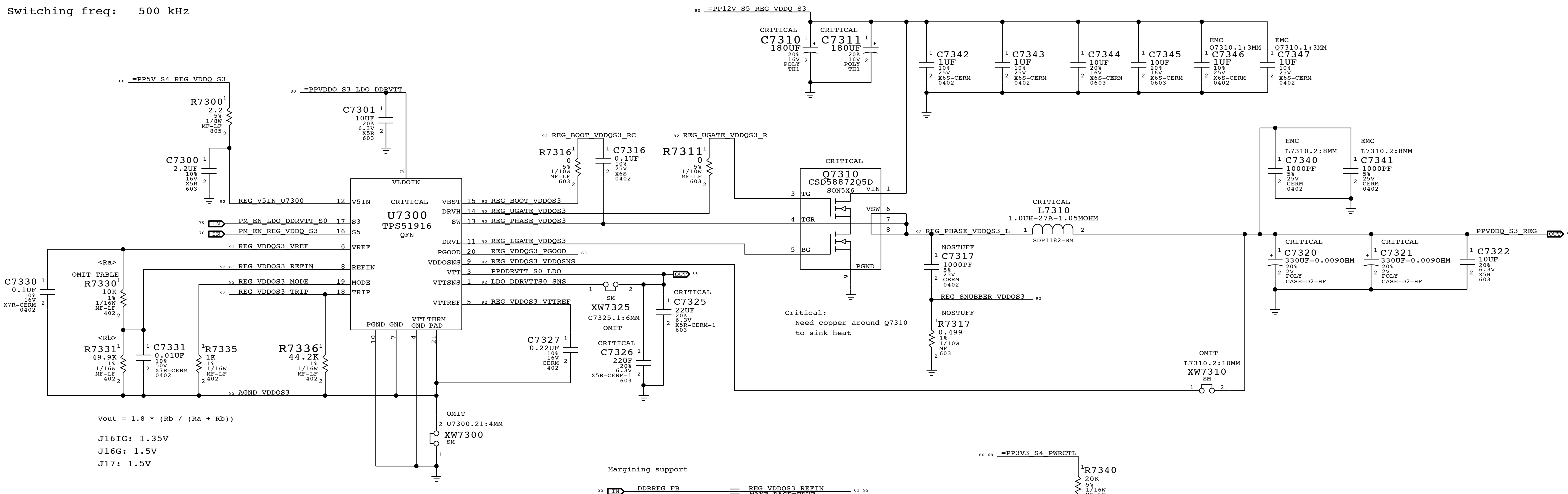
VDDQ (1.5V / 1.35V) S3 Regulator

OC trip point: 30.4 A VDDQ = $\frac{R7336}{8\ E5 * Rds(Q7310)} + \frac{0.65625}{L7310 * f(switch)}$

3 A VTT (FIXED)
10 mA VTTREF (FIXED)

Switching freq: 500 kHz



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
114S0335	1	RES,16.5K,402	R7330	VDDQ:P1V35
114S0315	1	RES,10K,402	R7330	VDDQ:P1V5

SYNC_MASTER=J16_ROSSANA SYNC_DATE=06/17/2013

PAGE TITLE: VReg VDDQ S3

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DRAWING NUMBER: 051-9889 SIZE: D

REVISION: 13.0.0

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PAGE: 73 OF 123

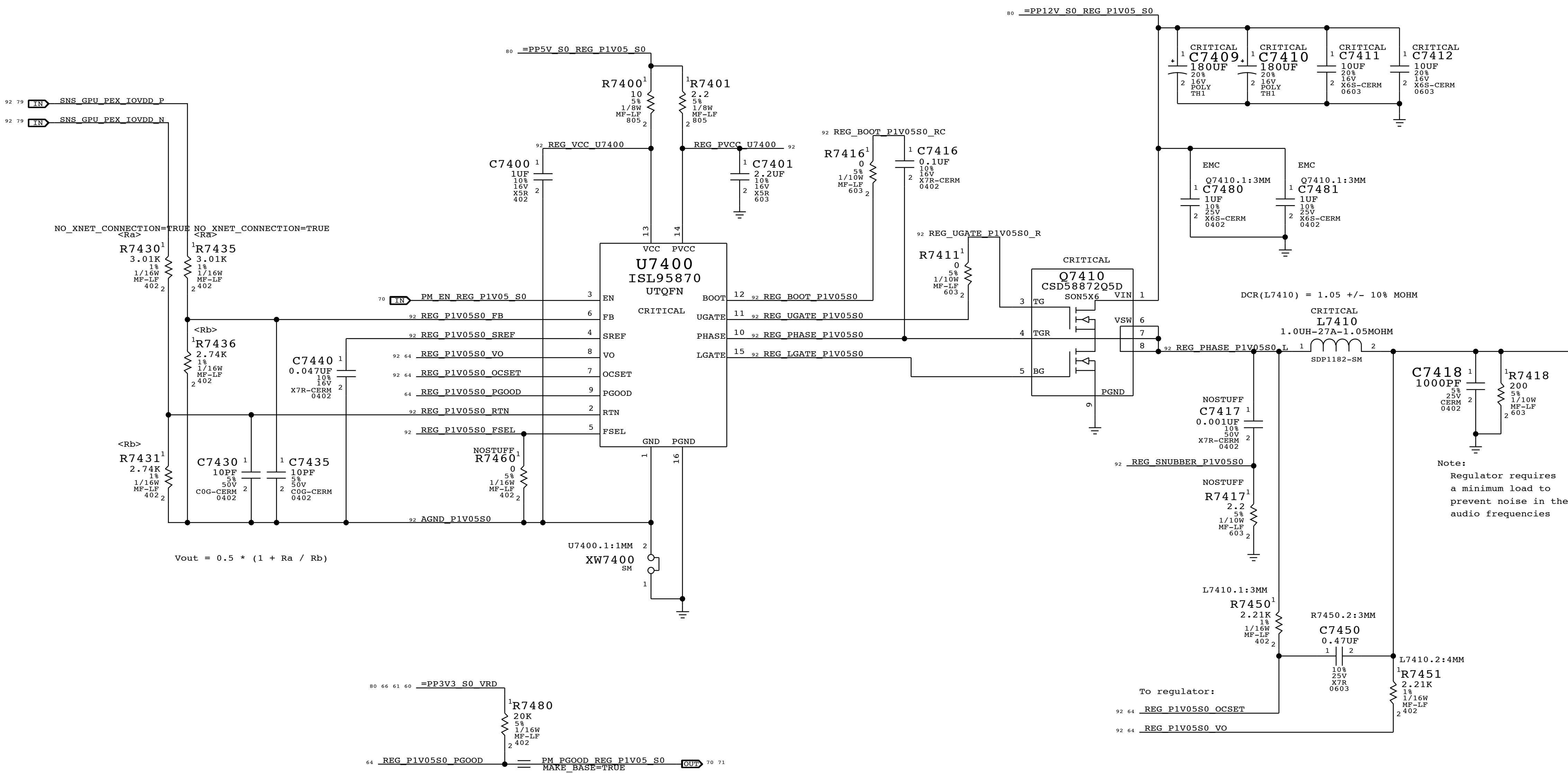
SHEET: 63 OF 97

PCH/GPU/TBT (1.05V) S0 Regulator

Switching freq: 500 kHz

OC trip point: $16.3\ \text{A} = \dfrac{R7450 * 8.5\ E{-}6}{DCR(L7410)}$

FSEL STRAP	SW FREQ
GND	300 kHz
VCC	1 MHz
100k to GND	600 kHz
FLOAT	500 kHz



SYNC_MASTER=J16_ROSSANA	SYNC_DATE=07/29/2013
PAGE TITLE	VReg PCH/GPU/TBT 1V05 S0
Apple Inc.	DRAWING NUMBER 051-9889 SIZE D
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	BRANCH
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	SHEET 64 OF 97

3.3V S5 Regulator

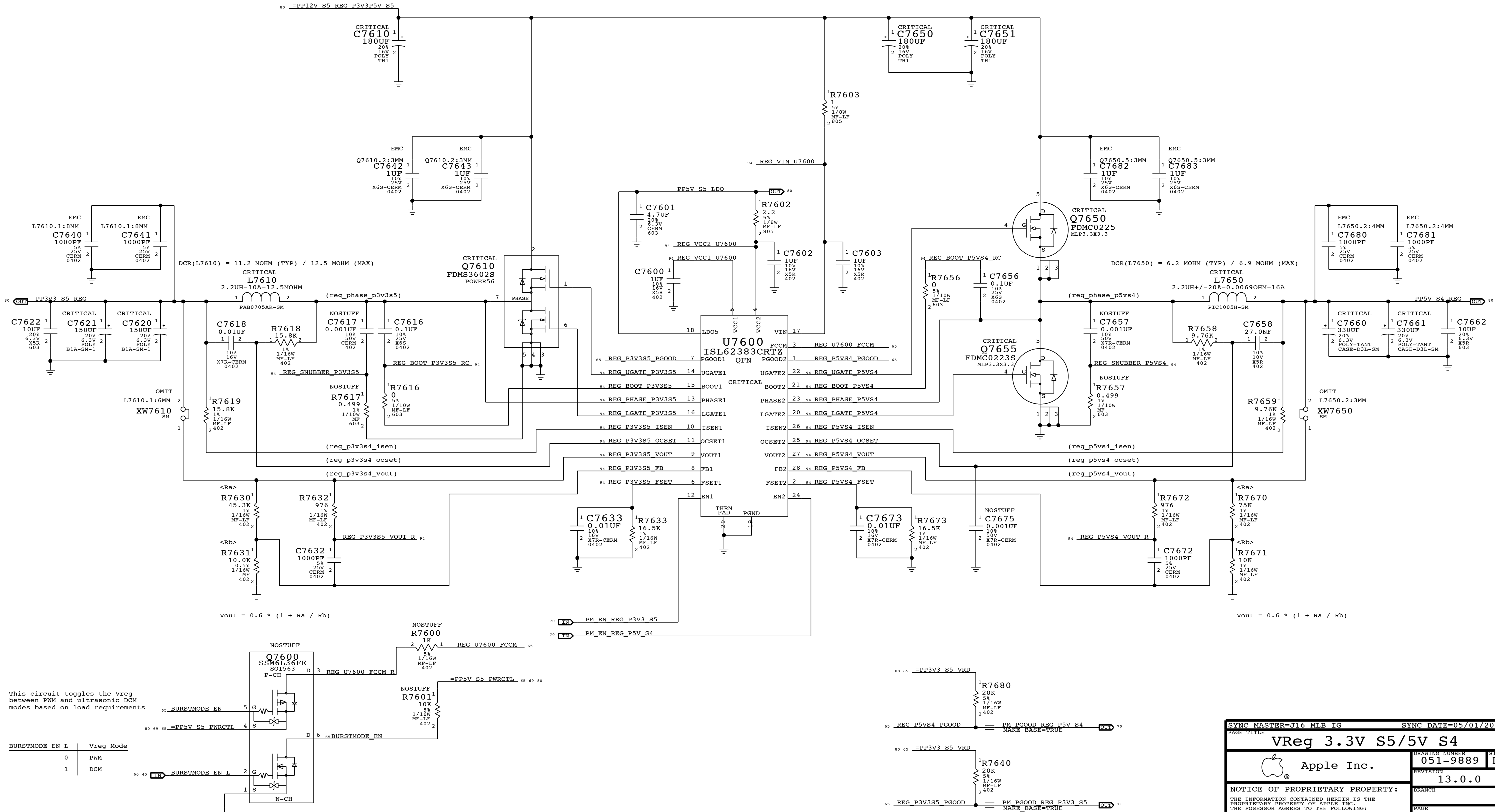
OC trip point: 12.5 A = $\frac{\text{R7618 * 10 E-6}}{\text{DCR(L7610)}}$

Switching freq: 356 kHz = $\frac{1}{\text{170 E-12 * R7633}}$

5V S4 Regulator

OC trip point: 14.1 A = $\frac{\text{R7658 * 10 E-6}}{\text{DCR(L7650)}}$

Switching freq: 356 kHz = $\frac{1}{\text{170 E-12 * R7673}}$



BURSTMODE_EN_L	Vreg Mode
0	PWM
1	DCM

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

VReg 3.3V S5/5V S4

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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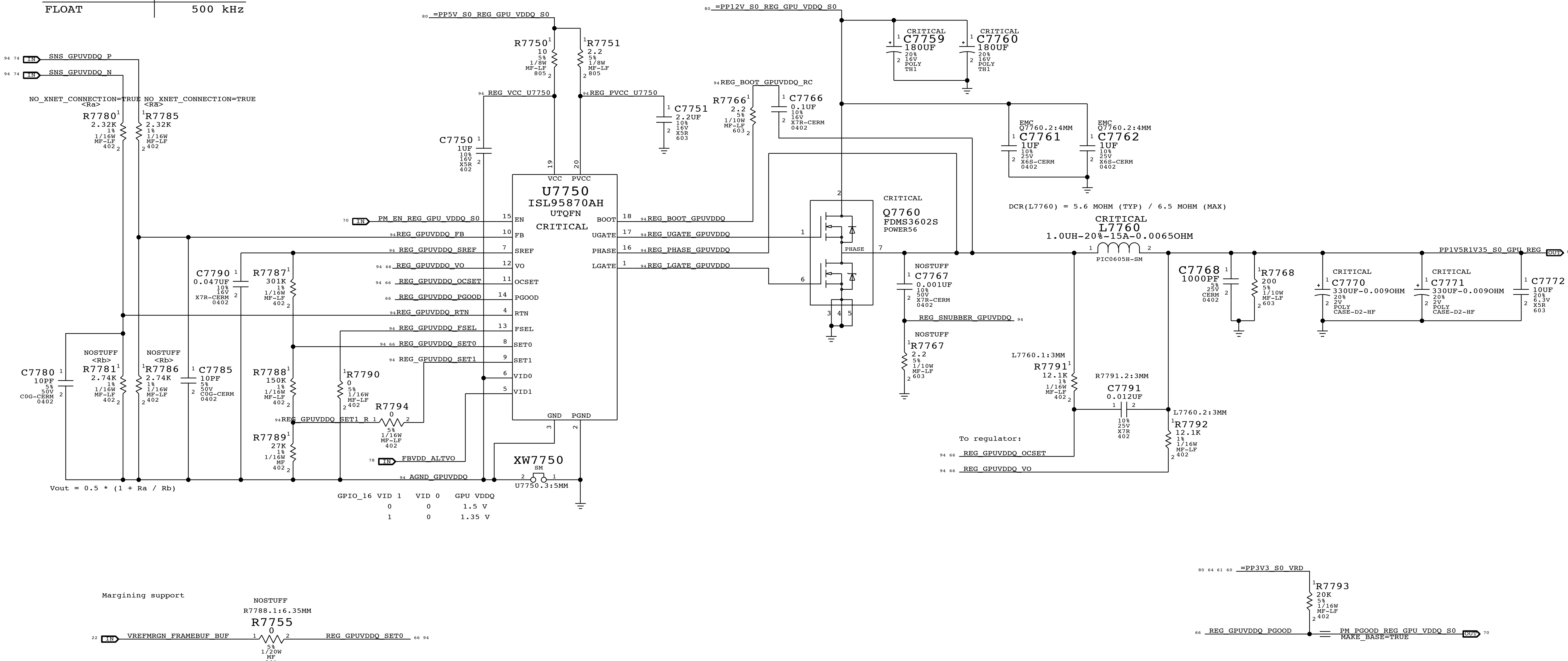
GPU VDDQ S0 REGULATOR

OC trip point: 15.8 A = R7791 * 8.5 E-6 / DCR(L7760)

Switching freq: 300 kHz

FSEL STRAP	SW FREQ
GND	300 kHz
VCC	1 MHz
100k to GND	600 kHz
FLOAT	500 kHz



SYNC_MASTER=J16_DG SYNC_DATE=04/21/2013

PAGE TITLE: VReg GPU VDDQ

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DRAWING NUMBER 051-9889 SIZE D

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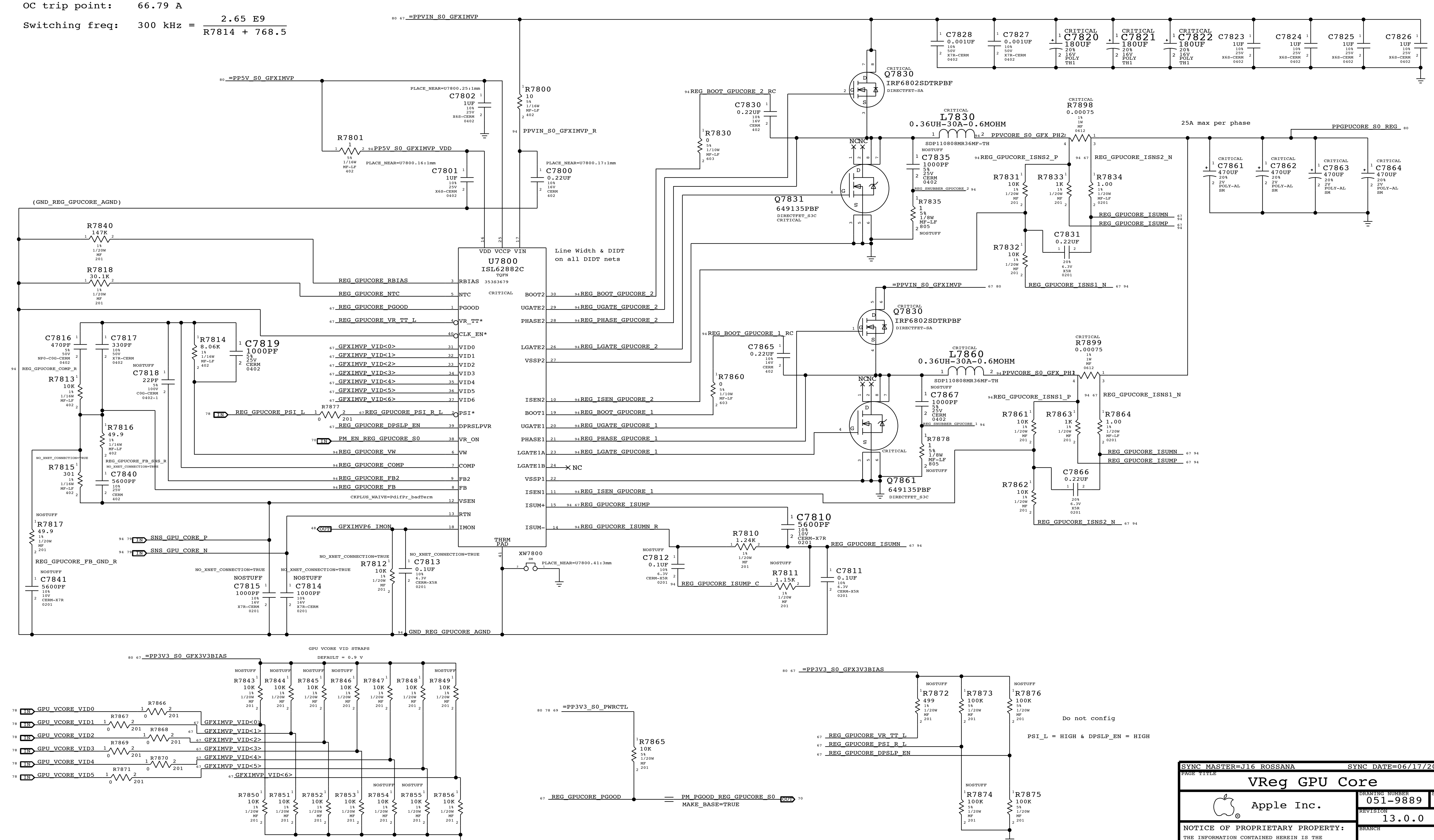
GPU Core S0 Regulator

OC trip point: 66.79 A

Switching freq: 300 kHz = $\frac{2.65\ E9}{R7814 + 768.5}$



SYNC_MASTER=J16_ROSSANA		SYNC_DATE=06/17/2013
PAGE TITLE: VReg GPU Core		
Apple Inc.	DRAWING NUMBER: 051-9889	SIZE: D
	REVISION: 13.0.0	
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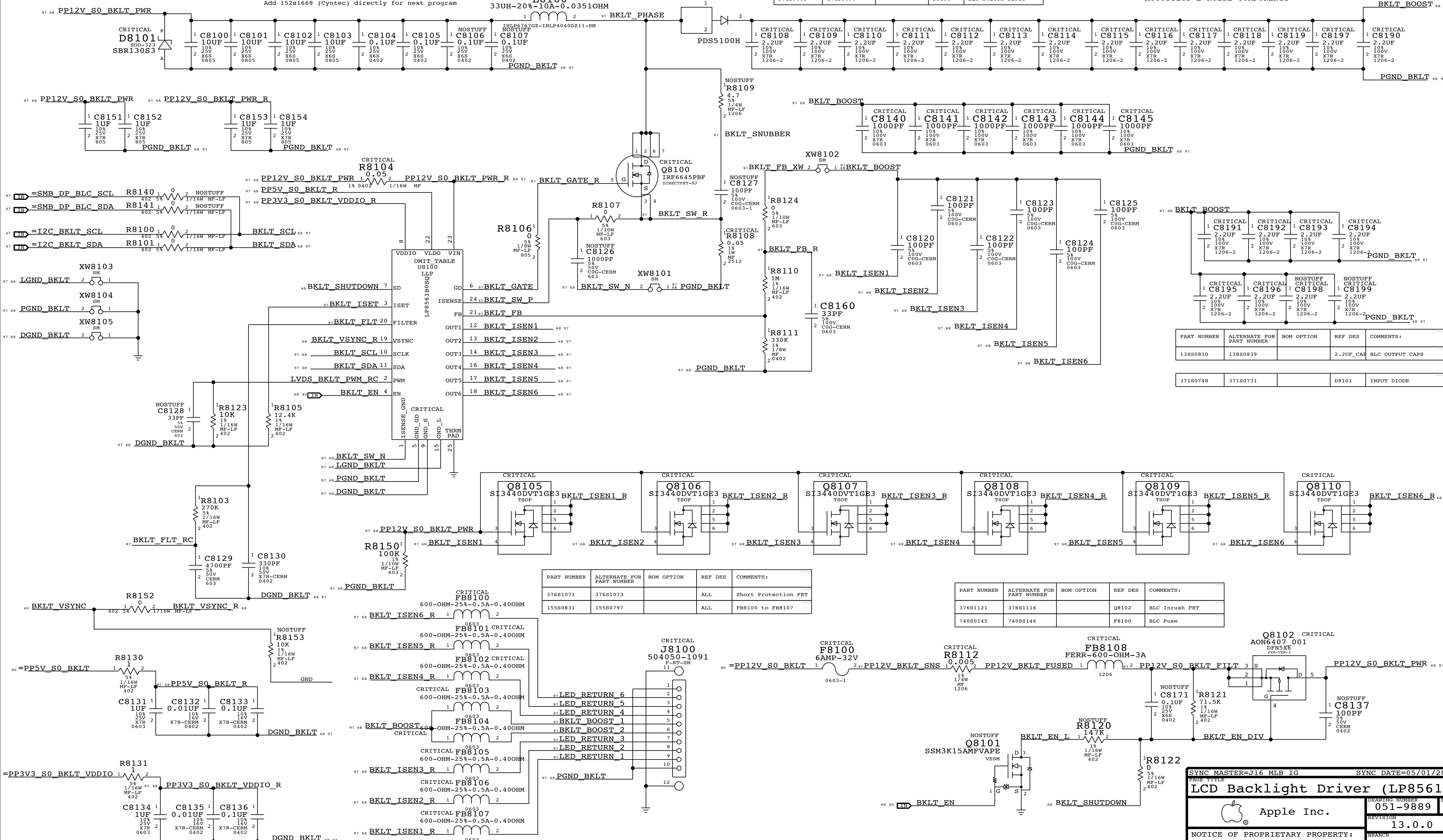
	SHEET: 67 OF 97	

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1668	1	IND,PWR,33UH,20%,10A,35MOHM	L8100	CRITICAL	

OMIT/table L8100 due to late sourcing change, did not want footprint change
Add 152s1668 (Cyntec) directly for next program

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S0648	371S0694		D8100	BLC Switch Diode

ACOUSTICS E-NOISE COMPONENTS



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0810	138S0839		2.2UF_CAP	BLC OUTPUT CAPS

371S0748	371S0731		D8101	INPUT DIODE

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1071	376S1073		ALL	Short Protection FET
155S0831	155S0797		ALL	FB8100 to FB8107

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1121	376S1116		Q8102	BLC Inrush FET
740S0145	740S0146		F8100	BLC Fuse

SYNC_MASTER=J16_MLB_IG SYNC_DATE=05/01/2013

PAGE TITLE

LCD Backlight Driver (LP8561)

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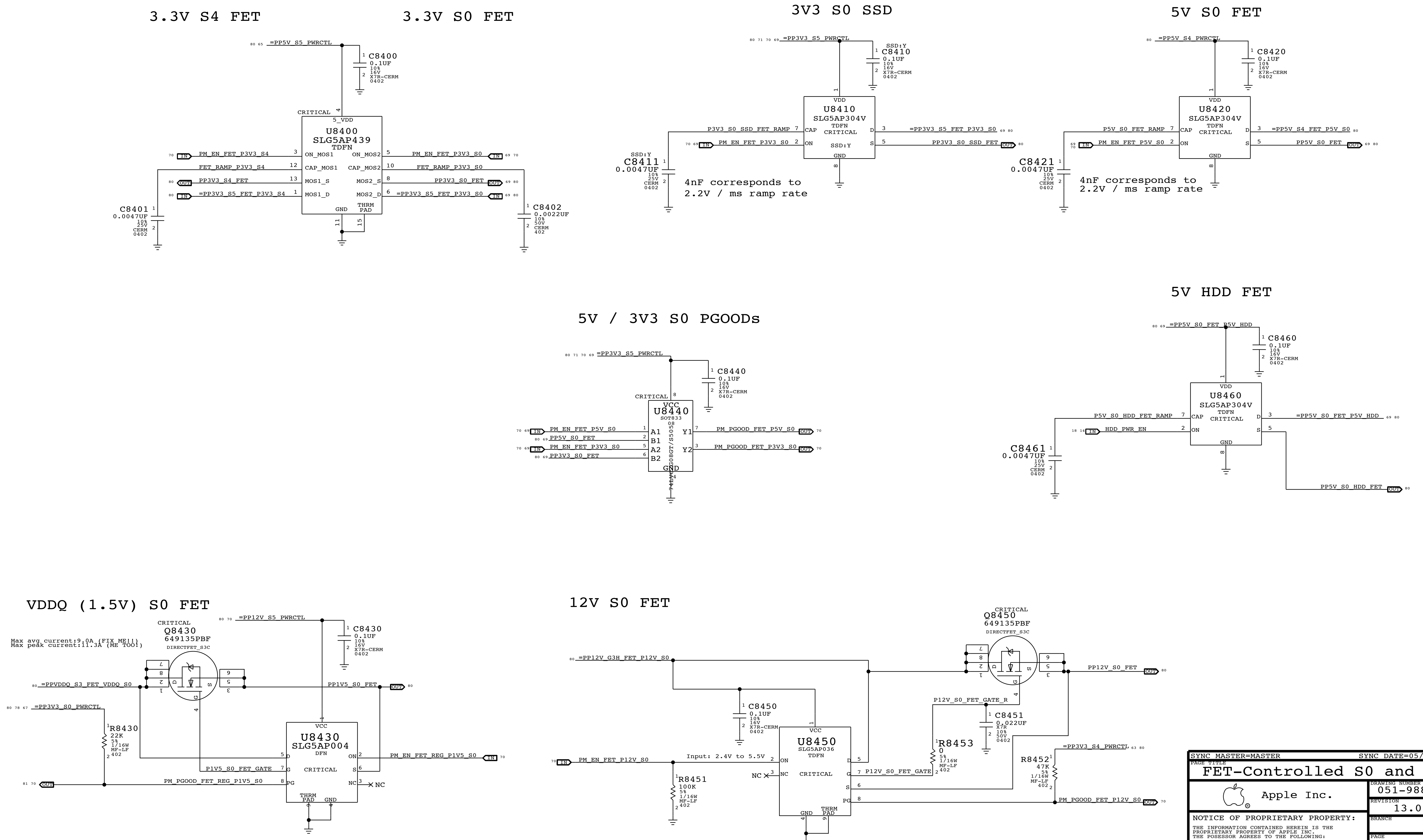

SYNC_MASTER=MASTER SYNC_DATE=05/01/2013

PAGE TITLE

FET-Controlled S0 and S4

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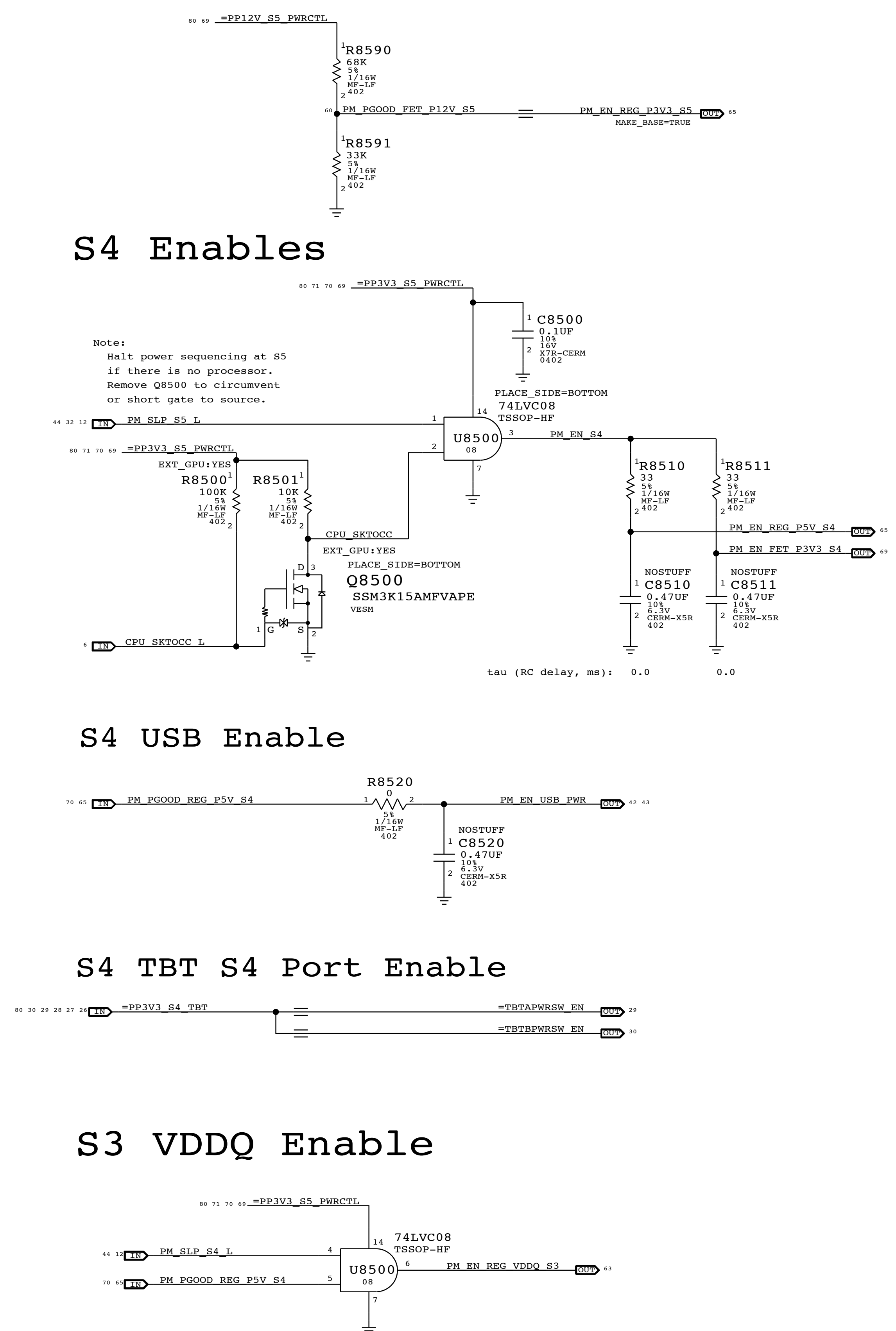
S5 Enable

S4 Enables

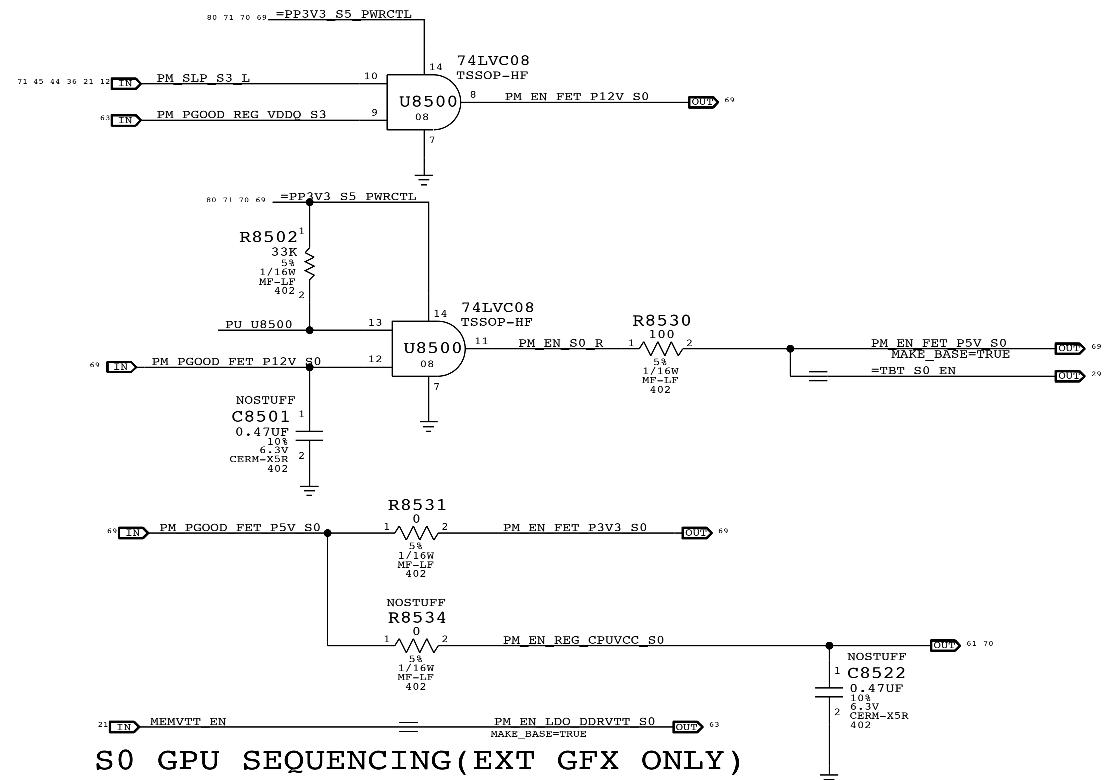
S4 USB Enable

S4 TBT S4 Port Enable

S3 VDDQ Enable

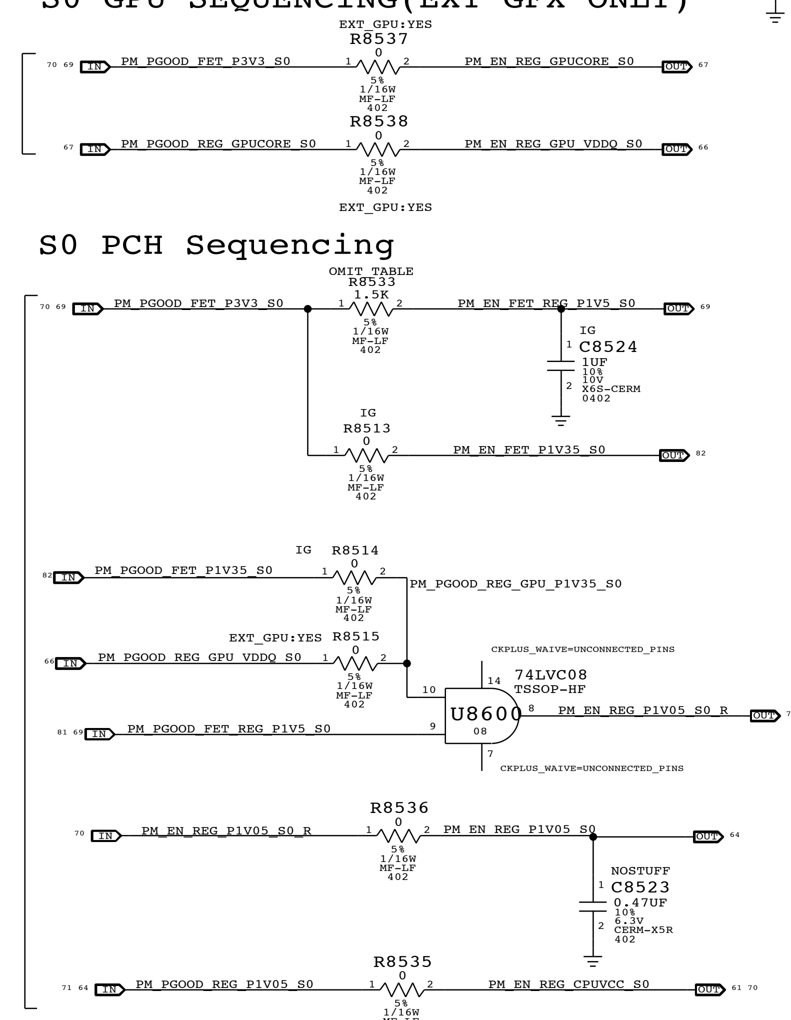


S0 Enables



S0 GPU SEQUENCING(EXT GFX ONLY)

S0 PCH Sequencing



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0070	1	RES,1.5K,0402,5%	R8533	IG
116S0004	1	RES,0OHM,0402,5%	R8533	EXT_GPU:YES

Rail definitions

Platform: All processor non-Core and non-Graphics (5 V, 3.3 V, 1.5 V, 1.05V for PCH/TBT/GPU)
Uncore: VDDQ

Notes on sequencing requirements

Intel:
1. No hard specification on platform rails
2. SMC guarantees timing on PCH DPWROK and PWROK
3. VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
4. VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
5. VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
6. VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

NVIDIA:
1. 3V3_S0 must ramp first
4. VDDQ MUST RAMP AFTER GPU_CORE
5. PEX_VDD with IFPC/D/E/F_IOVDD (1.05V) must ramp after VDDQ
6. All rails must reach their target voltages in more than 40 uS

SYNC_MASTER=MASTER SYNC_DATE=05/01/2013

PAGE TITLE: PM Regulator Enables

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REVISION: 13.0.0

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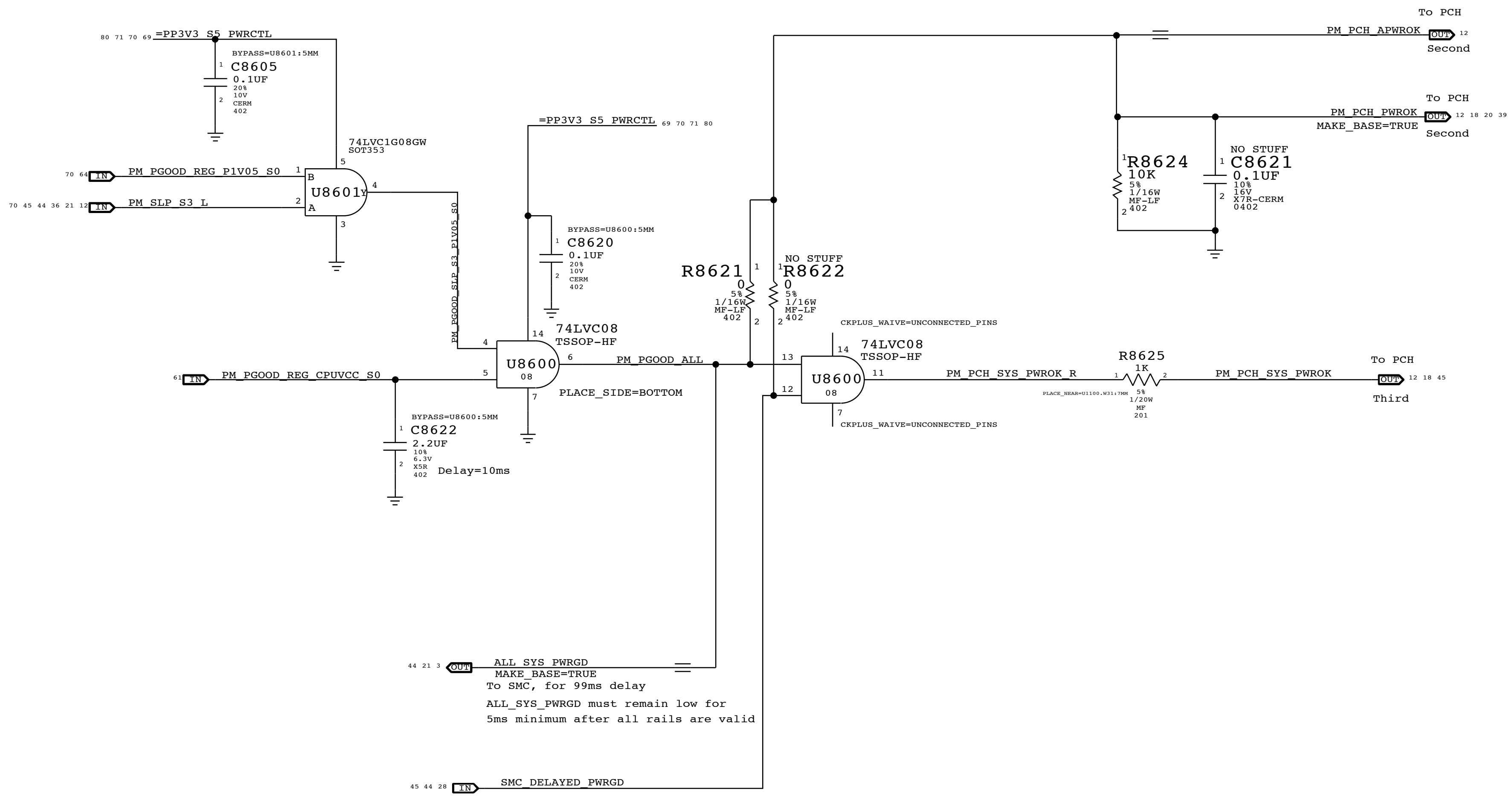
ALL_SYS_PWRGD,PCH_PWROK & SYS_PWROK Generation

PCH Power Goods



Resume Reset

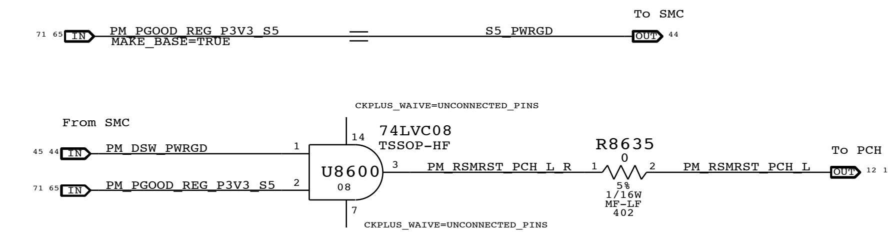
Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

Note:
The iMac J16/J17 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

Method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC_MASTER=J16_MLB_IG	SYNC_DATE=05/01/2013
PAGE TITLE: PM Power Good	
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	REVISION: 13.0.0
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	SHEET: 71 OF 97

Page Notes

Power aliases required by this page:

- =PP3V3_GPU_VDD33

Signal aliases required by this page:

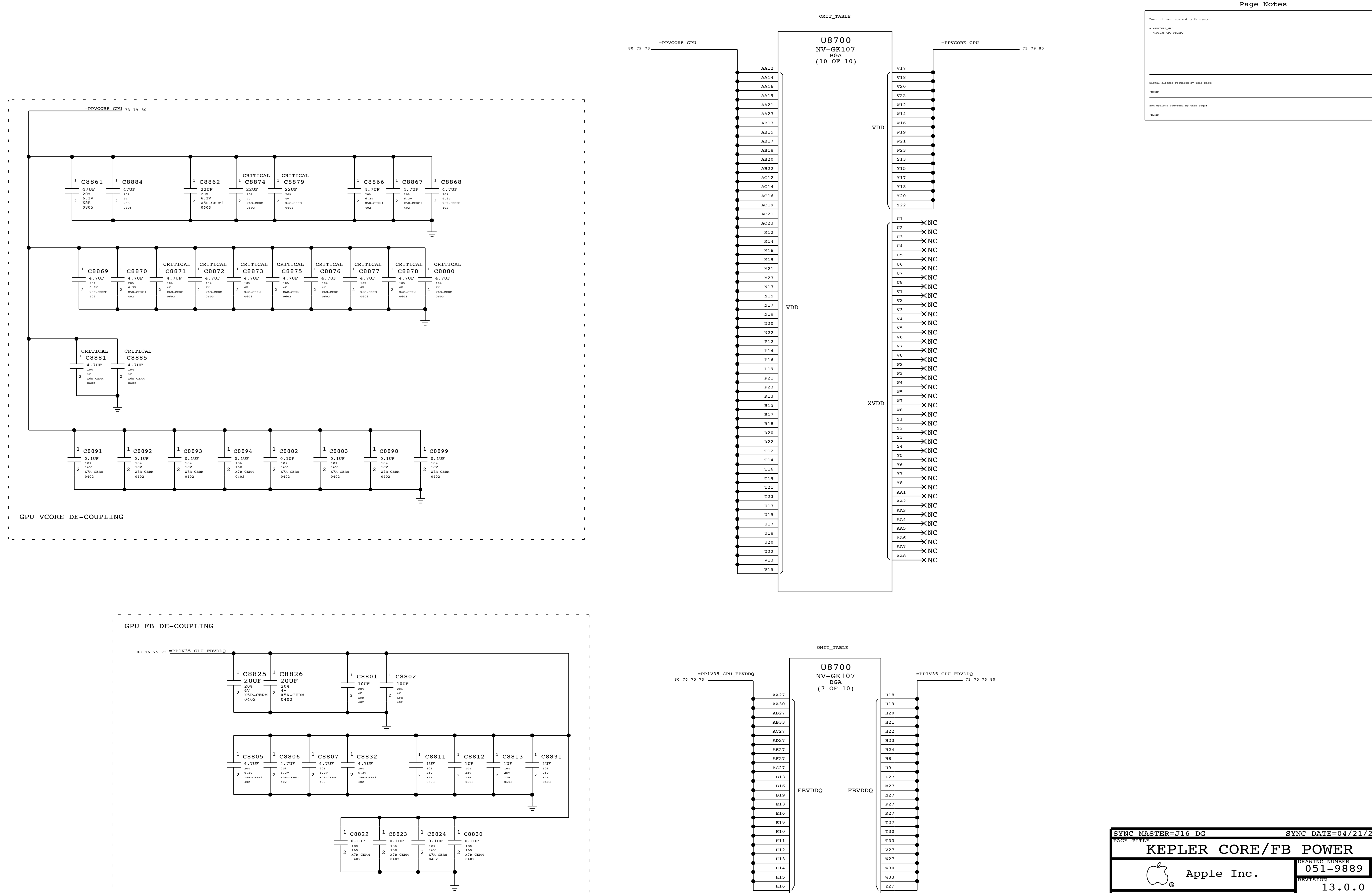
(NONE)

BOM options provided by this page:

(NONE)

POLARITY SWAPS INTENDED ON LANES,SEE NOTES AT DIFF PAIRS.
ALL LANES ARE ALSO REVERSED, SEE ALIASES ON CSA 102

SYNC MASTER=J16 DG	SYNC DATE=04/21/2013
PAGE TITLE	KEPLER PCI-E
Apple Inc.	DRAWING NUMBER 051-9889 · SIZE D
	REVISION 13.0.0
	BRANCH
	PAGE 87 OF 123
	SHEET 72 OF 97



SYNC_MASTER=J16_DG SYNC_DATE=04/21/2013

PAGE TITLE

KEPLER CORE/FB POWER

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DRAWING NUMBER	SIZE
051-9889	D

REVISION 13.0.0

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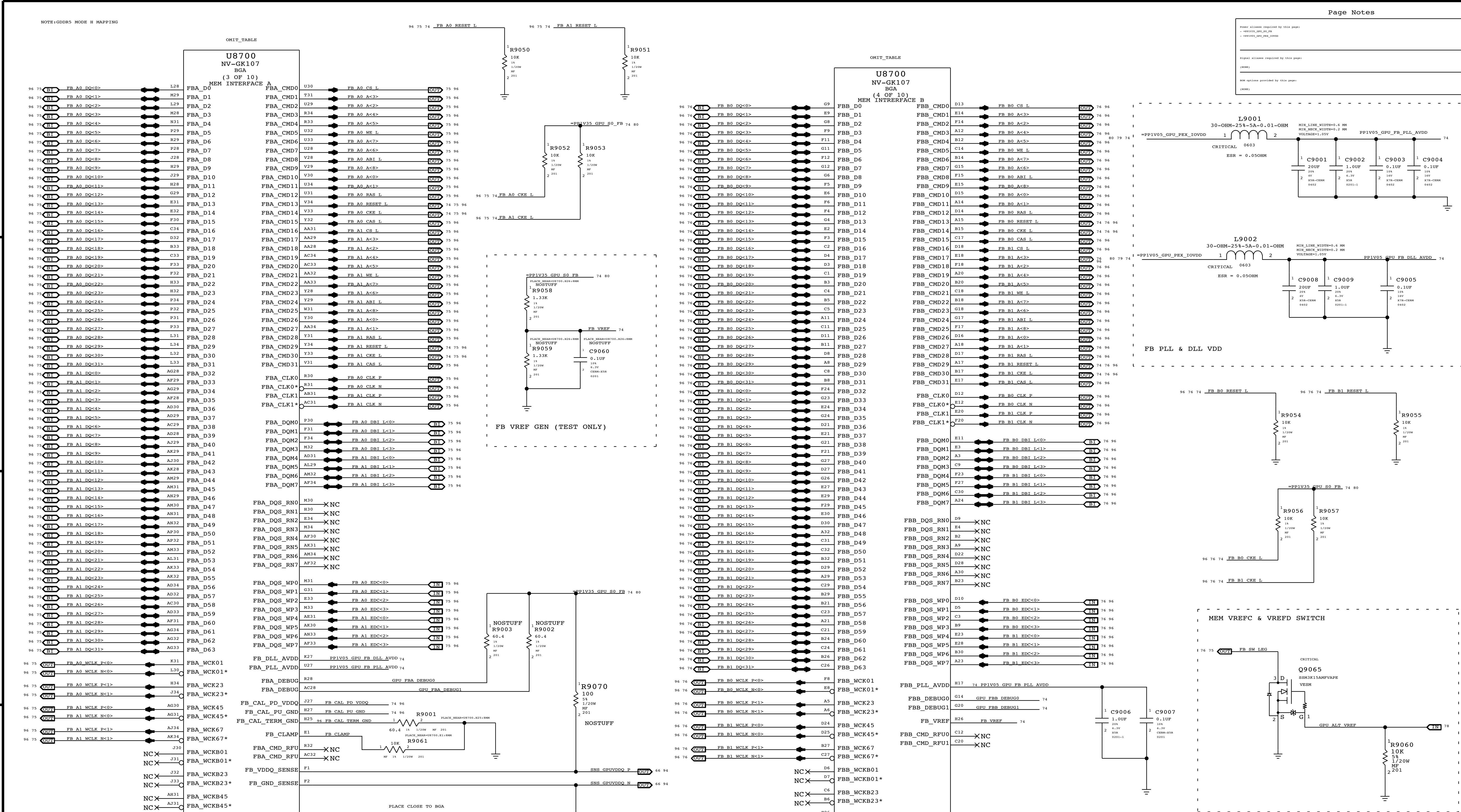

SYNC MASTER=J16 DG
SYNC DATE=04/21/2013

PAGE TITLE
KEPLER FRAME BUFFER I/F

DRAWING NUMBER 051-9889
SIZE D

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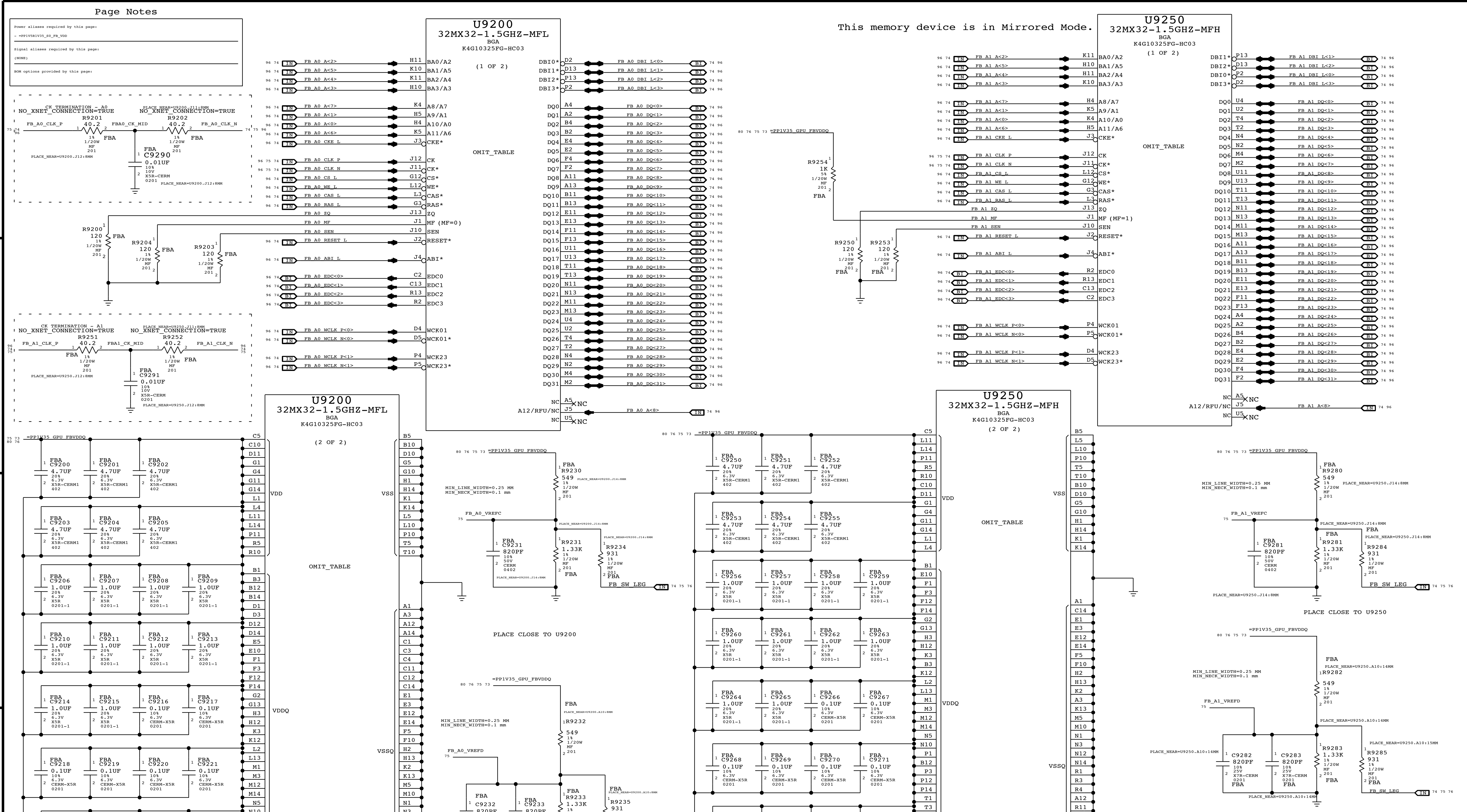

SYNC_MASTER=J16_DG SYNC_DATE=04/21/2013

PAGE TITLE

GDDR5 Frame Buffer A

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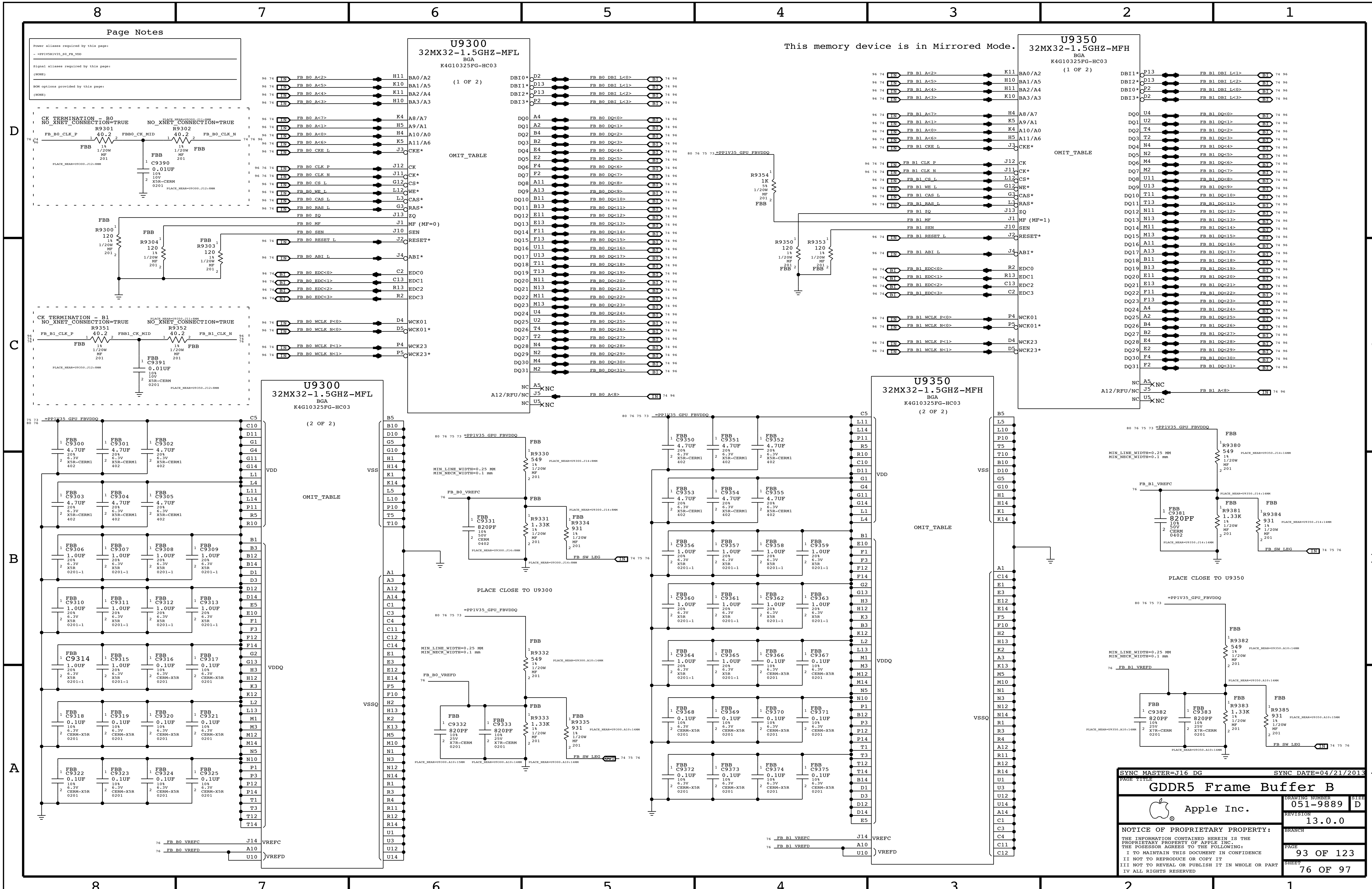

SYNC_MASTER=J16_DG SYNC_DATE=04/21/2013

PAGE TITLE

GDDR5 Frame Buffer B

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DRAWING NUMBER 051-9889 SIZE D

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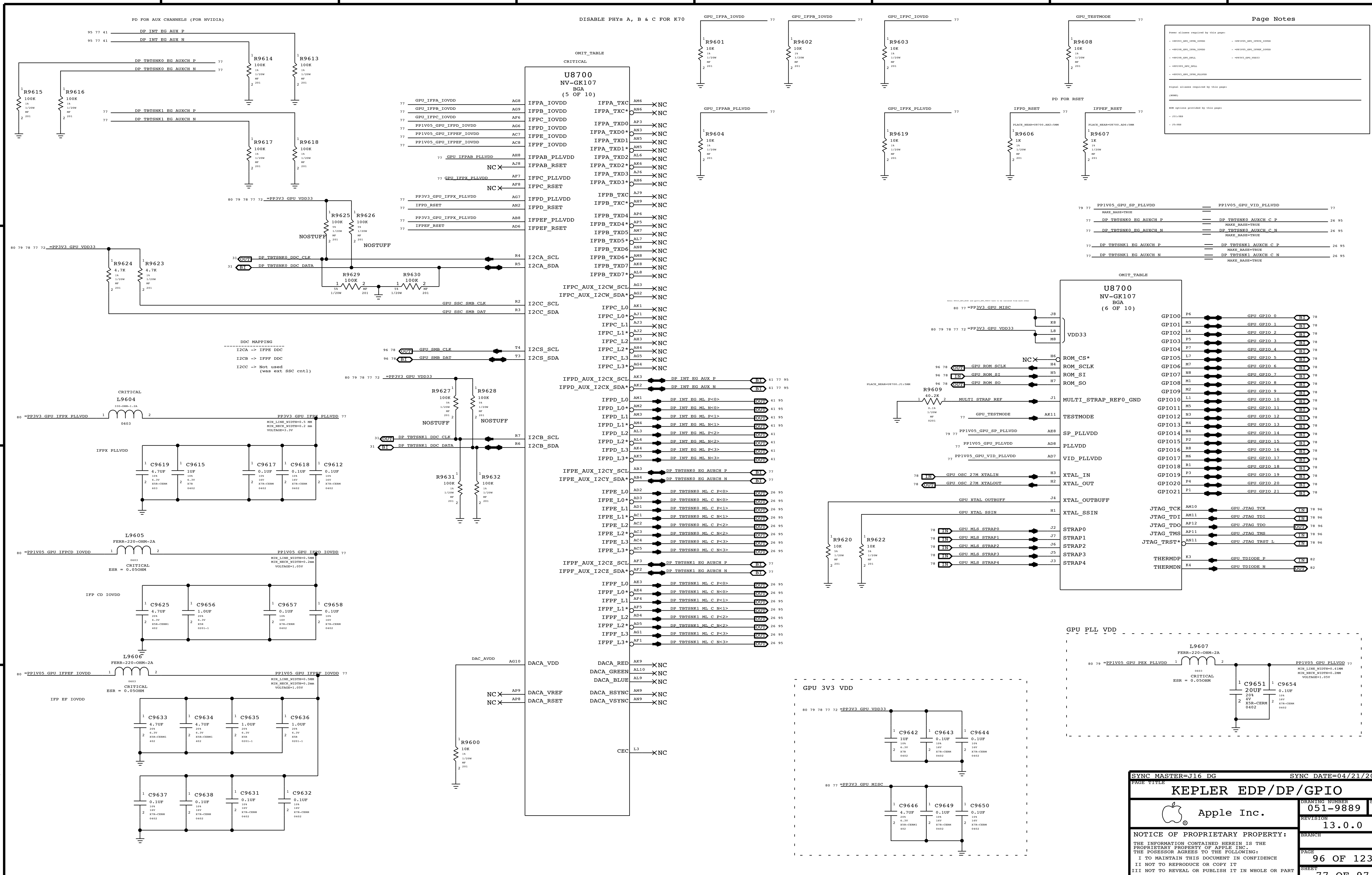

SYNC_MASTER=J16_DG SYNC_DATE=04/21/2013

PAGE TITLE

KEPLER EDP/DP/GPIO

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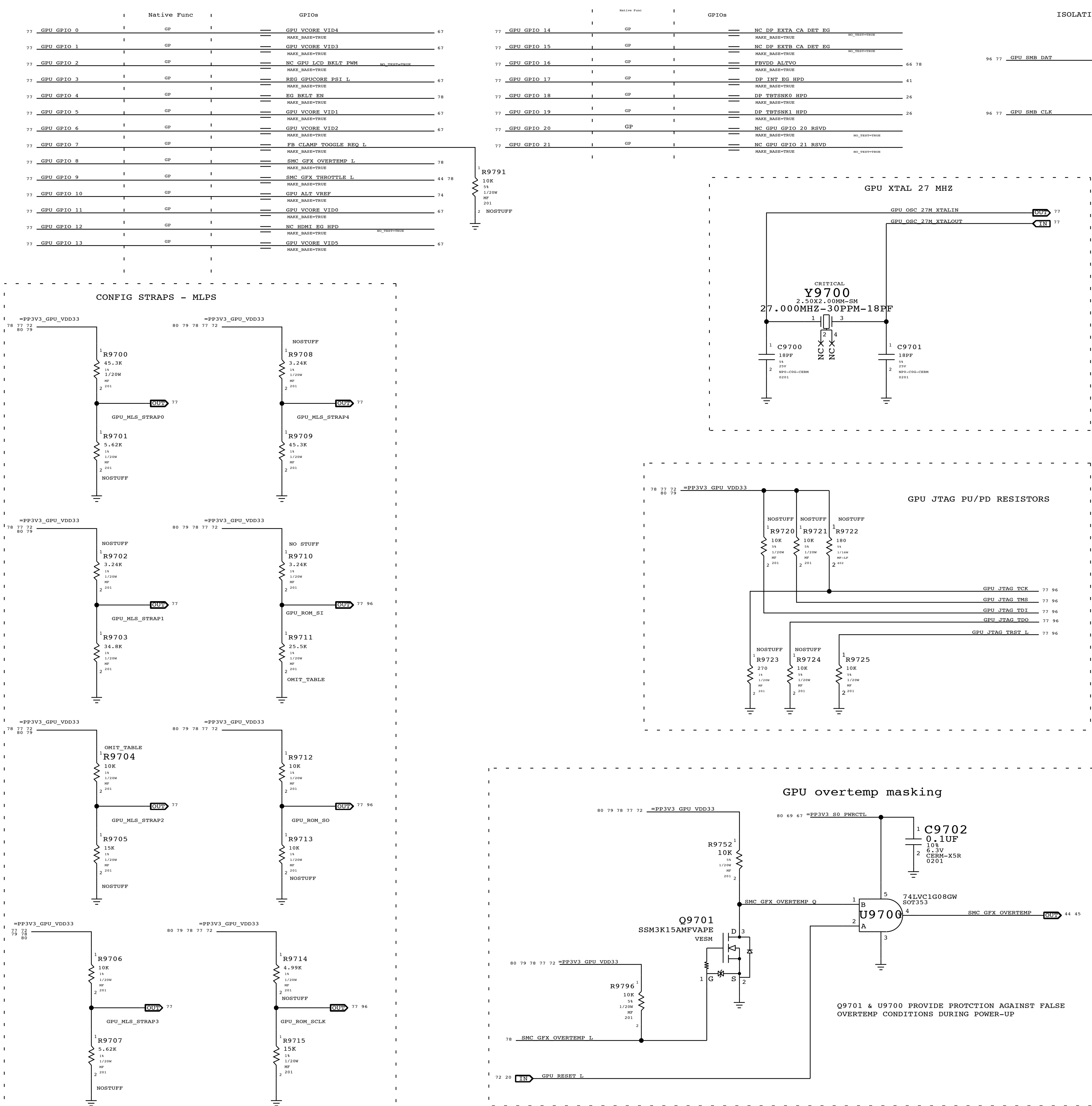
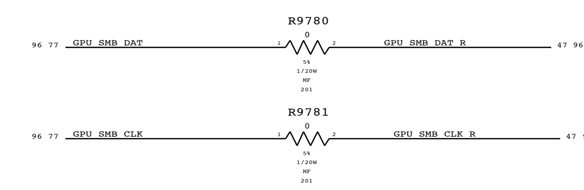
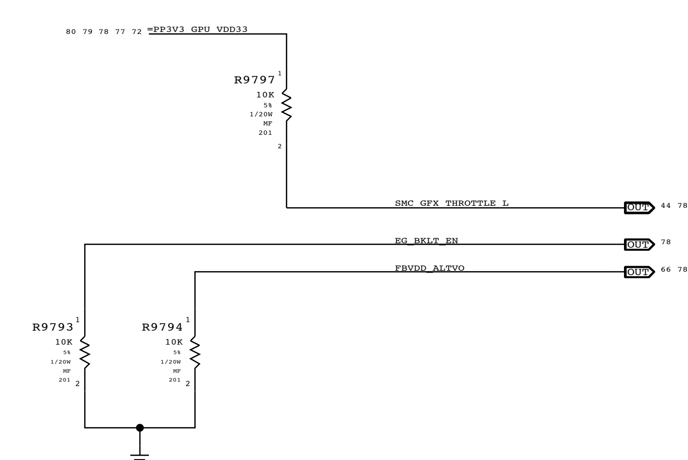




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0409	1	RES,MF,4.99k ,1,1/20W,0201	R9704		GPU:107GSXA
118S0409	1	RES,MF,4.99k ,1,1/20W,0201	R9704		GPU:107GSXB
118S0013	1	RES,MF,10.0K ,1,1/20W,0201	R9704		GPU:107GX
118S0230	1	RES,MF,24.9k ,1,1/20W,0201	R9711		FB:CH1_HYNIX
118S0230	1	RES,MF,24.9k ,1,1/20W,0201	R9711		FB:CH2_HYNIX
118S0230	1	RES,MF,24.9K ,1,1/20W,0201	R9711		FB:BOTH_HYNIX
118S0280	1	RES,MF,30.1k ,1,1/20W,0201	R9711		FB:CH1_ELPIDA
118S0280	1	RES,MF,30.1k ,1,1/20W,0201	R9711		FB:CH2_ELPIDA
118S0280	1	RES,MF,30.1k ,1,1/20W,0201	R9711		FB:BOTH_ELPIDA



SYNC MASTER=J16 DG SYNC DATE=04/21/2013

PAGE TITLE

KEPLER GPIOS,CLK & STRAPS

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REVISION 13.0.0

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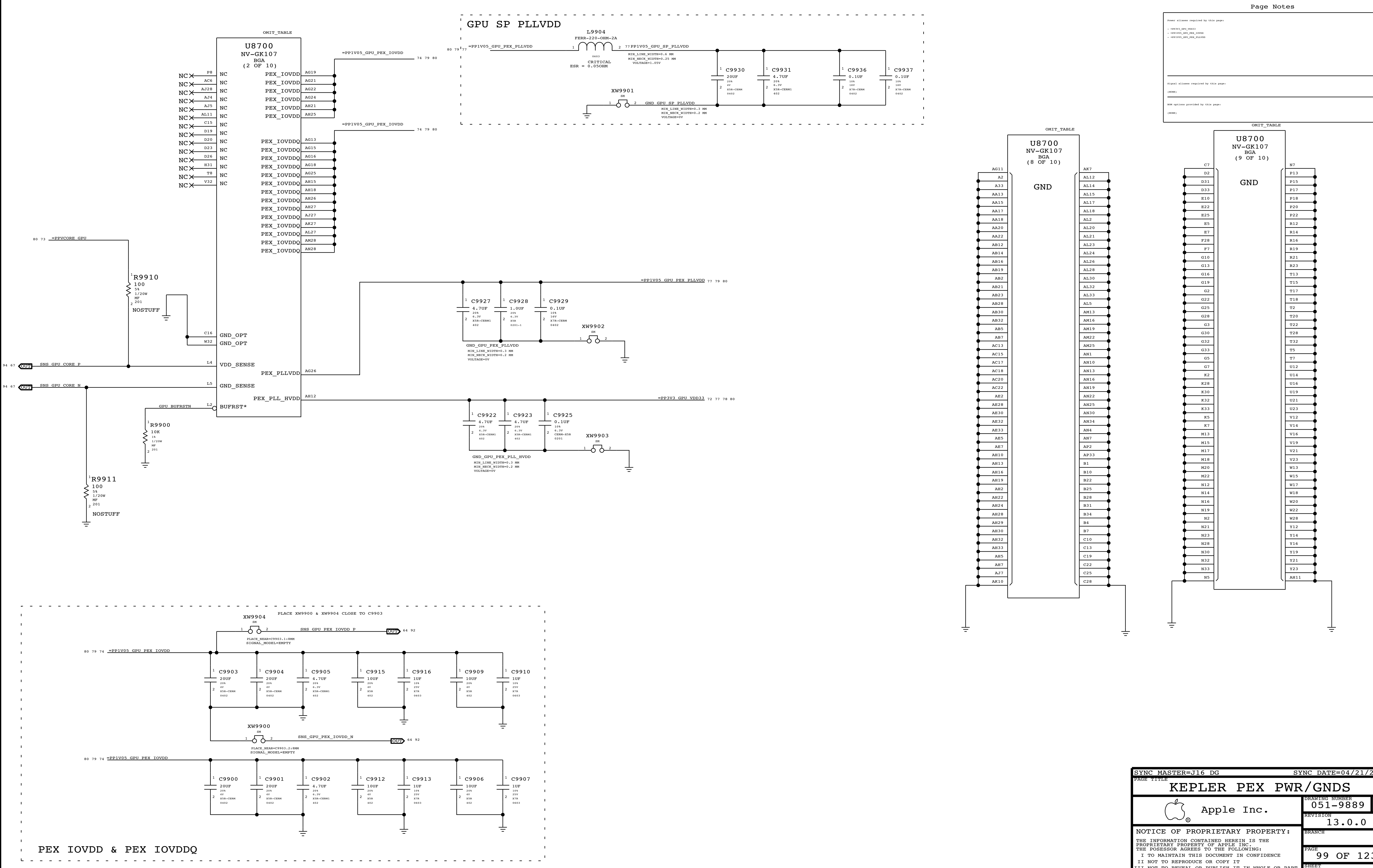

SYNC_MASTER=J16_DG SYNC_DATE=04/21/2013

PAGE TITLE

KEPLER PEX PWR/GNDS

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G3 Rails
Always on: Keeps the PCH RTC alive

94 PP3V3_G3 MAKE_BASE=TRUE
- PP3V3_G3_RTC 19
- =PP3V3_G3_PCH_RTC 15
- =PP3V3_G3_PCH 11 12

G3H Rails

94 PP12V_ACDC MAKE_BASE=TRUE
- PP12V_G3H_ACDC 60
- =PP12V_G3H_SNS_R 48

94 PP12V_G3H MAKE_BASE=TRUE
- PP12V_G3H_SNS 48
- =PP12V_G3H_REG_3V42_G3H 60
- =PP12V_G3H_FET_P12V_S5 60
- =PP12V_G3H_FET_P12V_S0 69

94 PP3V42_G3H MAKE_BASE=TRUE
- PP3V42_G3H_REG 60
- =PP3V3_G3H_BT 32
- =PP3V3_G3H_SMC 44 45 51
- =PP3V3_G3H_RTC_D 19
- =PPVIN_G3H_SMCVREF 45
- =PP3V3_G3H_SMC_USBMUX 42
- =PP3V3_G3H_LPCPLUS 46

Ground/Common

GND MAKE_BASE=TRUE

S5 Rails
Enabled when system has AC and is in S5

94 PP12V_S5 MAKE_BASE=TRUE
- PP12V_S5_FET 60
- =PP12V_S5_REG_P3V3P5V_S5 65
- =PP12V_S5_REG_VDDQ_S3 63
- =PP12V_S5_PWRCTL 69 70
- =PPHV_SW_TBTAPWRSW 29
- =PPHV_SW_TBTBPWRSW 30
- =PP12V_S5_SNS 48

94 PP5V_S5 MAKE_BASE=TRUE
- PP5V_S5_LDO 65
- =PP5V_S5_PWRCTL 65 69

94 PP3V3_S5 MAKE_BASE=TRUE
- PP3V3_S5_REG 65
- =PP3V3_S5_FET_P3V3_S4 69
- =PP3V3_S5_FET_P3V3_S0 69
- =PP3V3_S5_PWRCTL 69 70 71
- =PP3V3_S5_VRD 65
- =PP3V3_S5_LED 3
- =PP3V3_S5_PCH_VCCSUS_GPIO 11 12 13 14 15 17
- =PP3V3_S5_PCH_VCCDSW 15 17
- =PP3V3_S5_PCH_VCC_SPI 15 17
- =PP3V3_S5_PCH_VCCSUS_HDA 15 17
- =PP3V3_S5_PCH_VCCSUS_USB 15 17
- =PP3V3_S5_PCH_VCCSUS_RTC 15 17
- =PP3V3_S5_PCH_VCCSUS_USB3 15
- =PP3V3_S5_PCH 19 20
- =PP3V3_S5_ROM 46
- =PP3V3_S5_SENSE 48
- =PP3V3_S5_SMC 45
- =PP3V3_S5_XDP 18
- =PP3V3_S4_TBTAPWRSW 29
- =PP3V3_S4_TBTBPWRSW 30
- =PP3V3_S5_SNS 49

S4 Rails
Enabled when system has AC and is in run or sleep

94 PP5V_S4 MAKE_BASE=TRUE
- PP5V_S4_REG 65
- =PP5V_S4_REG_VDDQ_S3 63
- =PP5V_S4_FET_P5V_S0 69
- =PP5V_S4_MEMRESET 21
- =PP5V_S4_PWRCTL 69
- =PP5V_S4_USB 42 43
- =PP5V_S4_CAMERA 38

94 PP3V3_S4 MAKE_BASE=TRUE
- PP3V3_S4_FET 69
- =PP3V3_S4_FET_ENET 36
- =PP3V3_S4_PWRCTL 63 69
- =PP3V3_S4_ALS 38
- =PP3V3_S4_LED 3
- =PP3V3_S4_MEMRESET 21
- =PP3V3_S4_PCH 13
- =PP3V3_S4_PM 21
- =PP3V3_S4_SDCARD 37
- =PP3V3_S4_SMC 45
- =PP3V3_S4_SMBUS_SMC_2 47
- =PP3V3_S4_TBT 26 27 28 29 30 70
- =PP3V3_S4_CAMERA 38 39
- =PP3V3_S4_AUDIO_DIG 56
- =PP3V3_S4_AP 32 45
- =PP3V3_S3_VREFMRGN 22

S3 Rails
Enabled when system is in run or sleep

94 PP3V3_ENET MAKE_BASE=TRUE
- PP3V3_ENET_FET 36
- =PP3V3_ENET_PHY 35 36
- =PP3V3_ENET_SYSCLK 19
- =PPVDDIO_ENET_CLK 19

92 PPVDDQ_S3 MAKE_BASE=TRUE
- PPVDDQ_S3_REG 63
- =PPVDDQ_S3_FET_VDDQ_S0 69
- =PPVDDQ_S3_SNS_DDR_R 49

92 PPVDDQ_S3_DDR MAKE_BASE=TRUE
- PPVDDQ_S3_SNS_DDR 49
- =PPVDDQ_S3_LDO_DDRVTT 63
- =PPDDR_S3_MEMVREF 22
- =PPVDDQ_S3_MEM_A 23
- =PPVDDQ_S3_MEM_B 24
- =PPVDDQ_S3_MEMRESET 21

S0 Rails
Enabled when system is in run

94 PP12V_S0 MAKE_BASE=TRUE
- PP12V_S0_FET 69
- =PP12V_S0_REG_CPUVCC_S0 62
- =PP12V_S0_REG_P1V05_S0 64
- =PP12V_S0_SNS_GPUCORE_R 48
- =PP12V_S0_SNS_FBVDDQ_R 48
- =PP12V_S0_AUDIO_SPKRAMP 54 55
- =PP12V_S0_BKLT 68
- =PP12V_S0_FAN 51
- =PP12V_S0_LCD 40

94 PP12V_S0_GPUCORE MAKE_BASE=TRUE
- PP12V_S0_SNS_GPUCORE 48
- =PPVIN_S0_GFXIMVP 67

94 PP12V_S0_FBVDDQ MAKE_BASE=TRUE
- PP12V_S0_SNS_FBVDDQ 48
- =PP12V_S0_REG_GPU_VDDQ_S0 66

94 PP5V_S0_HDD MAKE_BASE=TRUE
- PPHDD_S0_SNS 48
- =PP5V_S0_SATA 34

94 PPHDD_S0 MAKE_BASE=TRUE
- PP5V_S0_HDD_FET 69
- =PPHDD_S0_SNS_R 48

94 PP5V_S0 MAKE_BASE=TRUE
- PP5V_S0_FET 69
- =PP5V_S0_REG_CPUVCC_S0 61 62
- =PP5V_S0_REG_P1V05_S0 64
- =PP5V_S0_REG_GPU_VDDQ_S0 66
- =PP5V_S0_AUDIO 52 59
- =PP5V_S0_ISENSE 48
- =PP5V_S0_LPCPLUS 46
- =PP5V_S0_BKLT 68
- =PP5V_S0_GFXIMVP 67
- =PP5V_S0_FET_P5V_HDD 69

94 PP3V3_S0 MAKE_BASE=TRUE
- PP3V3_S0_FET 69
- =PP3V3_S0_P3V3TBTFET 28
- =PP3V3_S0_PWRCTL 67 69 78 80
- =PP3V3_S0_VRD 60 61 64 66
- =PP3V3_S0_AUDIO 38 52 54 55 58
- =PP3V3_S0_AUDIO_DIG 56
- =PP3V3_S0_DP 31 40 47
- =PP3V3_S0_FAN 51
- =PP3V3_S0_INTDPMUX 41
- =PP3V3_S0_LED 3
- =PP3V3_S0_LED_SATA 33
- =PP3V3_S0_PCH 19
- =PP3V3_S0_ENET 35
- =PP3V3_S0_PCH_VCC3_3_GPIO 11 12 13 14 15 17
- =PP3V3_S0_PCH_VCC3_3_THRM 15 17
- =PP3V3_S0_PCH_VCCCLK3_3 15 17
- =PP3V3_S0_PCH_VCC3_3_HVCMOS 15 17
- =PP3V3_S0_PCH_VCC_FUSE 15 17
- =PP3V3_S0_PCH_VCC3_3_USB 15 17
- =PP3V3_S0_RSTBUF 20
- =PP3V3_S0_SDCARD 37
- =PP3V3_S0_SENSE 34 48 49 50
- =PP3V3_S0_SMBUS 47
- =PP3V3_S0_SMBUS_SMC_0 47
- =PP3V3_S0_SMBUS_SMC_1 47
- =PP3V3_S0_SMBUS_SMC_3 47
- =PP3V3_S0_SMC 45
- =PP3V3_S0_TBTPWRCTL 28
- =PP3V3_S0_BKLT_VDDIO 68
- =PP3V3_GPU_IFPX_PLLVDD 77
- =PP3V3_GPU_MISC 77
- =PP3V3_GPU_VDD33 72 77 78 79
- =PPSPD_S0_MEM_A 23
- =PPSPD_S0_MEM_B 24
- =PP3V3_S0_GFX3V3BIAS 67
- =PP3V3_S0_PWRCTL 67 69 78 80
- =PP3V3_S0_PCH_GPIO 14 28

94 PP3V3_S0_SSD MAKE_BASE=TRUE
- PP3V3_S0_SSD_FET 69
- =PPSSD_S0_SNS_R 49

94 PPSSD_S0 MAKE_BASE=TRUE
- PPSSD_S0_SNS 49
- =PP3V3_S0_SSD 33

92 PP1V5_S0 MAKE_BASE=TRUE
- PP1V5_S0_FET 69
- =PP1V5_S0_SNS_R 49

92 PP1V5_S0_PCH MAKE_BASE=TRUE
- PP1V5_S0_SNS 49
- =PP1V5_S0_CPU_MEM 6 8 10
- =PP1V5_S0_AUD_DIG 52
- =PP1V5_S0_SENSE 34
- =PP1V5_S0_PCH_VCCVRM 17
- =PP1V5_S0_PCH_RCOMP 12 13
- =PP1V5_S0_PCH_SATA 11
- =PP1V5_S0_PCH_CLK 11 19

92 PPDDRVTT_S0 MAKE_BASE=TRUE
- PPDDRVTT_S0_LDO 63
- =PPDDRVTT_S0_CLAMP 21
- =PPDDRVTT_S0_MEM_A 23
- =PPDDRVTT_S0_MEM_B 24

92 PP1V05_S0 MAKE_BASE=TRUE
- PP1V05_S0_REG 64
- =PP1V05_S0_P1V05TBTFET 28
- =PP1V05_S0_PCH_VCC 15 17
- =PP1V05_S0_PCH_VCCCLK_CLK100 15 17
- =PP1V05_S0_PCH_VCCASW 15 17
- =PP1V05_S0_PCH_VCC_CLK 17
- =PP1V05_S0_PCH_VCCIO_FDI 15 17
- =PP1V05_S0_PCH_VCCCLK_SSC 15 17
- =PP1V05_S0_PCH_VCCUSBPLL 15 17
- =PP1V05_S0_PCH_VCCIO 15 17
- =PP1V05_S0_PCH_VCCCLK_SSC100 15 17
- =PP1V05_S0_PCH_VCCCLK_CLK135 15 17
- =PP1V05_S0_PCH_VCCIO_USB2 15 17
- =PP1V05_S0_PCH_V_PROC_IO 14 15 17
- =PP1V05_S0_SNS 48
- =PP1V05_S0_SMC 45
- =PP1V05_S0_XDP 18
- =PP1V05_GPU_IFPCD_IOVDD 77
- =PP1V05_GPU_IFPEF_IOVDD 77
- =PP1V05_GPU_PEX_IOVDD 74 79
- =PP1V05_GPU_PEX_PLLVDD 77 79

93 PPCPUVCC_S0_CPU MAKE_BASE=TRUE
- PPCPUVCC_S0_REG 62
- =PPCPUVCC_S0_CPU 6 8 10 48 61

Thunderbolt Rails (S0)
Enabled when Thunderbolt cable is plugged in

94 PP3V3_TBTLC MAKE_BASE=TRUE
- =PP3V3_TBTLC_FET 28
- =PP3V3_TBT_PCH_GPIO 20
- =PP3V3_TBTLC_RTR 26 27 28
- =PPVDDIO_TBT_CLK 19

92 28 PP1V05_TBTLC MAKE_BASE=TRUE
- =PP1V05_TBTLC_FET 28
- =PP1V05_TBTLC_RTR 27

92 PP1V05_TBTCIO MAKE_BASE=TRUE
- =PP1V05_TBTCIO_FET 28
- =PP1V05_TBTCIO_RTR 27

GPU Rails (S0)
Enabled when system is in run

94 PPVCORE_S0_GPU MAKE_BASE=TRUE
- PPGPUCORE_S0_REG 67
- =PPVCORE_GPU 73 79
- =PPGPUCORE_S0_SNS 48

94 PPVDDQ_S0_GPU MAKE_BASE=TRUE
- PP1V5R1V35_S0_GPU_REG 66
- =PP1V35_GPU_FBVDDQ 73 75 76
- =PP1V35_GPU_S0_FB 74

SYNC_MASTER=MASTER	SYNC_DATE=04/21/2013
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PEG aliases

86 72 PEG_D2R_N<0..15> MAKE_BASE=TRUE = =PEG_D2R_N<15..0> 5

86 72 PEG_D2R_P<0..15> MAKE_BASE=TRUE = =PEG_D2R_P<15..0> 5

86 72 PEG_R2D_C_N<0..15> MAKE_BASE=TRUE = =PEG_R2D_C_N<15..0> 5

86 72 PEG_R2D_C_P<0..15> MAKE_BASE=TRUE = =PEG_R2D_C_P<15..0> 5

61 REG_ISENVCC_4_NR = AGND_CPU MAKE_BASE=TRUE 61 62 93

Memory PGOOD

21 =PM_PGOOD_MEM_S0 = PM_PGOOD_FET_REG_P1V5_S0 MAKE_BASE=TRUE 69 70

SYNC_MASTER=MASTER SYNC_DATE=04/21/2013

PAGE TITLE

Signal Aliases

Apple Inc.

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CPU Reserved

Page refs	Signal	Alias	Properties
88 18 6	CPU_CFG<15..12>	TP_CPU_CFG<15..12>	MAKE_BASE=TRUE

CPU Memory

Page refs	Signal	Alias	Properties
7	TP_MEM_A_DQS_N<8>	NC_MEM_A_DQSN<8>	MAKE_BASE=TRUE NO_TEST=TRUE
7	TP_MEM_A_DQS_P<8>	NC_MEM_A_DQSP<8>	MAKE_BASE=TRUE NO_TEST=TRUE
7	TP_MEM_B_DQS_N<8>	NC_MEM_B_DQSN<8>	MAKE_BASE=TRUE NO_TEST=TRUE
7	TP_MEM_B_DQS_P<8>	NC_MEM_B_DQSP<8>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_A_CLK_N<2..3>	NC_MEM_A_CLKN<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_A_CLK_P<2..3>	NC_MEM_A_CLKP<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_A_CS_L<2..3>	NC_MEM_A_CS_L<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_A_CKE<2..3>	NC_MEM_A_CKE<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_B_CLK_N<2..3>	NC_MEM_B_CLKN<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_B_CLK_P<2..3>	NC_MEM_B_CLKP<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_B_CS_L<2..3>	NC_MEM_B_CS_L<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_B_CKE<2..3>	NC_MEM_B_CKE<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_A_ODT<2..3>	NC_MEM_A_ODT<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE
85 7	MEM_B_ODT<2..3>	NC_MEM_B_ODT<2..3>	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Clocks

Page refs	Signal	Alias	Properties	Page refs
	TP_ITPXDP_CLK100MP	ITPXDP_CLK100M_P	MAKE_BASE=TRUE NO_TEST=TRUE	11 88
	TP_ITPXDP_CLK100MN	ITPXDP_CLK100M_N	MAKE_BASE=TRUE NO_TEST=TRUE	11 88
11	TP_PCIE_CLK100M_PE4N	NC_PCIE_CLK100M_PE4N	MAKE_BASE=TRUE NO_TEST=TRUE	
11	TP_PCIE_CLK100M_PE4P	NC_PCIE_CLK100M_PE4P	MAKE_BASE=TRUE NO_TEST=TRUE	
11	TP_PCIE_CLK100M_PE5N	NC_PCIE_CLK100M_PE5N	MAKE_BASE=TRUE NO_TEST=TRUE	
11	TP_PCIE_CLK100M_PE5P	NC_PCIE_CLK100M_PE5P	MAKE_BASE=TRUE NO_TEST=TRUE	
11	TP_PCIE_CLK100M_PE7N	NC_PCIE_CLK100M_PE7N	MAKE_BASE=TRUE NO_TEST=TRUE	
11	TP_PCIE_CLK100M_PE7P	NC_PCIE_CLK100M_PE7P	MAKE_BASE=TRUE NO_TEST=TRUE	

PCH GPIO

Page refs	Signal	Alias	Properties
11	TP_PCH_GPIO64_CLKOUTFLEX0	NC_PCH_GPIO64_CLKOUTFLEX0	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCH_GPIO65_CLKOUTFLEX1	NC_PCH_GPIO65_CLKOUTFLEX1	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCH_GPIO66_CLKOUTFLEX2	NC_PCH_GPIO66_CLKOUTFLEX2	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCH_GPIO67_CLKOUTFLEX3	NC_PCH_GPIO67_CLKOUTFLEX3	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Unused Display

Page refs	Signal	Alias	Properties
5	TP_DP_IG_B_MLN<3..0>	NC_DP_IG_B_MLN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
5	TP_DP_IG_B_MLP<3..0>	NC_DP_IG_B_MLP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
5	TP_DP_IG_C_MLN<3..0>	NC_DP_IG_C_MLN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
5	TP_DP_IG_C_MLP<3..0>	NC_DP_IG_C_MLP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
5	TP_DP_IG_D_MLN<3..0>	NC_DP_IG_D_MLN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
5	TP_DP_IG_D_MLP<3..0>	NC_DP_IG_D_MLP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE

PCH PCI

Page refs	Signal	Alias	Properties
13	TP_LPC_DREQ0_L	NC_LPC_DREQ0_L	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Miscellaneous

Page refs	Signal	Alias	Properties
11	TP_HDA_SDIN1	NC_HDA_SDIN1	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_HDA_SDIN2	NC_HDA_SDIN2	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_HDA_SDIN3	NC_HDA_SDIN3	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCI_CLK33M_OUT2	NC_PCI_CLK33M_OUT2	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_PCI_CLK33M_OUT3	NC_PCI_CLK33M_OUT3	MAKE_BASE=TRUE NO_TEST=TRUE

UNUSED GRAPHICS ALIASES

Page refs	Signal	Alias	Properties
77	GPU_TDIODE_P	NC_GPU_TDIODE_P	MAKE_BASE=TRUE NO_TEST=TRUE
77	GPU_TDIODE_N	NC_GPU_TDIODE_N	MAKE_BASE=TRUE NO_TEST=TRUE

UNUSED THUNDERBOLT ALIASES

Page refs	Signal	Alias	Properties
26	TP_TBT_PCIE_RESET0_L	NC_TBT_PCIE_RESET0_L	MAKE_BASE=TRUE NO_TEST=TRUE
26	TP_TBT_PCIE_RESET1_L	NC_TBT_PCIE_RESET1_L	MAKE_BASE=TRUE NO_TEST=TRUE
26	TP_TBT_PCIE_RESET2_L	NC_TBT_PCIE_RESET2_L	MAKE_BASE=TRUE NO_TEST=TRUE
26	TP_TBT_PCIE_RESET3_L	NC_TBT_PCIE_RESET3_L	MAKE_BASE=TRUE NO_TEST=TRUE
26	TP_TBT_THERM_DP	NC_TBT_THERM_DP	MAKE_BASE=TRUE NO_TEST=TRUE

UNUSED VREG ALIASES

Page refs	Signal	Alias	Properties
61	REG_PWM_CPUVCC_4	NC_REG_PWM_CPUVCC_4	MAKE_BASE=TRUE NO_TEST=TRUE
61	REG_ISENVCC_4_P	NC_REG_ISENVCC_4P	MAKE_BASE=TRUE NO_TEST=TRUE

POWER SEQUENCING

Page refs	Signal	Alias	Properties
70	PM_EN_FET_P1V35_S0	NC_PM_EN_FET_P1V35_S0	MAKE_BASE=TRUE NO_TEST=TRUE
70	PM_PGOOD_FET_P1V35_S0	NC_PM_PGOOD_FET_P1V35_S0	MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=MASTER SYNC_DATE=04/21/2013

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Unused Signal Aliases

Apple Inc.

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SYNC MASTER=J16 DG
SYNC DATE=04/21/2013

PAGE TITLE
Functional / ICT Test

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J16 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,BOTTOM	NO_TYPE,BGA	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	=50_OHM_SE	12.7 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.215 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.215 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.170 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP,BOTTOM	Y	0.170 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP,BOTTOM	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.138 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP,BOTTOM	Y	0.138 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.110 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.180 MM	0.085 MM	=STANDARD	0.140 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.180 MM	0.085 MM	=STANDARD	0.140 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.135 MM	0.085 MM	=STANDARD	0.160 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	0.160 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.089 MM	0.085 MM	=STANDARD	0.220 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.089 MM	0.085 MM	=STANDARD	0.220 MM	0.1 MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	*	0.076 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

Board Stack-up

Finished board thickness: 1.58 mm

Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 mm
2	Plane	1 oz
	Prepreg	0.076 mm
3	Signal	0.5 oz
	Prepreg	0.435 mm
4	Plane	1 oz
	Core	0.127 mm
5	Plane	1 oz
	Prepreg	0.435 mm
6	Signal	0.5 oz
	Prepreg	0.076 mm
2	Plane	1 oz
	Prepreg	0.071 mm
Btm	Signal	0.5 oz (Cu plated)

SYNC MASTER=J16 DG SYNC DATE=04/21/2013

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J16 RULE DEFINITIONS

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D

DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
DDR_COMP	*	Y	0.305 MM	0.25 MM	=STANDARD	=STANDARD	=STANDARD

Minimum diff spacing is 4 mil
Table 4-5, Intel Doc# 486712

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D
DDR_COMP_PHY	*	DDR_COMP

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

C

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2.5:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6.5:1_SPACING	?
DDR_COMP_ISO	*	0.2 MM	?

Main Segment Min Spacing Rules (mils) (Shark Bay PDG, Intel Doc# 486712)

Table	Trace	Design (diff)	Iso	Design	Comments
4-2	4	(diff)	15	19.69	CLK trace spacing controlled by =68_OHM_DIFF
4-3	8	9.84	12	13.78	
4-4	6	7.87	12	13.78	
4-5	8.5	7.87	12	11.81	DQ or DQS to other signals not in the same bytelane (but not ch)
					DQ to DQ in the same bytelane of the same channel
			10	11.81	DQ to DQS in the same bytelane of the same channel
			12	11.81	DQ or DQS in different bytelanes of the same channel
			25	25.59	DQ or DQS in different channels
			-	25.59	DDR3 to any other signal not DDR3

B

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_COMP	*	*	DDR_COMP_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_A_DQS*	*	*	DDR_DATA_ISO	
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_B_DQS*	*	*	DDR_DATA_ISO	
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ	See Note (3)
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL	See Note (3)
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL	
DDR_A_*	DDR_B_*	*	DDR_CH2CH	See Note (2)

Note (1):
Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to contraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):
Intel suggests 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):
In order for the constraints DDR_*_DQ_BYTE* to =SAME to win out over DDR_{A,B}_DQ_BYTE* to DDR_{A,B}_DQ_BYTE* so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL.

A

DDR3

	Electrical Contraint Set	Physical	Spacing		
	Channel A				
I178	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<1..0>	7 23
I179	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<1..0>	7 23
I255	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<3..2>	7 82
I256	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<3..2>	7 82
I180	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<1..0>	7 23
I181	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<1..0>	7 23
I182	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<1..0>	7 23
I247	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<3..2>	7 82
I248	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<3..2>	7 82
I249	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<3..2>	7 82
I183	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_A<15..0>	7 23
I184	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_BA<2..0>	7 23
I185	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_RAS_L	7 23
I186	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_CAS_L	7 23
I187	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_WE_L	7 23
I188	DDR_A_DQ_BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0	MEM_A_DQ<7..0>	7 25
I189	DDR_A_DQ_BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1	MEM_A_DQ<15..8>	7 25
I191	DDR_A_DQ_BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2	MEM_A_DQ<23..16>	7 25
I190	DDR_A_DQ_BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3	MEM_A_DQ<31..24>	7 25
I192	DDR_A_DQ_BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4	MEM_A_DQ<39..32>	7 25
I193	DDR_A_DQ_BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5	MEM_A_DQ<47..40>	7 25
I194	DDR_A_DQ_BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6	MEM_A_DQ<55..48>	7 25
I195	DDR_A_DQ_BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7	MEM_A_DQ<63..56>	7 25
I196	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_P<0>	7 25
I197	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_N<0>	7 25
I198	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_P<1>	7 25
I199	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_N<1>	7 25
I200	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_P<2>	7 25
I201	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_N<2>	7 25
I203	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_P<3>	7 25
I202	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_N<3>	7 25
I204	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_P<4>	7 25
I205	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_N<4>	7 25
I206	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_P<5>	7 25
I207	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_N<5>	7 25
I208	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_P<6>	7 25
I209	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_N<6>	7 25
I210	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_P<7>	7 25
I211	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_N<7>	7 25
	Channel B				
I212	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<1..0>	7 24
I213	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<1..0>	7 24
I254	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<3..2>	7 82
I253	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<3..2>	7 82
I214	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<1..0>	7 24
I215	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<1..0>	7 24
I216	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<1..0>	7 24
I250	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<3..2>	7 82
I251	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<3..2>	7 82
I252	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<3..2>	7 82
I217	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_A<15..0>	7 24
I218	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_BA<2..0>	7 24
I219	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_RAS_L	7 24
I220	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_CAS_L	7 24
I221	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_WE_L	7 24
I222	DDR_B_DQ_BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0	MEM_B_DQ<7..0>	7 25
I223	DDR_B_DQ_BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1	MEM_B_DQ<15..8>	7 25
I225	DDR_B_DQ_BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2	MEM_B_DQ<23..16>	7 25
I224	DDR_B_DQ_BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3	MEM_B_DQ<31..24>	7 25
I226	DDR_B_DQ_BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4	MEM_B_DQ<39..32>	7 25
I227	DDR_B_DQ_BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5	MEM_B_DQ<47..40>	7 25
I228	DDR_B_DQ_BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6	MEM_B_DQ<55..48>	7 25
I229	DDR_B_DQ_BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7	MEM_B_DQ<63..56>	7 25
I230	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_P<0>	7 25
I231	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_N<0>	7 25
I232	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_P<1>	7 25
I233	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_N<1>	7 25
I234	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_P<2>	7 25
I235	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_N<2>	7 25
I237	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_P<3>	7 25
I236	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_N<3>	7 25
I238	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_P<4>	7 25
I239	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_N<4>	7 25
I240	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_P<5>	7 25
I241	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_N<5>	7 25
I242	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_P<6>	7 25
I243	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_N<6>	7 25
I244	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_P<7>	7 25
I245	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_N<7>	7 25
	Reset				
I246		DDR_50S		MEM_RESET_L	21 23 24
	SM COMP				
I257		DDR_COMP_PHY	DDR_COMP	CPU_SM_RCOMP<0..2>	6

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PAGE TITLE
DDR3 Constraints

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D
REVISION 13.0.0
BRANCH

8 7 6 5 4 3 2 1

PCI Express/DMI

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
PCIE_COMP	*	Y	0.305 MM	0.25 MM	=STANDARD	=STANDARD	=STANDARD
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE3_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	PCIE_COMP
CPU_ASYNC_PHY	*	CPU_50S

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=2:1_SPACING	?
CPU_ASYNC_ISO	*	=2.5:1_SPACING	?
CPU_MS_ISO	TOP,BOTTOM	=4.5:1_SPACING	?
CPU_MS_ISO	*	=3:1_SPACING	?

Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO
PEG_R2D	PEG_R2D	*	PEG_SAME_DIR
PEG_D2R	PEG_D2R	*	PEG_SAME_DIR
PEG_D2R	PEG_R2D	*	PEG_ALT_DIR
PEG_D2R	*	*	PEG_ISO
PEG_R2D	*	*	PEG_ISO
CPU_ASYNC	*	*	CPU_ASYNC_ISO
CPU_ASYNC_MS	*	*	CPU_MS_ISO

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PEG_SAME_DIR	*	=3.5X_DIELECTRIC	?
PEG_ALT_DIR	*	=7X_DIELECTRIC	?
PEG_ISO	*	=4:1_SPACING	?

PEG Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	80	16	15.75	PCIe Gen3. Allow looser spacing for same direction on stripline per Anil

PCIe (CPU)

x16 Graphics

	Electrical Contraint Set	Physical	Spacing	Net	
I440	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<15>	72
I441	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<15>	72
I442		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<15>	72 81
I443		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<15>	72 81
I444	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<15>	72 81
I445	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<15>	72 81
I446		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<15>	72
I447		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<15>	72
I449	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<14>	72
I448	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<14>	72
I450		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<14>	72 81
I452		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<14>	72 81
I451	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<14>	72 81
I453	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<14>	72 81
I455		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<14>	72
I454		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<14>	72
I457	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<13>	72
I456	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<13>	72
I458		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<13>	72 81
I460		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<13>	72 81
I459	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<13>	72 81
I461	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<13>	72 81
I463		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<13>	72
I462		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<13>	72
I464	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<12>	72
I465	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<12>	72
I467		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<12>	72 81
I466		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<12>	72 81
I468	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<12>	72 81
I470	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<12>	72 81
I469		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<12>	72
I471		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<12>	72
I472	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<11>	72
I474	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<11>	72
I473		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<11>	72 81
I475		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<11>	72 81
I477	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<11>	72 81
I476	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<11>	72 81
I478		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<11>	72
I479		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<11>	72
I480	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<10>	72
I482	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<10>	72
I481		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<10>	72 81
I484		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<10>	72 81
I483	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<10>	72 81
I485	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<10>	72 81
I487		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<10>	72
I486		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<10>	72
I489	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<9>	72
I488	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<9>	72
I491		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<9>	72 81
I490		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<9>	72 81
I492	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<9>	72 81
I494	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<9>	72 81
I493		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<9>	72
I496		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<9>	72
I495	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<8>	72
I497	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<8>	72
I498		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<8>	72 81
I499		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<8>	72 81
I501	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<8>	72 81
I500	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<8>	72 81
I502		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<8>	72
I503		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<8>	72

CPU ASYNCHRONOUS

	Electrical Contraint Set	Physical	Spacing	Net	
I573		CPU_ASYNC_PHY	CPU_ASYNC	CPU_PROCHOT_L	6 44 45 61
I574		CPU_ASYNC_PHY	CPU_ASYNC	CPU_PROCHOT_R_L	6
I575	PECI	SMC_3xCTRL	SMC_GEN	CPU_PECI	6 14 44 45
I583	PECI	CPU_ASYNC_PHY	CPU_ASYNC_MS	CPU_PECI_R	45
I577		CPU_ASYNC_PHY	CPU_ASYNC	CPU_CATERR_L	6 45
I584		CPU_ASYNC_PHY	CPU_ASYNC	SMC_CPU_CATERR_L	44 45
I578		CPU_ASYNC_PHY	CPU_ASYNC	CPU_PWRGD	6 14 18
I585		CPU_ASYNC_PHY	CPU_ASYNC	XDP_CPU_PWRGD	18
I586		CPU_ASYNC_PHY	CPU_ASYNC	PCH_PROCPWRGD	14
I579		CPU_ASYNC_PHY	CPU_ASYNC	PM_SYNC	6 12
I580		CPU_ASYNC_PHY	CPU_ASYNC	PM_THRMTRIP_L	14 45
I587		CPU_ASYNC_PHY	CPU_ASYNC	PM_THRMTRIP_L_R	14
I581		CPU_ASYNC_PHY	CPU_ASYNC	CPU_RESET_L	6 14 18
I588		CPU_ASYNC_PHY	CPU_ASYNC	XDP_CPURST_L	18
I582	XDP_BPM_L	CPU_ASYNC_PHY	CPU_ASYNC	XDP_BPM_L<1..0>	6 18

PCIe (CPU)

x16 Graphics

	Electrical Contraint Set	Physical	Spacing	Net	
I504	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<7>	72
I506	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<7>	72
I505		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<7>	72 81
I508		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<7>	72 81
I507	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<7>	72 81
I509	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<7>	72 81
I511		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<7>	72
I510		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<7>	72
I513	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<6>	72
I512	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<6>	72
I515		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<6>	72 81
I514		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<6>	72 81
I516	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<6>	72 81
I518	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<6>	72 81
I517		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<6>	72
I519		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<6>	72
I521	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<5>	72
I520	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<5>	72
I522		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<5>	72 81
I523		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<5>	72 81
I524	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<5>	72 81
I525	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<5>	72 81
I526		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<5>	72
I527		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<5>	72
I528	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<4>	72
I529	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<4>	72
I530		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<4>	72 81
I531		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<4>	72 81
I532	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<4>	72 81
I533	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<4>	72 81
I534		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<4>	72
I535		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<4>	72
I536	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<3>	72
I538	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<3>	72
I537		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<3>	72 81
I539		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<3>	72 81
I541	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<3>	72 81
I540	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<3>	72 81
I543		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<3>	72
I542		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<3>	72
I544	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<2>	72
I546	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<2>	72
I545		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<2>	72 81
I548		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<2>	72 81
I547	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<2>	72 81
I549	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<2>	72 81
I551		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<2>	72
I550		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<2>	72
I552	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_P<1>	72
I554	PCIE_GEN3_R2D	PCIE3_PHY	PEG_R2D	PEG_R2D_N<1>	72
I553		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<1>	72 81
I555		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<1>	72 81
I557	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_P<1>	72 81
I556	PCIE_GEN3_D2R	PCIE3_PHY	PEG_D2R	PEG_D2R_N<1>	72 81
I559		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<1>	72
I558		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<1>	72
I560	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_P<0>	72
I562	PCIE_GEN3_R2D_RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D_N<0>	72
I561		PCIE3_PHY	PEG_R2D	PEG_R2D_C_P<0>	72 81
I564		PCIE3_PHY	PEG_R2D	PEG_R2D_C_N<0>	72 81
I563	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_P<0>	72 81
I565	PCIE_GEN3_D2R_RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R_N<0>	72 81
I567		PCIE3_PHY	PEG_D2R	PEG_D2R_C_P<0>	72
I566		PCIE3_PHY	PEG_D2R	PEG_D2R_C_N<0>	72

CPU PCIe Clocks

I568	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_P	11 72
I569	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_N	11 72

CPU PCIe Compensation

I570		COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_RCOMP	5

CPU eDP Compensation

I571		COMP_PCIE_PHY	COMP_PCIE	CPU_EDP_RCOMP	5
I572		COMP_PCIE_PHY	COMP_PCIE	CPU_CFG_RCOMP	6

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CPU PCIe Constraints

Apple Inc.

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REVISION 13.0.0

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Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_85D
COMP_DMI_PHY	*	DMI_COMP

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_SAME_DIR	*	=3.5X_DIELECTRIC	?
PCIE_ALT_DIR	*	=7X_DIELECTRIC	?
PCIE_ISO	*	=4:1_SPACING	?

TBT x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_TBT_R2D	PCIE_TBT_R2D	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_D2R	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_R2D	*	PCIE_ALT_DIR
PCIE_TBT_D2R	*	*	PCIE_ISO
PCIE_TBT_R2D	*	*	PCIE_ISO

PCH x1 PCIE Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO

DMI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
DMI_SAME_DIR	*	=4X_DIELECTRIC	?
DMI_ALT_DIR	*	=5X_DIELECTRIC	?
DMI_ISO	*	=4X_DIELECTRIC	?

DMI x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_N2S	DMI_N2S	*	DMI_SAME_DIR
DMI_S2N	DMI_S2N	*	DMI_SAME_DIR
DMI_N2S	DMI_S2N	*	DMI_ALT_DIR
DMI_N2S	*	*	DMI_ISO
DMI_S2N	*	*	DMI_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_COMP	*	Y	0.2032 MM	0.2032 MM	3 MM	=STANDARD	=STANDARD

PCIe (PCH)

	Electrical Contraint Set	Physical	Spacing		
	x4 Thunderbolt				
I132	PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_P<3..0>	26
I133	PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_N<3..0>	26
I134		PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_C_P<3..0>	13 26
I135		PCIE_PHY	PCIE_TBT_R2D	PCIE_TBT_R2D_C_N<3..0>	13 26
I136	PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_P<3..0>	13 26
I137	PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_N<3..0>	13 26
I138		PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_C_P<3..0>	26
I139		PCIE_PHY	PCIE_TBT_D2R	PCIE_TBT_D2R_C_N<3..0>	26
I140	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_P	11 26
I141	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_N	11 26
	x1 AirPort				
I142	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_P	32
I143	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_N	32
I144		PCIE_PHY	PCIE	PCIE_AP_R2D_C_P	13 32
I146		PCIE_PHY	PCIE	PCIE_AP_R2D_C_N	13 32
I145	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_P	13 32
I147	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_N	13 32
I148	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_P	11 32
I149	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_N	11 32
	x1 Caesar IV				
I151	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_P	35
I150	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_N	35
I152		PCIE_PHY	PCIE	PCIE_ENET_R2D_C_P	13 35
I154		PCIE_PHY	PCIE	PCIE_ENET_R2D_C_N	13 35
I153	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_P	13 35
I155	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_N	13 35
I156		PCIE_PHY	PCIE	PCIE_ENET_D2R_C_P	35
I157		PCIE_PHY	PCIE	PCIE_ENET_D2R_C_N	35
I158	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_P	11 35
I159	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_N	11 35
	x2 SSD				
I172	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_SSD_P	11 33
I173	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_SSD_N	11 33
	PCH PCIE Compensation				
I178		COMP_DMI_PHY	COMP_PCIE	PCH_PCIE_RCOMP	13

DMI

	Electrical Contraint Set	Physical	Spacing		
	DMI				
I166	DMI_N2S	PCIE_PHY	DMI_N2S	DMI_N2S_P<3..0>	5 12
I165	DMI_N2S	PCIE_PHY	DMI_N2S	DMI_N2S_N<3..0>	5 12
I167	DMI_S2N	PCIE_PHY	DMI_S2N	DMI_S2N_P<3..0>	5 12
I168	DMI_S2N	PCIE_PHY	DMI_S2N	DMI_S2N_N<3..0>	5 12
I169	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_P	6 11
I170	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_N	6 11
	DMI Compensation				
I171		COMP_DMI_PHY	COMP_PCIE	PCH_DMI_RCOMP	12

CPU DP REF CLK

	Electrical Contraint Set	Physical	Spacing		
	CPU DP REF CLK				
I175	CPU_CLK135_PLL	PCIE_PHY	CLK_PCIE	CPU_CLK135M_DPLLREF_N	6 11
I174	CPU_CLK135_PLL	PCIE_PHY	CLK_PCIE	CPU_CLK135M_DPLLREF_P	6 11
I176	CPU_CLK135_PLL	PCIE_PHY	CLK_PCIE	CPU_CLK135M_DPLLSS_N	6 11
I177	CPU_CLK135_PLL	PCIE_PHY	CLK_PCIE	CPU_CLK135M_DPLLSS_P	6 11

SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_85D
COMP_SATA_PHY	*	PCIE_COMP
SATA_PHY_90	*	SATA_90D

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=6:1_SPACING	?
COMP_SATA_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

SATA Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
15.2.1	90	95	20	23.62	SATA Gen2, SATA Gen3

SATA Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
15-3	50	50	15	15.75	SATA Gen2, SATA Gen3

FDI

FDI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FDI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI	*	Y	0.25 MM	0.25 MM	3 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
FDI_SE_PHY	*	FDI_50S
COMP_FDI_PHY	*	COMP_FDI

FDI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FDI_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
FDI	*	*	FDI_ISO
COMP_FDI	*	*	COMP_FDI_ISO

FDI Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
6-1/6-2	85	85	12	11.81	FDI main length

FDI Compensation Rules (mils)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_55S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
CLK_JTAG_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	CLK_JTAG_ISO

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	55	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

SATA

	Electrical Contraint Set	Physical	Spacing		
	PCH SATA Port 0 (HDD)				
I90	SATA_R2D	SATA_PHY_90	SATA	SATA_HDD_R2D_P	33
I92	SATA_R2D	SATA_PHY_90	SATA	SATA_HDD_R2D_N	33
I88		SATA_PHY_90	SATA	SATA_HDD_R2D_C_P	11 33
I89		SATA_PHY_90	SATA	SATA_HDD_R2D_C_N	11 33
I94	SATA_D2R	SATA_PHY_90	SATA	SATA_HDD_D2R_P	11 33
I95	SATA_D2R	SATA_PHY_90	SATA	SATA_HDD_D2R_N	11 33
I91		SATA_PHY_90	SATA	SATA_HDD_D2R_C_P	33
I93		SATA_PHY_90	SATA	SATA_HDD_D2R_C_N	33
	PCH SATA Port 1 (SSD)				
I96	SATA_SSD_R2D	SATA_PHY	SATA	SSD_R2D_P<0..1>	11 33
I98	SATA_SSD_R2D	SATA_PHY	SATA	SSD_R2D_N<0..1>	11 33
I115	SATA_SSD_R2D	SATA_PHY	SATA	SSD_R2D_C_P<0..1>	33
I116	SATA_SSD_R2D	SATA_PHY	SATA	SSD_R2D_C_N<0..1>	33
I103	SATA_SSD_D2R	SATA_PHY	SATA	SSD_D2R_P<0..1>	11 33
I102	SATA_SSD_D2R	SATA_PHY	SATA	SSD_D2R_N<0..1>	11 33
	PCH SATA Compensation				
I104		COMP_SATA_PHY	COMP_SATA	PCH_SATA_RCOMP	11

FDI

	Electrical Contraint Set	Physical	Spacing		
	FDI				
I112		FDI_SE_PHY	FDI	FDI_CSYNC	5 12
I113		FDI_SE_PHY	FDI	FDI_INT	5 12
	FDI Compensation				
I114		COMP_FDI_PHY	COMP_FDI	PCH_FDI_RCOMP	12

XDP

	Electrical Contraint Set	Physical	Spacing		
	CPU XDP				
I37		XDP_PHY	XDP	XDP_BPM_L<7..2>	6 18
I38	CPU_CFG	XDP_PHY	XDP	CPU_CFG<17..4>	6 18 82
I117	CPU_CFG_3	XDP_PHY	XDP	CPU_CFG<3>	6 18
I118	CPU_CFG	XDP_PHY	XDP	CPU_CFG<2..0>	6 18
I39	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_P	
I40	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_N	
I41		CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_P	11 82
I43		CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_N	11 82
I42		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_P	
I44		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_N	
I45		XDP_PHY	CLK_JTAG	XDP_CPU_TCK	6 18
I46		XDP_PHY	XDP	XDP_CPU_TMS	6 18
I47		XDP_PHY	XDP	XDP_CPU_TDI	6 18
I48		XDP_PHY	XDP	XDP_CPU_TDO	6 18
	PCH XDP				
I49		XDP_PHY	CLK_JTAG	XDP_PCH_TCK	11 18
I50		XDP_PHY	XDP	XDP_PCH_TMS	11 18
I51		XDP_PHY	XDP	XDP_PCH_TDI	11 18
I52		XDP_PHY	XDP	XDP_PCH_TDO	11 18

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PAGE TITLE
SATA/FDI/XDP Constraints

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

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PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	=2:1_SPACING	?

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI	*	=2:1_SPACING	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5:1_SPACING	?
CLK_LPC	*	=2:1_SPACING	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

Crystal

Crystal-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=2:1_SPACING	?

PCI

Electrical Contraint Set	Physical	Spacing	Net	Pages
PCI Clock				
I333	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIN	11 19
I334	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIOUT	11 19

LPC

Electrical Contraint Set	Physical	Spacing	Net	Pages
LPC				
I336	LPC_55S	LPC	LPC_AD<3..0>	13 44 46
I335	LPC_55S	LPC	LPC_AD_R<3..0>	13
I338	LPC_55S	LPC	LPC_FRAME_L	13 44 46
I337	LPC_55S	LPC	LPC_FRAME_R_L	13
LPC Clocks				
I339	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	19 46
I340	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	11 19
I341	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	19 44
I342	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	11 19

PCH Clocks

Electrical Contraint Set	Physical	Spacing	Net	Pages
PCH Reference Clock				
I385	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB	11 19
I406	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB_R	11
PCH RTC 32K				
I347	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	11 19
I348	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	11 19
I352	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	19
SMC 32K				
I349	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R	12 45
I350	CLK_PCH_55S	CLK_PCH	SMC_CLK32K	44 45

25 MHz Reference Clocks

Electrical Contraint Set	Physical	Spacing	Net	Pages
25M Reference Crystal				
I379	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1	19
I380	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2	19
I381	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R	19
25M Reference Clocks				
I384	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET	19 35
I386	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET_R	19
I388	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT	19 26
I387	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT_R	26

HDA

Electrical Contraint Set	Physical	Spacing	Net	Pages
HDA				
I353	HDA_55S	HDA	HDA_BIT_CLK	11 52
I361	HDA_55S	HDA	HDA_BIT_CLK_R	11
I363	HDA_55S	HDA	HDA_RST_L	11 52
I362	HDA_55S	HDA	HDA_RST_R_L	11
I364	HDA_55S	HDA	HDA_SDOUT	11 52
I367	HDA_55S	HDA	HDA_SDOUT_R	11 19
I366	HDA_55S	HDA	HDA_SYNC	11 52
I368	HDA_55S	HDA	HDA_SYNC_R	11
I365	HDA_55S	HDA	HDA_SDIN0	11 52
I369	HDA_55S	HDA	AUD_SDI_R	52
SPDIF				
I370		HDA	AUD_SPDIF_CHIP	52
I371		HDA	AUD_SPDIF_OUT	52 56

SPI Bootrom

Electrical Contraint Set	Physical	Spacing	Net	Pages
SPI ROM				
I391	SPI_50S	SPI	SPI_CLK_R	13 46
I392	SPI_50S	SPI	SPI_CLK	46
I393	SPI_50S	SPI	SPI_ALT_CLK	46
I408	SPI_50S	SPI	SPI_SMC_CLK	44 46
I407	SPI_50S	SPI	SPI_MLB_CLK	46
I394	SPI_50S	SPI	SPI_CS0_R_L	13 46
I395	SPI_50S	SPI	SPI_CS0_L	46
I396	SPI_50S	SPI	SPI_ALT_CS_L	46
I409	SPI_50S	SPI	SPI_SMC_CS_L	44 46
I397	SPI_50S	SPI	SPI_MLB_CS_L	46
I399	SPI_50S	SPI	SPI_MOSI_R	13 46
I400	SPI_50S	SPI	SPI_MOSI	46
I401	SPI_50S	SPI	SPI_ALT_MOSI	46
I410	SPI_50S	SPI	SPI_SMC_MOSI	44 46
I411	SPI_50S	SPI	SPI_MLB_MOSI	46
I404	SPI_50S	SPI	SPI_MISO	13 46
I403	SPI_50S	SPI	SPI_ALT_MISO	46
I412	SPI_50S	SPI	SPI_SMC_MISO	44 46
I413	SPI_50S	SPI	SPI_MLB_MISO	46
I405	SPI_50S	SPI	SPIROM_USE_MLB	14 46

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PAGE TITLE
PCH and BR Constraints

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USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D

USB-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=3:1_SPACING	?
USB2_ISO	TOP,BOTTOM	=3:1_SPACING	?
USB3_ISO	*	=5.5:1_SPACING	?
USB3_ISO	TOP,BOTTOM	=5.5:1_SPACING	?

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
12.2.1	90	90	12	11.81	USB 2.0
13.3.1	85	85	20	21.65	USB 3.0

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB3	*	*	USB3_ISO

Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S
CIV_SPI	*	SPI_55S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kV isolation

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

Camera Processor-to-Camera Sensor I/F (SMIA/MIPI)

Camera Processor's SMIA Interface Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMIA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D

Camera Processor's SMIA Interface Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF

USB 3.0 and USB 2.0 Trixies Muxing

	Electrical Contraint Set	Physical	Spacing	Net	Pages
External Port A (J4600)					
I320	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_P	42
I321	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_N	42
I391		USB3_PHY	USB3	USB3_EXTA_RX_F_P	13 42
I392		USB3_PHY	USB3	USB3_EXTA_RX_F_N	13 42
I413	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_P	13 42
I414	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_N	13 42
I415		USB3_PHY	USB3	USB3_EXTA_TX_F_P	42
I416		USB3_PHY	USB3	USB3_EXTA_TX_F_N	42
I493		USB3_PHY	USB3	USB3_EXTA_TX_C_P	42
I494		USB3_PHY	USB3	USB3_EXTA_TX_C_N	42
I417	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_EXTA_0_P	13 42
I418	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_EXTA_0_N	13 42
I419		USB2_PHY	USB2	USB2_EXTA_MUXED_P	42
I420		USB2_PHY	USB2	USB2_EXTA_MUXED_N	42
I393		USB2_PHY	USB2	USB2_EXTA_P	42
I394		USB2_PHY	USB2	USB2_EXTA_N	42
External Port B (J4610)					
I421	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P	42
I422	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N	42
I423		USB3_PHY	USB3	USB3_EXTB_RX_F_P	13 42
I424		USB3_PHY	USB3	USB3_EXTB_RX_F_N	13 42
I425	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P	13 42
I427	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N	13 42
I426		USB3_PHY	USB3	USB3_EXTB_TX_F_P	42
I428		USB3_PHY	USB3	USB3_EXTB_TX_F_N	42
I495		USB3_PHY	USB3	USB3_EXTB_TX_C_P	42
I496		USB3_PHY	USB3	USB3_EXTB_TX_C_N	42
I429	USB2_CONN	USB2_PHY	USB2	USB_EXTB_8_P	13 42
I430	USB2_CONN	USB2_PHY	USB2	USB_EXTB_8_N	13 42
I434		USB2_PHY	USB2	USB2_EXTB_P	42
I433		USB2_PHY	USB2	USB2_EXTB_N	42
External Port C (J4700)					
I437	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_P	43
I439	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_N	43
I438		USB3_PHY	USB3	USB3_EXTC_RX_F_P	13 43
I440		USB3_PHY	USB3	USB3_EXTC_RX_F_N	13 43
I441	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_P	13 43
I442	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_N	13 43
I444		USB3_PHY	USB3	USB3_EXTC_TX_F_P	43
I443		USB3_PHY	USB3	USB3_EXTC_TX_F_N	43
I498		USB3_PHY	USB3	USB3_EXTC_TX_C_P	43
I497		USB3_PHY	USB3	USB3_EXTC_TX_C_N	43
I446	USB2_CONN	USB2_PHY	USB2	USB_EXTC_1_P	13 43
I445	USB2_CONN	USB2_PHY	USB2	USB_EXTC_1_N	13 43
I450		USB2_PHY	USB2	USB2_EXTC_P	43
I452		USB2_PHY	USB2	USB2_EXTC_N	43
External Port D (J4710)					
I453	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_P	43
I455	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_N	43
I454		USB3_PHY	USB3	USB3_EXTD_RX_F_P	13 43
I456		USB3_PHY	USB3	USB3_EXTD_RX_F_N	13 43
I457	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_P	13 43
I458	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_N	13 43
I460		USB3_PHY	USB3	USB3_EXTD_TX_F_P	43
I459		USB3_PHY	USB3	USB3_EXTD_TX_F_N	43
I499		USB3_PHY	USB3	USB3_EXTD_TX_C_P	43
I500		USB3_PHY	USB3	USB3_EXTD_TX_C_N	43
I462	USB2_CONN	USB2_PHY	USB2	USB_EXTD_9_P	13 43
I461	USB2_CONN	USB2_PHY	USB2	USB_EXTD_9_N	13 43
I466		USB2_PHY	USB2	USB2_EXTD_P	43
I468		USB2_PHY	USB2	USB2_EXTD_N	43
Camera (J3510)					
I324	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_P	13 38
I326	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_N	13 38
PCH USB Compensation					
I384		PCH_55S	COMP_PCH	PCH_USB_RBIAS	13

RMH Love

	Electrical Contraint Set	Physical	Spacing	Net	Pages
I409	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_P	13 32
I410	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_N	13 32
I505	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_MUX_P	32
I506	USB2_MUXED_BT	USB2_PHY	USB2	USB_BT_MUX_N	32

Et tu Brute?

	Electrical Contraint Set	Physical	Spacing	Net	Pages
Ethernet					
I474	ENET_MDI	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_P<3..0>	35 36
I475	ENET_MDI	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_N<3..0>	35 36
I476		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_P<3..0>	36
I477		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_N<3..0>	36
I519			ENET_TRANS	ENETCONN_MCT0	36
I520			ENET_TRANS	ENETCONN_MCT1	36
I521			ENET_TRANS	ENETCONN_MCT2	36
I522			ENET_TRANS	ENETCONN_MCT3	36
I523			ENET_TRANS	ENETCONN_MCT_BS	36
I478		ENET_COMP_PHY	COMP_ENET	ENET_RDAC	35
SD					
I479	SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0>	35
I480		SD_PHY	SD	SDCONN_DATA<7..0>	35 37
I515		SD_PHY	SD	SDCONN_DATA_R_<7..0>	
I481	SD_CMD	SD_PHY	SD	ENET_SD_CMD	35
I482		SD_PHY	SD	SDCONN_CMD	35 37
I516		SD_PHY	SD	SDCONN_CMD_R	
I483	SD_CLK	SD_PHY	SD	ENET_SD_CLK	35
I484		SD_PHY	SD	SDCONN_CLK	35 37
I486		SD_PHY	SD	SDCONN_CLK_R	37
I485		SD_PHY	SD	ENET_MEDIA_SENSE	11 35
I488		SD_PHY	SD	ENET_SD_DETECT_L	35 37
CIV SPI					
I490		CIV_SPI	SPI	ENET_SCLK	35
I489		CIV_SPI	SPI	ENET_MISO	35
I492		CIV_SPI	SPI	ENET_MOSI	35
I491		CIV_SPI	SPI	ENET_CS_L	35

Camera Processor-Camera Sensor I/F

	Electrical Contraint Set	Physical	Spacing	Net	Pages
I501	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_P	38
I502	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_N	38
I504	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_P	38
I503	SMIA_DP	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_N	38
I507		SPI_50S	SPI	CAM_SF_CLK	38
I508		SPI_50S	SPI	CAM_SF_CLK_R	38
I509		SPI_50S	SPI	CAM_SF_DIN	38
I510		SPI_50S	SPI	CAM_SF_DIN_R	38
I511		SPI_50S	SPI	CAM_SF_CS_L	38
I513		SPI_50S	SPI	CAM_SF_WP_L	38
I512		SPI_50S	SPI	CAM_SF_DOUT	38
I514		SPI_50S	SPI	CAM_SF_DOUT_R	38
I517		SMB_PHY	SMB	I2C_CAMSENSOR_SDA	38
I518		SMB_PHY	SMB	I2C_CAMSENSOR_SCL	38

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USB/Ethernet/SD Constraints

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SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=1.5:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PWR_P2MM
SENSE	GND	*	GND_P2MM

SMC Generic Control Line Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMC_ISO	*	=1:1_SPACING	?
SMC_3xISO	*	=3:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMC_CTRL	*	*	SMC_ISO
SMC_3XCTRL	*	*	SMC_3xISO

SMC Generic Control Line Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMC_GEN	*	SMC_50S

Current/Voltage Sense

Electrical Contraint Set		Physical	Spacing	Net	Pages
Common					
I3			SENSE	GND_SMC_AVSS	44 45 48 49
12V S5 (System Total)					
I4	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VG3H_P	48
I5	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VG3H_N	48
I6			SENSE	ISNS_P12VG3H_R	48
I7			SENSE	ISNS_P12VG3H	45 48
I8			SENSE	VSNS_P12VG3H	45 48
12V S0 (GPU Core)					
I22	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUCORE_P	48
I23	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUCORE_N	48
I24			SENSE	ISNS_P12VS0_GPUCORE_R	48
I25			SENSE	ISNS_P12VS0_GPUCORE	45 48
I26			SENSE	VSNS_P12VS0_GPUCORE	45 48
HDD					
I80	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_HDD_P	48
I82	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_HDD_N	48
I81			SENSE	ISNS_HDDS0_R	48
I83			SENSE	ISNS_HDDS0	45 48
I84			SENSE	VSNS_HDDS0	45 48
GPU Core (alternate low side sense)					
I119	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNS_GPUCORE_P	
I120	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNS_GPUCORE_N	
I118			SENSE	ISNS_GPUCORE_ALT	45 48
I122			SENSE	VSNS_GPUCORE_ALT	45 48
SSD					
I85	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_SSD_P	49
I87	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_SSD_N	49
I86			SENSE	ISNS_SSDS0_R	49
I88			SENSE	ISNS_SSDS0	45 49
I89			SENSE	VSNS_P3V3S5	45 49
VDDQ S3 (DDR)					
I33	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_VDDQS3_DDR_P	49
I32	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_VDDQS3_DDR_N	49
I35			SENSE	ISNS_VDDQS3_DDR_R	49
I34			SENSE	ISNS_VDDQS3_DDR	45 49
I36			SENSE	VSNS_VDDQS3_DDR	45 49
VDDQ S0 (GPU FBVDDQ)					
I37	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_FBVDDQ_P	48
I38	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P12VS0_FBVDDQ_N	48
I39			SENSE	ISNS_P12VS0_FBVDDQ_R	48
I40			SENSE	ISNS_P12VS0_FBVDDQ	45 48
I130			SENSE	VSNS_P12VS0_FBVDDQ	45 48
CPU Core					
I44			SENSE	ISNS_CPUVCC_FB	48
I45			SENSE	ISNS_CPUVCC	45 48
I46			SENSE	VSNS_CPUVCC	45 48
PP1V05_S0_PCH					
I110			SENSE	VSNS_P1V05S0_PCH	45 48
PP1V5_S0					
I111	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P1V5S0_P	49
I113	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P1V5S0_N	49
I112			SENSE	ISNS_P1V5S0_R	49
I114			SENSE	ISNS_P1V5S0	45 49
I115			SENSE	VSNS_P1V5S0	45 49
Airport					
I102	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P3V3S4_AP_P	
I103	SNS_CURRENT	SNS_DIFF_PHY	SENSE	SNS_P3V3S4_AP_N	
I104			SENSE	ISNS_P3V3S4_AP_R	
I105			SENSE	ISNS_P3V3S4_AP	

SMC

Electrical Contraint Set	Physical	Spacing	Net	Pages
SMC				
I125	CLK_XTAL	XTAL	SMC_XTAL_R	45
I1	CLK_XTAL	XTAL	SMC_XTAL	44 45
I2	CLK_XTAL	XTAL	SMC_EXTAL	44 45
I91	SMC_GEN	SMC_CTRL	SMC_LRESET_L	20 44
I90	SMC_GEN	SMC_CTRL	SMC_RUNTIME_SCI_L	14 44
I92	SMC_GEN	SMC_CTRL	SMC_WAKE_SCI_L	14 44
I93	SMC_GEN	SMC_CTRL	SMC_FAN_0_CTL	44 51
I94	SMC_GEN	SMC_CTRL	SMC_FAN_0_TACH	44 51
I95	SMC_GEN	SMC_3XCTRL	SMC_PECI_L	44 45
I96	SMC_GEN	SMC_3XCTRL	SMC_PECI_L_R	45

SMBus

Electrical Contraint Set	Physical	Spacing	Net	Pages
SMC				
I10	SMB_PHY	SMB	SMBUS_SMC_0_S0_SCL	44 47
I9	SMB_PHY	SMB	SMBUS_SMC_0_S0_SDA	44 47
I11	SMB_PHY	SMB	SMBUS_SMC_1_S0_SCL	44 47
I12	SMB_PHY	SMB	SMBUS_SMC_1_S0_SDA	44 47
I13	SMB_PHY	SMB	SMBUS_SMC_2_S4_SCL	44 47
I14	SMB_PHY	SMB	SMBUS_SMC_2_S4_SDA	44 47
I15	SMB_PHY	SMB	SMBUS_SMC_3_SCL	44 47
I16	SMB_PHY	SMB	SMBUS_SMC_3_SDA	44 47
I19	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SCL	44 45
I20	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SDA	44 45
I126	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SCL	44 45
I127	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SDA	44 45
PCH				
I21	TBT_I2C_55S	TBT_I2C	SMBUS_PCH_CLK	13 47
I52	TBT_I2C_55S	TBT_I2C	SMBUS_PCH_DATA	13 47
I53	SMB_PHY	SMB	SML_PCH_0_CLK	13 47
I54	SMB_PHY	SMB	SML_PCH_0_DATA	13 47
I128	SMB_PHY	SMB	SML_PCH_1_CLK	13 47
I129	SMB_PHY	SMB	SML_PCH_1_DATA	13 47
Display TCon				
I55	SMB_PHY	SMB	SMB_DP_TCON_SCL	40 47
I56	SMB_PHY	SMB	SMB_DP_TCON_SDA	40 47

Temperature Sense

Electrical Contraint Set		Physical	Spacing	Net	Pages
EMC1414-1 (Production)					
I57	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_1_P	50
I58	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_1_N	50
I59	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_3_P	50
I60	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_3_N	50
I65	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_ACDC_P	50 60
I66	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_ACDC_N	50 60
I123		SNS_DIFF_PHY	SENSE	SNS_T1_2_P	50
I124		SNS_DIFF_PHY	SENSE	SNS_T1_2_N	50
TMP423 (Development)					
I67	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_1_P	50
I68	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_1_N	50
I69	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_2_P	50
I70	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_2_N	50
I71	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_3_P	50
I72	SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_3_N	50
HDD Out-of-Band					
I73			SENSE	HDD_OOB1_D2R_L	34
I74			SENSE	HDD_OOB1_D2R_F_L	34
I75			SENSE	HDD_OOB1_D2R_R_L	34
I117			SENSE	SMC_OOB1_D2R_L	34 44
I76			SENSE	SMC_OOB1_R2D_L	34 44
I116			SENSE	SMC_OOB1_R2D_R_L	34
SSD Out-of-Band					
I77			SENSE	SMC_OOB2_R2D_L	33 45
I78			SENSE	SMC_OOB2_D2R_L	33 45

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SMBus/Sensor Constraints

Apple Inc.

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DC-DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GND	*	GND_P5MM
GND	BGA	GND_P3MM
POWER	*	POWER_P6MM
POWER	BGA	POWER_P3MM
VR_CTL_PHY	*	POWER_P3MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_DIDT_PHY	*	POWER_P6MM
VR_DIDT_PHY	BGA	STANDARD

Power-specific Spacing Definitions
Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?
GND_ISO	*	=STANDARD	?

Constraints
Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?

DC-DC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_ISO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO

VDDQ S3 (1.5V)/VTT S0

	Physical	Spacing	Voltage	DIDT	NO_TEST		
Input Bus							
I43	POWER	POWER	5V			REG_V5IN_U7300	63
Local Ground							
I44	GND	GND	0V			AGND_VDDQS3	63
VDDQ S3							
I45	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3	63
I47	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L	63
I46	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VDDQS3	63
I48	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VDDQS3_RC	63
I49	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3	63
I50	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3_R	63
I52	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_VDDQS3	63
I51	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3	63
I53	VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSNS	63
I55	VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF	63
I56	VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN	63
I57	VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE	63
I58	VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP	63
I59	VR_CTL_PHY	VR_CTL				LDO_DDRVTTS0_SNS	63
I61	VR_CTL_PHY	VR_CTL				REG_VDDQS3_VTTREF	63
Output Bus							
I60	POWER	POWER	1.5V			PPVDDQ_S3	80
I62	POWER_DDR	POWER_DDR	0.75V			PPDDRVTT_S0	80
FET Switched							
I63	POWER	POWER	1.5V			PP1V5_S0	80
Sensed							
I64	POWER	POWER	1.5V			PPVDDQ_S3_DDR	80
I67	POWER	POWER	1.5V			PP1V5_S0_PCH	80

PCH/GPU/TBT 1.05V S0

Electrical Contraint Set		Physical	Spacing	Voltage	DIDT	NO_TEST		
Input Bus								
I1		POWER	POWER	5V			REG_VCC_U7400	64
I2		POWER	POWER	5V			REG_PVCC_U7400	64
Local Ground								
I3		GND	GND	0V			AGND_P1V05S0	64
1.05V S0								
I5		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0	64
I4		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0_L	64
I7		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P1V05S0	64
I6		VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P1V05S0_RC	64
I8		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0	64
I9		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0_R	64
I10		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P1V05S0	64
I12		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P1V05S0	64
I11		VR_CTL_PHY	VR_CTL				REG_P1V05S0_OCSET	64
I14		VR_CTL_PHY	VR_CTL				REG_P1V05S0_VO	64
I16	VSNS_GPU_P1V05	SNS_DIFF_PHY	SENSE				SNS_GPU_PEX_IOVDD_P	64 79
I17	VSNS_GPU_P1V05	SNS_DIFF_PHY	SENSE				SNS_GPU_PEX_IOVDD_N	64 79
I19			SENSE				REG_P1V05S0_FB	64
I18			SENSE				REG_P1V05S0_RTN	64
I21		VR_CTL_PHY	VR_CTL				REG_P1V05S0_SREF	64
I20		VR_CTL_PHY	VR_CTL				REG_P1V05S0_FSEL	64
Output Bus								
I22		POWER	POWER	1.05V			PP1V05_S0	80
FET Switched								
I23		POWER	POWER	1.05V			PP1V05_TBTLC	28 80
I66		POWER	POWER	1.05V			PP1V05_TBTCIO	80

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VReg Constraints

Apple Inc.

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CPU VCC Phases

	Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		
	Input Bus							
I836		POWER	POWER	12V			PP12V_S0_CPUVCC_FLT	61 62
I113		POWER	POWER	5V			REG_VCC_U7000	61
	Local Ground							
I126		GND	GND	0V			AGND_CPU	61 62 81
	Phase 1							
I883		VR_CTL_PHY	VR_CTL				REG_TD_1	62
I884		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1	61 62
I885		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1_R	61
I887		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUVCC_1	62
I888		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1	62
I890		VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUVCC_1_RC	62
I889		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUVCC_1	62
I891		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_1	62
I892		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_1	62
I893		POWER	POWER	1.8V			PPCPUVCC_S0_SENSE_1	62
I894	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_1_P	61 62
I895	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_1_N	62
I896			SENSE				REG_ISENVCC_1_NR	61 62
	Phase 2							
I113		VR_CTL_PHY	VR_CTL				REG_TD_2	62
I114		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2	61 62
I114		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2_R	61
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUVCC_2	62
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2	62
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUVCC_2_RC	62
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUVCC_2	62
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_2	62
I114		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_2	62
I114		POWER	POWER	1.8V			PPCPUVCC_S0_SENSE_2	62
I114	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_2_P	61 62
I115	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_2_N	62
I115			SENSE				REG_ISENVCC_2_NR	61 62
	Phase 3							
I115		VR_CTL_PHY	VR_CTL				REG_TD_3	62
I115		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3	61 62
I115		VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3_R	61
I115		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUVCC_3	62
I115		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_3	62
I115		VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUVCC_3_RC	62
I116		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUVCC_3	62
I116		VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_3	62
I116		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_3	62
I116		POWER	POWER	1.8V			PPCPUVCC_S0_SENSE_3	62
I116	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_3_P	61 62
I116	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENVCC_3_N	62
I116			SENSE				REG_ISENVCC_3_NR	61 62

CPU VCC Controller

	Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		
	ISL6372							
I102		VR_CTL_PHY	VR_CTL				REG_CPUVCC_DVC	61
I102		VR_CTL_PHY	VR_CTL				CPUVCC_DVC_RC	61
I102		VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC_2	61
I102		VR_CTL_PHY	VR_CTL				REG_CPUVCC_COMP	61
I102		VR_CTL_PHY	VR_CTL				CPUVCC_COMP_RC	61
I102		VR_CTL_PHY	VR_CTL				REG_CPUVCC_FB	61
I102		VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC	61
I103		VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_1	61
I103		VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_2	61
I103		VR_CTL_PHY	VR_CTL				CPUVCC_PSICOMP_RC	61
I104		VR_CTL_PHY	VR_CTL				REG_CPUVCC_PSICOMP	61
I104		VR_CTL_PHY	VR_CTL				REG_CPUVCC_HFCOMP	61
I103	VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_P	8 61
I104	VSNS_CPU_CORE	SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_N	9 61
I103		SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_R_P	61
I103		SNS_DIFF_PHY	SENSE				CPU_VCCSENSE_R_N	61
I103		SNS_DIFF_PHY	SENSE	1.8V			SNS_VCC_XW_P	61
I103		SNS_DIFF_PHY	SENSE	0V			SNS_VCC_XW_N	61
I103			SENSE				REG_CPUVCC_VSEN	61
I103			SENSE				REG_CPUVCC_RGND	61
I104			SENSE				REG_CPUVCC_VIN	61
I104		VR_CTL_PHY	VR_CTL				REG_CPUVCC_IMON	48 61
I104		VR_CTL_PHY	VR_CTL				CPUVCC_IMON_R	61
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_TM	61
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_IMX	61
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_NPSI	61
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_FDVID	61
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_TMX	61
I105		VR_CTL_PHY	VR_CTL				REG_CPUVCC_MEMVRSEL	61
I106		VR_CTL_PHY	VR_CTL				REG_CPUVCC_RSET	61
I106	CPU_VIDSCLK	VR_VID_PHY	VR_VID				CPU_VIDSCLK	8 61
I106		VR_VID_PHY	VR_VID				CPU_VIDSCLK_R	8
I106	CPU_VIDALERT_L	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	8 61
I106		VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	8
I106	CPU_VIDSOUT	VR_VID_PHY	VR_VID				CPU_VIDSOUT	8 61
I106		VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	8
	Output Bus							
I106		POWER	POWER	1.8V			PPCPUVCC_S0_CPU	80

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CPU VReg Constraints

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GPU Core Phases and Controller

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus							
I126	POWER	POWER	12V			PPVIN_S0_GFXIMVP_R	67
I126	POWER	POWER	5V			PP5V_S0_GFXIMVP_VDD	67
Phase 1							
I127	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_1	67
I127	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_1	67
I127	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_1_RC	67
I127	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_1	67
I127	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_1	67
I127	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_ISEN_GPUCORE_1	67
I127	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_GPUCORE_1	67
Phase 2							
I129	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_2	67
I129	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_2	67
I130	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_2_RC	67
I130	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_2	67
I130	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_2	67
I130	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_ISEN_GPUCORE_2	67
I130	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_GPUCORE_2	67
ISL62882C							
I147	VR_CTL_PHY	VR_CTL				REG_GPUCORE_ISUMN	67
I129	VR_CTL_PHY	VR_CTL				REG_GPUCORE_ISUMN_R	67
I147	VR_CTL_PHY	VR_CTL				REG_GPUCORE_ISUMP	67
I147	VR_CTL_PHY	VR_CTL				REG_GPUCORE_ISUMP_C	67
I129	VR_CTL_PHY	VR_CTL				REG_GPUCORE_COMP	67
I129	VR_CTL_PHY	VR_CTL				REG_GPUCORE_COMP_R	67
I147	VR_CTL_PHY	VR_CTL				REG_GPUCORE_VW	67
I131	VR_CTL_PHY	VR_CTL				REG_GPUCORE_FB	67
I147	VR_CTL_PHY	VR_CTL				REG_GPUCORE_FB2	67
I130 VSNS_GPU_CORE	SNS_DIFF_PHY	SENSE				SNS_GPU_CORE_P	67 79
I130 VSNS_GPU_CORE	SNS_DIFF_PHY	SENSE				SNS_GPU_CORE_N	67 79
I146 ISNS_GPU_CORE_PH1	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS1_P	67
I146 ISNS_GPU_CORE_PH1	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS1_N	67
I146 ISNS_GPU_CORE_PH2	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS2_P	67
I146 ISNS_GPU_CORE_PH2	SNS_DIFF_PHY	SENSE				REG_GPUCORE_ISNS2_N	67
I146	GND	GND	0V			GND_REG_GPUCORE_AGND	67
Output Bus							
I129	POWER	POWER	1.0V			PPVCORE_S0_GPU	80
I146	POWER	POWER	1.0V			PPVCORE_S0_GFX_PH2	67
I146	POWER	POWER	1.0V			PPVCORE_S0_GFX_PH1	67

GPU VDDQ

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus							
I141	POWER	POWER	5V			REG_VCC_U7750	66
I143	POWER	POWER	5V			REG_PVCC_U7750	66
Local Ground							
I141	GND	GND	0V			AGND_GPUVDDQ	66
GPU VDDQ							
I141	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUVDDQ	66
I142	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUVDDQ	66
I142	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUVDDQ_RC	66
I142	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUVDDQ	66
I142	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUVDDQ	66
I147	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_GPUVDDQ	66
I142	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_OCSET	66
I142	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_VO	66
I143 VSNS_GPU_VDDQ	SNS_DIFF_PHY	SENSE				SNS_GPUVDDQ_P	66 74
I143 VSNS_GPU_VDDQ	SNS_DIFF_PHY	SENSE				SNS_GPUVDDQ_N	66 74
I143		SENSE				REG_GPUVDDQ_FB	66
I143		SENSE				REG_GPUVDDQ_RTN	66
I143	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_SREF	66
I143	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_FSEL	66
I144	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_SET0	66
I144	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_SET1	66
I144	VR_CTL_PHY	VR_CTL				REG_GPUVDDQ_SET1_R	66
Output Bus							
I143	POWER	POWER	1.5V			PPVDDQ_S0_GPU	80

3.3V S5/5V S4

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus							
I843	POWER	POWER	12V			REG_VIN_U7600	65
I125	POWER	POWER	5V			REG_VCC1_U7600	65
I125	POWER	POWER	5V			REG_VCC2_U7600	65
3.3V S5							
I849	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P3V3S5	65
I847	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P3V3S5	65
I851	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P3V3S5_RC	65
I848	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P3V3S5	65
I850	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P3V3S5	65
I852	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P3V3S5	65
I107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_ISEN	65
I107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_OCSET	65
I113	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FSET	65
I108	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT	65
I107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT_R	65
I107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FB	65
5V S3							
I113	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P5VS4	65
I113	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P5VS4	65
I113	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P5VS4_RC	65
I114	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P5VS4	65
I114	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P5VS4	65
I114	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P5VS4	65
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_ISEN	65
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_OCSET	65
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_FSET	65
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT	65
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT_R	65
I114	VR_CTL_PHY	VR_CTL				REG_P5VS4_FB	65
Output Bus							
I115	POWER	POWER	5V			PP5V_S5	80
I115	POWER	POWER	5V			PP5V_S4	80
I114	POWER	POWER	3.3V			PP3V3_S5	80
FET Switched							
I123	POWER	POWER	5V			PP5V_S0	80
I121	POWER	POWER	3.3V			PP3V3_S4	80
I123	POWER	POWER	3.3V			PP3V3_S0	80
I145	POWER	POWER	3.3V			PP3V3_S0_SSD	80
I121	POWER	POWER	3.3V			PP3V3_ENET	80
I138	POWER	POWER	3.3V			PP3V3_TBTLC	80
Sensed							
I145	POWER	POWER	3.3V			PPSSD_S0	80
I147	POWER	POWER	3.3V			PP3V3_S4_AP	

3.42V G3H

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
3.42V G3H							
I144	POWER	VR_SWITCH	12V			P3V42G3H_BOOST	60
I144	POWER	VR_SWITCH	12V			P3V42G3H_SW	60
I144	VR_CTL_PHY	VR_CTL				P3V42G3H_FB	60
I144	VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L	60
Output Bus							
I144	POWER	POWER	3.425V			PP3V42_G3H	80

3.3V G3

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
I147	POWER	POWER	3.3V			PP3V3_G3	80

HDD S0

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
FET Switched							
I139	POWER	POWER	5V			PPHDD_S0	80
Sensed							
I146	POWER	POWER	5V			PP5V_S0_HDD	80

12V

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus							
I144	POWER	POWER	12V			PP12V_ACDC	80
FET Switched							
I144	POWER	POWER	12V			PP12V_S5	80
I145	POWER	POWER	12V			PP12V_S0	80
Sensed							
I145	POWER	POWER	12V			PP12V_G3H	80
I145	POWER	POWER	12V			PP12V_S0_GPUCORE	80
I145	POWER	POWER	12V			PP12V_S0_FBVDDQ	80

4V5_*

	Physical	Spacing	Netname	Page
I147	POWER	POWER	4V5_REG_IN	52

Ground/Common

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net
Common						
I139	GND	GND	0V			GND

SYNC MASTER=MASTER SYNC DATE=04/21/2013

PAGE TITLE
Platform VReg Constraints

Apple Inc.

DRAWING NUMBER 051-9889 SIZE D

REVISION 13.0.0

BRANCH

Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTDP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_I2C	*	=2x_DIELECTRIC	?
TBT_SPI	*	=2x_DIELECTRIC	?
TBTDP	*	=5x_DIELECTRIC	?
TBTDP	TOP,BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

Pairs should be within 100 mils of clock length.
Max length of DisplayPort traces: 12 inches

DisplayPort intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX channel intra-pair matching should be 5 ps. No relationship to other signals.

TBT IC Net Properties

	Electrical Contraint Set	Physical	Spacing	Netname	
I511		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	26 77
I513		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	26 77
I512	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	26
I514	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	26
I516		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P	26 77
I515		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N	26 77
I517	DP_TBTSNK0_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_P	26
I518	DP_TBTSNK0_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_N	26
I519		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	26 77
I521		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	26 77
I520	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	26
I523	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	26
I522		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P	26 77
I524		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N	26 77
I526	DP_TBTSNK1_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_P	26
I525	DP_TBTSNK1_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_N	26
I603	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_P<3..0>	41 *
I604	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_N<3..0>	41 *
I686	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	41 *
I687	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	41 *
I529	DP_INTPNL_TBT_AUX_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_P	41 *
I530	DP_INTPNL_TBT_AUX_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_N	41 *
I601		DP_85D	DISPLAYPORT	DP_TBTSRC_AUX_C_P	41 *
I602		DP_85D	DISPLAYPORT	DP_TBTSRC_AUX_C_N	41 *
I531		TBT_I2C_55S	TBT_I2C	I2C_TBTRTR_SCL	26
I532		TBT_I2C_55S	TBT_I2C	I2C_TBTRTR_SDA	26
I533	TBT_SPI_CLK	TBT_SPI_55S	TBT_SPI	TBT_SPI_CLK	26
I534	TBT_SPI_MOSI	TBT_SPI_55S	TBT_SPI	TBT_SPI_MOSI	26
I538	TBT_SPI_MISO	TBT_SPI_55S	TBT_SPI	TBT_SPI_MISO	26
I535	TBT_SPI_CS_L	TBT_SPI_55S	TBT_SPI	TBT_SPI_CS_L	26

*: Only used on hosts supporting T29 video-in

DisplayPort

	Electrical Contraint Set	Physical	Spacing	Netname	
	Graphics Source				
I642	DP_INTPNL_EG_ML_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_ML_P<1..0>	41 77
I643	DP_INTPNL_EG_ML_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_ML_N<1..0>	41 77
I644	DP_INTPNL_EG_AUX_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_P	41 77
I645	DP_INTPNL_EG_AUX_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_N	41 77
I646		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_C_P	41
I647		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_C_N	41
	Internal Panel				
I605		DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_P<3..0>	41
I606		DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_N<3..0>	41
I607	DP_INTPNL_ML_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_ML_P<3..0>	40 41
I608	DP_INTPNL_ML_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_ML_N<3..0>	40 41
I609	DP_INTPNL_AUX_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_P	40 41
I610	DP_INTPNL_AUX_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_N	40 41
	Internal DP SPDIF				
I611			HDA	DP_INT_SPDIF_AUDIO	40 52

TBT/DP Net Properties

	Electrical Contraint Set	Physical	Spacing	Netname	
	Port A				
I541	TBT_A_R2D1	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<1>	26 29
I542	TBT_A_R2D1	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<1>	26 29
I689	TBT_A_R2D0	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<0>	26 29
I688	TBT_A_R2D0	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<0>	26 29
I544		TBTDP_90D	TBTDP	TBT_A_R2D_P<1..0>	29
I543		TBTDP_90D	TBTDP	TBT_A_R2D_N<1..0>	29
I595	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	26 29
I596	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	26 29
I680	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	26 29
I681	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	26 29
I545		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	29
I547		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	29
I678		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	29
I679		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	29
I546	DP_A_LSX	DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>	29
I548	DP_A_LSX	DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>	29
I549		TBTDP_90D	TBTDP	TBT_A_D2R_C_P<1..0>	29
I550		TBTDP_90D	TBTDP	TBT_A_D2R_C_N<1..0>	29
I564	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_P<1>	26 29
I563	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_N<1>	26 29
I675	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_P<0>	26 29
I674	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_N<0>	26 29
I599	TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_P	26 29
I600	TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_N	26 29
I565		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_P	29
I566		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_N	29
I648	DP_A_AUXCH_DDC	DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_P	29
I649	DP_A_AUXCH_DDC	DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_N	29
I650		TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_P	29
I651		TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_N	29
	Port B				
I652	TBT_B_R2D1	TBTDP_90D	TBTDP	TBT_B_R2D_C_P<1>	26 30
I653	TBT_B_R2D1	TBTDP_90D	TBTDP	TBT_B_R2D_C_N<1>	26 30
I690	TBT_B_R2D0	TBTDP_90D	TBTDP	TBT_B_R2D_C_P<0>	26 30
I691	TBT_B_R2D0	TBTDP_90D	TBTDP	TBT_B_R2D_C_N<0>	26 30
I655		TBTDP_90D	TBTDP	TBT_B_R2D_P<1..0>	30
I654		TBTDP_90D	TBTDP	TBT_B_R2D_N<1..0>	30
I657	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	26 30
I658	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	26 30
I685	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	26 30
I684	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	26 30
I683		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	30
I682		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	30
I656		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	30
I660		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	30
I659	DP_B_LSX	DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>	30
I662	DP_B_LSX	DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>	30
I661		TBTDP_90D	TBTDP	TBT_B_D2R_C_P<1..0>	30
I663		TBTDP_90D	TBTDP	TBT_B_D2R_C_N<1..0>	30
I665	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_P<1>	26 30
I664	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_N<1>	26 30
I676	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_P<0>	26 30
I677	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_N<0>	26 30
I666	TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_P	26 30
I667	TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_N	26 30
I669		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_P	30
I670		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_N	30
I668	DP_B_AUXCH_DDC	DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_P	30
I672	DP_B_AUXCH_DDC	DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_N	30
I671		TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_P	30
I673		TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_N	30

JTAG_*

	Physical	Spacing	Netname	
I692	XDP_PHY	XDP	JTAG_TBT_TDI	20 26
I693	XDP_PHY	XDP	JTAG_TBT_TDO	20 26
I694	XDP_PHY	XDP	JTAG_TBT_TMS	20 26

DP_*

	Physical	Spacing	Netname	
I695	TBT_I2C_55S	TBT_I2C	DP_TBTPA_DDC_CLK	29 31
I696	TBT_I2C_55S	TBT_I2C	DP_TBTPA_DDC_DATA	29 31
I697	TBT_I2C_55S	TBT_I2C	DP_TBTPB_DDC_CLK	30 31
I698	TBT_I2C_55S	TBT_I2C	DP_TBTPB_DDC_DATA	30 31

TBT_*

	Physical	Spacing	Voltage	Netname	
I700	TBT_I2C_55S	TBT_I2C		TBT_A_CONFIG1_RC	29
I699	TBT_I2C_55S	TBT_I2C		TBT_B_CONFIG1_RC	30

SYNC_MASTER=MASTER SYNC_DATE=04/21/2013

PAGE TITLE
TBT/DP Constraints

Apple Inc.

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BRANCH

GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

	Trace-to-Trace				Isolation				
Table	Micro	Design	Strip	Design	Micro	Design	Strip	Design	Comments
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Memory address (MA). Implented 4.5 Gbps or less rules for K70.
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Address dynamic bus inversion (ADBI)
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Control (CTRL)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Clock (CLK)
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data (DQ)
5-6/5-7	7:1	7:1	7:1	7:1	7:1	7:1	7:1	7:1	Error detection pins (EDC). Using larger isolation rules,
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data dynamic bus inversion (DBI)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Forwarded clock (WCK)

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=5x_DIELECTRIC	?
GDDR_ISO	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=5x_DIELECTRIC	?
GDDR_CLK2CLK	TOP,BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP,BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=7x_DIELECTRIC	?
GDDR_EDC_ISO	TOP,BOTTOM	=7x_DIELECTRIC	?
GDDR_EDC2EDC	*	=7x_DIELECTRIC	?
GDDR_EDC2EDC	TOP,BOTTOM	=7x_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP,BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=5x_DIELECTRIC	?
GDDR_WCK2WCK	TOP,BOTTOM	=5x_DIELECTRIC	?

Constraints (x in {A, B}, y in {0, 1})

Memory Address: MAxy[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

Data: DQxy[31:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DQ	*	*	GDDR_ISO
GDDR_*_*_DQ	=SAME	*	GDDR_DQ2DQ

Error Detection: EDCxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_EDC	*	*	GDDR_EDC_ISO
GDDR_*_*_EDC	=SAME	*	GDDR_EDC2EDC

Data Dynamic Bus Inversion: DDBIxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DBI	*	*	GDDR_ISO
GDDR_*_*_DBI	=SAME	*	GDDR_DBI2DBI

Forwarded Clock: WCKxy[1:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_WCK	*	*	GDDR_ISO
GDDR_*_*_WCK	=SAME	*	GDDR_WCK2WCK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

	Electrical Contraint Set	Physical	Spacing		
	Memory Address				
I642	GDDR_A0_MA	GDDR_MA_PHY	GDDR_A_0_MA	FB_A0_A<8..0>	74 75
I643	GDDR_A1_MA	GDDR_MA_PHY	GDDR_A_1_MA	FB_A1_A<8..0>	74 75
	Address Dynamic Bus Inv				
I644	GDDR_A0_ADBI	GDDR_ADBI_PHY	GDDR_A_0_ADBI	FB_A0_ABI_L	74 75
I645	GDDR_A1_ADBI	GDDR_ADBI_PHY	GDDR_A_1_ADBI	FB_A1_ABI_L	74 75
	Control				
I647	GDDR_A0_CKE	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CKE_L	74 75
I648	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CS_L	74 75
I649	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_WE_L	74 75
I650	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CAS_L	74 75
I651	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_RAS_L	74 75
I652	GDDR_A1_CKE	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CKE_L	74 75
I654	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CS_L	74 75
I653	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_WE_L	74 75
I655	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CAS_L	74 75
I656	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_RAS_L	74 75
	Clock				
I657	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_P	74 75
I658	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_N	74 75
I660	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_P	74 75
I659	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_N	74 75
	Data				
I661	GDDR_A0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<7..0>	74 75
I663	GDDR_A0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<15..8>	74 75
I662	GDDR_A0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<23..16>	74 75
I664	GDDR_A0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<31..24>	74 75
I666	GDDR_A1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<7..0>	74 75
I665	GDDR_A1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<15..8>	74 75
I667	GDDR_A1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<23..16>	74 75
I668	GDDR_A1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<31..24>	74 75
	Error Detection				
I670	GDDR_A0_EDC0	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<0>	74 75
I669	GDDR_A0_EDC1	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<1>	74 75
I671	GDDR_A0_EDC2	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<2>	74 75
I672	GDDR_A0_EDC3	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<3>	74 75
I673	GDDR_A1_EDC0	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<0>	74 75
I675	GDDR_A1_EDC1	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<1>	74 75
I674	GDDR_A1_EDC2	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<2>	74 75
I676	GDDR_A1_EDC3	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<3>	74 75
	Data Dynamic Bus Inv				
I677	GDDR_A0_DBI0	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<0>	74 75
I678	GDDR_A0_DBI1	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<1>	74 75
I680	GDDR_A0_DBI2	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<2>	74 75
I679	GDDR_A0_DBI3	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<3>	74 75
I682	GDDR_A1_DBI0	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<0>	74 75
I681	GDDR_A1_DBI1	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<1>	74 75
I683	GDDR_A1_DBI2	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<2>	74 75
I684	GDDR_A1_DBI3	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<3>	74 75
	Forwarded Clock				
I685	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<0>	74 75
I687	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<0>	74 75
I686	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<1>	74 75
I688	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<1>	74 75
I689	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<0>	74 75
I690	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<0>	74 75
I692	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<1>	74 75
I691	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<1>	74 75

GDDR5 Frame Buffer B

	Electrical Contraint Set	Physical	Spacing		
	Memory Address				
I695	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	74 76
I694	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	74 76
	Address Dynamic Bus Inv				
I696	GDDR_B0_ADBI	GDDR_ADBI_PHY	GDDR_B_0_ADBI	FB_B0_ABI_L	74 76
I697	GDDR_B1_ADBI	GDDR_ADBI_PHY	GDDR_B_1_ADBI	FB_B1_ABI_L	74 76
	Control				
I698	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	74 76
I701	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	74 76
I700	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	74 76
I702	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	74 76
I703	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	74 76
I704	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	74 76
I705	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	74 76
I706	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	74 76
I708	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	74 76
I707	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	74 76
	Clock				
I693	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	74 76
I710	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	74 76
I709	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	74 76
I711	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	74 76
	Data				
I713	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	74 76
I712	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	74 76
I714	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	74 76
[illegible]	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	74 76
[illegible]	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	74 76
[illegible]	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	74 76
[illegible]	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	74 76
[illegible]	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	74 76
	Error Detection				
[illegible]	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	74 76
[illegible]	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	74 76
[illegible]	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	74 76
[illegible]	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	74 76
[illegible]	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	74 76
[illegible]	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	74 76
[illegible]	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	74 76
[illegible]	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	74 76
	Data Dynamic Bus Inv				
[illegible]	GDDR_B0_DBI0	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<0>	74 76
[illegible]	GDDR_B0_DBI1	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<1>	74 76
[illegible]	GDDR_B0_DBI2	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<2>	74 76
[illegible]	GDDR_B0_DBI3	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<3>	74 76
[illegible]	GDDR_B1_DBI0	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<0>	74 76
[illegible]	GDDR_B1_DBI1	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<1>	74 76
[illegible]	GDDR_B1_DBI2	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<2>	74 76
[illegible]	GDDR_B1_DBI3	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<3>	74 76
	Forwarded Clock				
[illegible]	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	74 76
[illegible]	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	74 76
[illegible]	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	74 76
[illegible]	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	74 76
[illegible]	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	74 76
[illegible]	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<0>	74 76
[illegible]	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<1>	74 76
[illegible]	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<1>	74 76

GPU

	Electrical Contraint Set	Physical	Spacing		
	Clocks				
I756		CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_O_PL	72
I757		CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_O_NG	72
I759		CLK_PCIE_PHY	CLK_PCIE	GPU_PEX_TERMP	72
I768		CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK	77 78
I769		CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK	77 78
I780	SPI_50S	SPI		GPU_ROM_SI	77 78
I781	SPI_50S	SPI		GPU_ROM_SO	77 78
I771		CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_IN	
I772		CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_OUT	
I782	XDP_PHY	XDP		GPU_JTAG_TDI	77 78
I783	XDP_PHY	XDP		GPU_JTAG_TDO	77 78
I784	XDP_PHY	XDP		GPU_JTAG_TMS	77 78
I785	XDP_PHY	XDP		GPU_JTAG_TRST_L	77 78
	SMB				
I773		SMB_PHY	SMB	GPU_SMB_CLK	77 78
I774		SMB_PHY	SMB	GPU_SMB_DAT	77 78
I775		SMB_PHY	SMB	GPU_SMB_CLK_R	47 78
I776		SMB_PHY	SMB	GPU_SMB_DAT_R	47 78
	PCIe Compensation				
I777		PCIE_50S	COMP_PCIE	FB_CAL_PD_VDDQ	74
I778		PCIE_50S	COMP_PCIE	FB_CAL_PU_GND	74
I779		PCIE_50S	COMP_PCIE	FB_CAL_TERM_GND	74

Frame Buffer Reset

	Electrical Contraint Set	Physical	Spacing		
	Reset				
[illegible]	GDDR_A0_RESET	GDDR_50S	GDDR_CTRL	FB_A0_RESET_L	74 75
[illegible]	GDDR_A1_RESET	GDDR_50S	GDDR_CTRL	FB_A1_RESET_L	74 75
[illegible]	GDDR_B0_RESET	GDDR_50S	GDDR_CTRL	FB_B0_RESET_L	74 76
[illegible]	GDDR_B1_RESET	GDDR_50S	GDDR_CTRL	FB_B1_RESET_L	74 76

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GDDR5/GPU Constraints

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Backlight Controller

BLC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BLC_P6MM	*	Y	0.600 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD
BLC_P3MM	*	Y	0.300 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_BLC	*	BLC_P6MM
POWER_BLC_RET	*	BLC_P3MM
BLC_CTL_PHY	*	BLC_P3MM

BLC-specific Spacing Definitions
BLC High Voltage Output

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_HV_ISO	*	0.45mm	1000

Constraints
BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_CTL	*	BLC_CTL_ISO
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

BLC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PHASE_ISO	*	=8:1_SPACING	2000
PHASE_SW2SW	*	=1:1_SPACING	?
PHASE_SW2PWR	*	=2:1_SPACING	?
PHASE_SW2GND	*	=2:1_SPACING	?

BLC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_CTL_ISO	*	=3:1_SPACING	?

BLC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Is it chel'oh or sel'oh?

	Physical	Spacing	Voltage	DIDT	NO_TEST		
Input Bus							
I802	POWER	POWER	12V			PP12V_BKLT_SNS	68
I749	POWER	POWER	12V			PP12V_BKLT_FUSED	68
I750	POWER	POWER	12V			PP12V_S0_BKLT_FILT	68
I772	POWER	POWER	12V			PP12V_S0_BKLT_PWR	68
I794	POWER	POWER	12V			PP12V_S0_BKLT_PWR_R	68
I795	POWER	POWER	5V			PP5V_S0_BKLT_R	68
I796	POWER	POWER	3.3V			PP3V3_S0_BKLT_VDDIO_R	68
Local Ground							
I751	BLC_CTL_PHY	BLC_PHASE	0V			PGND_BKLT	68
I799	BLC_CTL_PHY	BLC_PHASE	0V			DGND_BKLT	68
I800	BLC_CTL_PHY	BLC_PHASE	0V			LGND_BKLT	68
Backlight							
I752	POWER_BLC	BLC_PHASE	80V	TRUE		BKLT_PHASE	68
I753	BLC_CTL_PHY	BLC_PHASE	80V	TRUE		BKLT_GATE	68
I754	BLC_CTL_PHY	BLC_PHASE	80V	TRUE		BKLT_GATE_R	68
I755	BLC_CTL_PHY	BLC_PHASE	80V	TRUE		BKLT_SNUBBER	68
I757	BLC_CTL_PHY	BLC_PHASE	12V	TRUE		BKLT_SW_R	68
I759	BLC_CTL_PHY	BLC_CTL				BKLT_ISET	68
I763	BLC_CTL_PHY	BLC_CTL				BKLT_FLT	68
I765	BLC_CTL_PHY	BLC_CTL				BKLT_FLT_RC	68
I797	SNS_DIFF_PHY	SENSE				BKLT_SW_P	68
I798	SNS_DIFF_PHY	SENSE				BKLT_SW_N	68
I762		SENSE				BKLT_FB	68
I801		BLC_HV	67V			BKLT_FB_XW	68
I764		BLC_HV	67V			BKLT_FB_R	68
I776	POWER_BLC_RET	BLC_CTL				BKLT_ISEN1	68
I775	POWER_BLC_RET	BLC_CTL				BKLT_ISEN2	68
I778	POWER_BLC_RET	BLC_CTL				BKLT_ISEN3	68
I777	POWER_BLC_RET	BLC_CTL				BKLT_ISEN4	68
I780	POWER_BLC_RET	BLC_CTL				BKLT_ISEN5	68
I779	POWER_BLC_RET	BLC_CTL				BKLT_ISEN6	68
I789	POWER_BLC_RET	BLC_HV				BKLT_ISEN1_R	68
I788	POWER_BLC_RET	BLC_HV				BKLT_ISEN2_R	68
I790	POWER_BLC_RET	BLC_HV				BKLT_ISEN3_R	68
I791	POWER_BLC_RET	BLC_HV				BKLT_ISEN4_R	68
I792	POWER_BLC_RET	BLC_HV				BKLT_ISEN5_R	68
I793	POWER_BLC_RET	BLC_HV				BKLT_ISEN6_R	68
I783	POWER_BLC_RET	BLC_HV				LED_RETURN_1	68
I782	POWER_BLC_RET	BLC_HV				LED_RETURN_2	68
I784	POWER_BLC_RET	BLC_HV				LED_RETURN_3	68
I785	POWER_BLC_RET	BLC_HV				LED_RETURN_4	68
I786	POWER_BLC_RET	BLC_HV				LED_RETURN_5	68
I787	POWER_BLC_RET	BLC_HV				LED_RETURN_6	68
Output Bus							
I770	POWER_BLC	BLC_HV	67V			BKLT_BOOST	68
I773	POWER_BLC	BLC_HV	67V			BKLT_BOOST_1	68
I774	POWER_BLC	BLC_HV	67V			BKLT_BOOST_2	68

Cello Miscellaneous

Electrical Contraint Set	Physical	Spacing		
SPI				
I564	SMB_PHY	SMB	BKLT_SCL	68
I563	SMB_PHY	SMB	BKLT_SDA	68

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BLC Constraints

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